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Harnessing the Power of 2D Materials for Flexible Energy Harvesting Applications

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ABSTRACT

Capturing of ambient energy is emerging as a transformative area in energy technology, potentially replacing batteries or significantly extending their lifespan. Harnessing of energy from ambient sources presents a significant opportunity to support sustainable development while mitigating environmental issues. Repurposing energy that would otherwise be wasted from high-consumption systems such as engines and industrial furnaces is essential for reducing ecological footprints and moving toward carbon-neutral goals. Furthermore, compact energy harvesting technologies will play a pivotal role in powering the rapidly expanding Internet of Things, enabling innovative advancements in smart homes, cities, industries, and health care that elevate our living standards. To achieve significant advancements in energy harvesting technologies, the development of innovative materials is crucial for converting ambient energy into electricity. In this regard, two-dimensional (2D) materials, a rising star in the material world, are profoundly and technologically intriguing for energy harvesting. The exceptional atomic thickness, high surface-to-volume ratio, flexibility, and tunable band gap effectively enhance their electronic, optical, and chemical properties, making them a potential candidate for use in flexible electronics and wearable energy harvesting technologies. Consequently, these unique properties of 2D materials remarkably enhance their energy harvesting capabilities, including photovoltaic, triboelectric, thermoelectric, and piezoelectric energy harvesting. Here, we present a tutorial-style review of 2D materials for harvesting energy from different ambient sources (aimed particularly at guiding and educating researchers, especially those new to the field), which starts with a brief overview of the promising properties of 2D materials for energy harvesting, then looks deeply into its advantages as compared to traditional materials along with their 3D counterparts, followed by providing insight into the mechanisms and performance of 2D material-based energy harvesters in portable/wearable electronics, and finally, based on current progress, an overview of the challenges along with corresponding strategies are identified and discussed.

1 | Introduction

Significant attempts have been made to produce sustainable and easily integrable power sources, and “smart living” is generally referred to as the Internet of Things (IoT), driven by the paradigm shift in energy, facilitated by the rapid development of

flexible, portable gadgets, wearable electronics, and micro-electromechanical systems [1, 2]. In this context, many autonomous microelectronic devices and sensors spread throughout the human body and all over the planet will form a massive communication network in it. This emphasizes how urgently a safe and reliable micro-sized power supply for such

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microelectronic devices must be provided [3]. Therefore, a lot of attention has been paid in the past for advancing these micro-sized energy devices like micro-supercapacitors (SCs) and micro-batteries. However, considering their low energy density and the challenge of replacing batteries when installed in far-flung areas, it is impractical to power these widespread micro-electronic devices only with traditional energy storage devices. Consequently, making a self-powered system by miniaturizing energy harvesting modules has been viable for realizing affordable, robust, and compact microelectronic devices. Therefore, integration of energy harvesters is highly anticipated to harness diverse forms of energy and power smart systems anytime and anywhere, effectively utilizing waste energy for electricity generation. Solar energy, wind power, mechanical energy from human activity, and thermal energy from the human body and surroundings are a few examples of readily available energy acquired from the ambient environment [4]. Many efficient energy harvesting methods have been demonstrated recently, thanks to the development of nanotechnology and nanoscale materials. Moreover, the utilization of innovative materials and methods has significantly enhanced the efficiency of solar and thermal energy harvesting systems. As a result of these ambient energy resources, recently, many energy harvesting technologies like thermoelectric, piezoelectric, triboelectric, and photovoltaic (PV) have emerged [5]. In terms of mechanical energy, triboelectric nanogenerators (TEGs) and piezoelectric nanogenerators (PENGs), known as low-frequency energy harvesters, effectively harvest energy from wind and water waves, respectively, and also provide distinctive advantages like being lightweight, portable, and flexible [6–8]. Compared to PENGs and TEGs, PVs have high current generation ability, and silicon-based PVs are a constantly growing semiconductor technology. Presently, commercial PV panels have a power conversion efficiency (PCE) of approximately 25% [9].

A swift succession of wearable smart devices has emerged since the launch of the Linux-powered IBM Watch Pad 1.0 smart-watch in 2000 [10]. The numerous applications that these wearable smart gadgets provide while adhering to strict size restrictions present a significant challenge for their power supply components. Based on the lightweight, thickness, greater conformity to the human body's natural contours, and greater adaptability to body motions, wearable electronics are currently developing quickly, necessitating that the power supply components match these requirements [11, 12]. Generally, wearable energy sources that are thin, flexible, light in weight, and have high power and energy densities, along with long-term stability, reasonable cost, and operational safety, are urgently required. Therefore, the development of wearable energy harvesters is a viable strategy to satisfy the energy requirements.

Considerable efforts are underway to probe appropriate materials capable of effectively harnessing different types of ambient energy to produce high-performance small energy harvesting modules. The emergence of two-dimensional (2D) materials has revolutionized the field of material science and energy technology, offering unprecedented opportunities for flexible energy harvesting applications. Hence, 2D materials garnered significant interest as potential options to further extend Moore's law

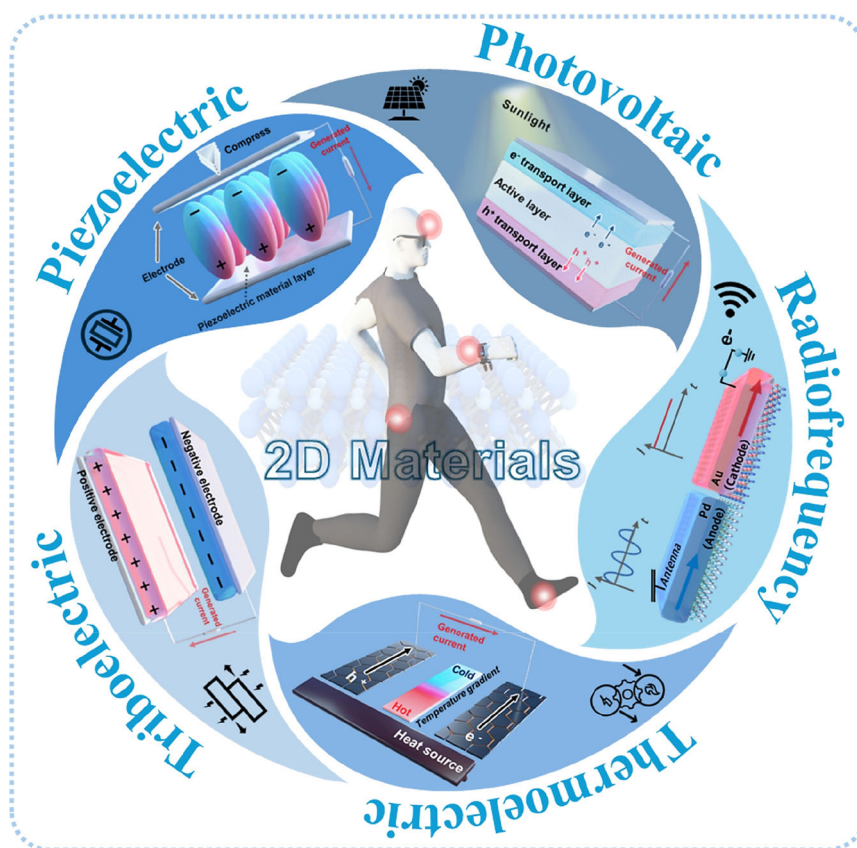
in the domain of microscale energy harvesting [13], and the unveiling of graphene (GR) through graphite exfoliation has provided new avenues for researchers to delve deeper into this emerging field. Since that time, 2D materials have drawn considerable attention from the research community for various applications like flexible electronics, nanoelectromechanical systems, electromagnetic shielding, sensors, and nanogenerators [14–16]. The enormous surface area, transparency, and notable mechanical and electrical capabilities of 2D materials have ignited substantial interest in their potential applications. Additionally, van der Waals (vdW) interactions between layers and in-plane covalent bonding are common features of these innovative 2D materials. Moreover, with quantum confinement, they display fascinating band structures and charge transport, which confers them with attractive optoelectronic and electrical properties [17]. The atomically thin and layered structures of MXenes (transition-metal carbides, nitrides or carbonitrides) [18, 19], transition metal dichalcogenides (TMDCs) [20, 21], BP (black phosphorus or phosphorene) [22, 23], hexagonal boron nitrides (h-BN) [24], and GR [25] demonstrate distinctive optical, thermal, and mechanical properties, making them right candidates for energy harvesting technology. It is thought that the flexibility in these 2D materials stems from their atomic-scale thickness, which is an essential feature for use in wearable energy devices. Therefore, these 2D materials' exceptional mechanical strength, flexibility, and high ionic/electrical conductivities enable the development of ultrathin and flexible energy harvesting and storage systems [26]. For instance, 2D BP has been explored extensively for energy harvesting along with energy storage such as PVs, nanogenerators, batteries, and SCs [27, 28]. In batteries, their 2D crystalline structure can effectively improve the Li storing ability at the surface/interface, making ion intercalation and deintercalation easier by buffering the volume expansion of electrode materials; the 2D structure of MXenes provides a high surface area, which enables the resulting nanostructures to have a high surface-to-volume ratio. Also, they offer a redox-active surface, fast electron supply, and abundant proton availability for SCs. In solar cells, due to high tunable band gap ability, the 2D TMDCs, as compared to traditional Si or gallium arsenide (GaAs) (with the same thickness)-based solar cells, can absorb more incident sunlight. MoS₂ shows layer-dependent enhancement of piezoelectricity compared to its bulk counterparts in PENGs. Adding reduced graphene oxide (rGO) or MoS₂ to the triboelectric layer can aid electron trapping and increase the triboelectric charge density in TENGs.

A limited number of reviews have explored the role of 2D materials in energy harvesting applications [12, 29–31]. For example, the review by Ali et al. [12] focuses on the application of 2D materials in textile-based energy harvesting. Similarly, Rana et al. [29] highlight only piezoelectric, triboelectric, and hybrid nanogenerators utilizing 2D materials and their vdW heterostructures. However, the scope of these reviews is restricted to specific materials or individual energy harvesting applications, leaving a gap for a broader, interdisciplinary perspective. This review aims to provide a comprehensive perspective on 2D materials for flexible energy harvesting applications, addressing the critical gaps in the literature. In this review, for the first time, beyond traditional 2D materials (such as GR, TMDCs, etc.), we explore emerging topics like

bio-inspired materials and Moiré superlattices, highlighting recent advancements. We integrate diverse mechanisms like PV, piezoelectric, triboelectric, and thermoelectric, along with unexplored radiofrequency (RF), within the context of wearable and flexible energy harvesting (Scheme 1). Also, by discussing advanced topics like interface and defect engineering, this tutorial-style review bridges foundational knowledge with cutting-edge insights that will serve as a valuable guide for both early-career and experienced researchers in advancing the field of flexible energy technologies. To establish a foundation for understanding how the structure of various 2D materials influences their electrical, thermal, optical, and mechanical properties for energy harvesting, Section 2 begins by discussing their advantages over three-dimensional (3D) counterparts. Section 3 provides an overview of the synthetic protocols for 2D materials, examining their merits and limitations in the context of energy harvesting applications. Section 4 focuses on the characteristics of state-of-the-art 2D materials, highlighting their relevance to this field. Sections 5–8 delve into advanced topics, including interface engineering, moiré superlattices, bio-inspired designs, and defect engineering, emphasizing their significance in optimizing energy harvesting performance. In Section 9, the mechanisms and performance of various energy harvesting technologies are explored, with a particular focus on the integration of 2D materials. Finally, we look ahead to the research challenges and potentially attractive avenues of exploration, along with abundant topics of research.

2 | 2D Materials: Rising Star for the Future

Why 2D materials? Do they possess distinctive features that make them superior to traditional materials like 0D, 1D, and 3D or their bulk counterparts? This section presents several examples from different perspectives to delve into the exceptional characteristics of 2D materials. As we know, the landscape of energy harvesting is rapidly evolving, particularly with the integration of 2D materials, which offer unique properties that enhance their functionality in flexible applications. In the realm of materials, 2D materials stand out as the “superstars” due to their remarkable electrical, chemical, and physical properties, stemming from their unique ultrathin size, potentially revolutionizing everything spanning from digitalization to energy harvesting (Figure 1). Furthermore, these materials show robust spin–orbit coupling, undergo indirect-to-direct band gap transformation, and manifest phenomena such as quantum Hall and quantum valley Hall effects, which are impossible to achieve with traditional 3D bulk materials [32]. These crystalline materials are almost entirely surface-based, with layers that are only a few atoms thick. Because of this feature, they are susceptible to quantum phenomena and they have a wide surface area that facilitates interaction with molecules and allows them to respond to physical forces, conferring them with attributes that can distinctly deviate from those observed in their 3D counterparts. Generally, the following are the various distinctive properties of 2D materials compared to their 3D counterparts, making them superior in many applications.



SCHEME 1 | Schematic overview of 2D materials used in energy harvesting applications, showcasing their integration into different portable and wearable electronics for sustainable power generation.

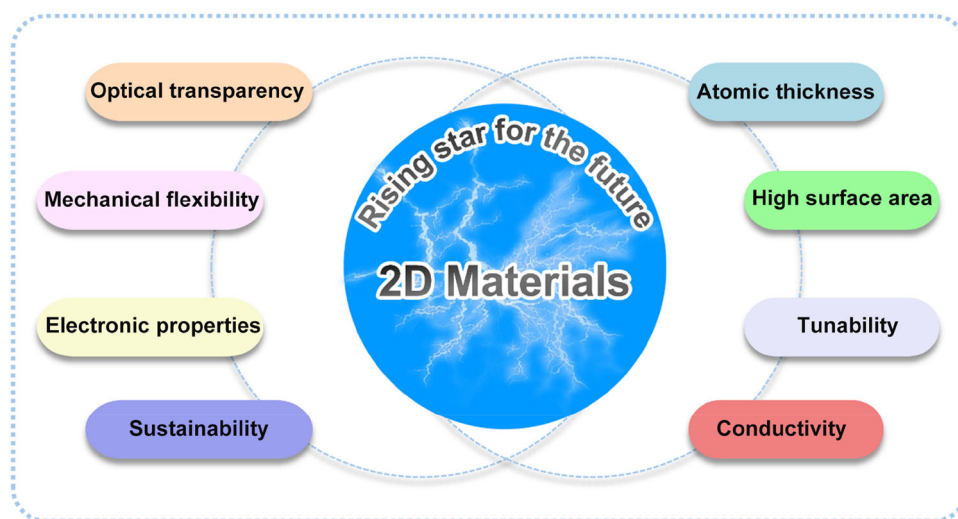


FIGURE 1 | Distinctive features that make 2D materials superior to traditional materials or their bulk counterparts.

- I. 2D materials without interlayer interactions, particularly monolayer nanosheets, are attractive options for fundamental condensed matter research and applications in electronic devices because of their 2D electron confinement, producing highly attractive electronic properties compared to other materials. Moreover, their band shape is also largely determined by this absence of interlayer contact (monolayer nanosheets) [33].
- II. Due to the quantum size effect resulting from their nanoscale thickness, they have exceptional mechanical flexibility and optical transparency, making them well-suited for producing transparent and flexible electrical and optoelectronic devices [34, 35].
- III. They possess an exceptionally large specific surface area and surface-to-volume ratio (the large surface-to-volume ratio leads to high-energy transfer efficiency and fast reaction time because of swift carrier mobility), along with massive lateral size and extremely thinness which makes them ideal for surface-active applications [36].
- IV. These materials have high thermal conductivity, significant carrier mobility at low temperatures as well as at room temperature, and chiral charge carrier properties [37–39].

The 2D massless Dirac fermion with exceptional mobility in GR, extraordinary direct band gap phenomena in single-layer MoS_2 , Ising superconductivity in MoS_2 and NbSe_2 , quantum spin Hall phenomena in single-layer WTe_2 , and 2D ferromagnetism in single-layer CrI_3 are just a few examples of the qualitative differences that 2D materials with nanosized thickness show from bulk 3D materials [40–44]. Furthermore, due to the perfect crystal structure and defect-free in GR, electrons can move across a few micrometers without scattering, resulting in exceptionally high charge carrier mobility along with superior electrical conductivity at ambient temperature [45]. Therefore, GR is perfect for electronic devices due to its large charge carrier mobility and good conductivity.

The mechanical behavior of 2D materials reveals rich physical phenomenon, providing valuable insights into the interactions

of lattice electrons in low-dimensional systems. The exceptional mechanical capabilities of 2D materials as compared to other dimensional materials can be attributed to their atomic thickness, in-plane covalent bonding, and grain boundaries. Generally, the mechanical strength of 2D materials varies widely due to their inherent diversity and distinct interlayer interactions; typically, they have an effective strain of 10% and Young's modulus between 150 and 400 GPa [46]. For instance, Falin et al. compared the mechanical properties of h-BN (1–8 layers) with GR (1–9 layers) and observed that a few layers of h-BN show mechanical behavior reasonably different from that of a few layers of GR. For instance, the former's mechanical strength remains intact with increasing layer thickness, whereas the strength of the latter decreases by approximately 30% with increasing number of layers. Consequently, layered h-BN has demonstrated excellent thermal stability, making it a promising candidate for mechanical reinforcement [47]. Remarkably, to date, among the 2D materials, GR is considered the strongest and thinnest substance, showing breaking strength and Young's modulus of 42 N m^{-1} and 1000 GPa, respectively, with elastic deformation sustainability of more than 20%. TMDCs (MoS_2 , MoSe_2 , WS_2 , WSe_2) represent another crucial category of 2D materials that have gained significant interest in energy harvesting, owing to direct band gap phenomena, quantum confinement, and atomic-scale thickness [48, 49]. MoS_2 is the most extensively explored member of the TMDCs family, effectively utilized as a charge-trapping material, due to its high electrical conductivity. Its 1T phase is used as an electrode in TENGs, and its monolayer sheets display a notable Young's modulus (270 GPa), surpassing that of steel, which stands at 205 GPa [50]. MXenes (2D transition-metal carbide, nitride, or carbonitride) are a recent addition to the 2D material family and because of their many benefits like excellent absorbance, substantial photogenerated charge carriers, high conductivity, and extensive charge storage abilities, they hold great promise for harvesting of a variety of ambient energy sources. For instance, due to exceptional electrical conductivity, their captured energy loss during current flow is minimal; they have a high surface-to-volume ratio; they have better mechanical stability than monolayer $\text{Ti}_3\text{C}_2\text{T}_x$ and $\text{Nb}_4\text{C}_3\text{T}_x$, with Young's

modulus of 330 and 390 GPa, respectively, which can be further increased with the addition of ionic/hydrogen bonding agents; and they have a hydrophilic nature due to numerous functional groups on their surface, because of which they have even more tribonegative properties than Teflon and accelerated electric current production in aqueous media [51–56]. In polycrystalline 2D materials, grain boundaries (interfaces that separate adjacent crystal grains, resulting in a line defect network) also substantially affect their mechanical and electrical properties. Generally, a grain boundary is formed when two grains with different orientations and edges meet each other. In one study, the authors documented the unique strength properties of grain boundaries in GR, showing that compared to low-angle grain boundaries with fewer defects, the larger angle grain boundary with high defects has greater strength, and even its strength is comparable to that of pristine sheets [57]. Within the family of 2D materials, the discovery of new members consistently generates excitement, as they often exhibit unique properties. In particular, 2D materials provide exceptional flexibility for tuning electronic characteristics, surpassing that of their 3D counterparts. Owing to exceptional electrical properties due to reduced dimensionality, 2D materials are superior candidates in the electrical and electronic fields. For instance, GR provides excellent electrical properties due to its highly modifiable Fermi level and its linearly dispersed band structure at the Dirac point. Moreover, as the crystal structure of 2D materials is normally free of defects, it increases the electrical conductivity and charge carrier mobility due to reduced distance for electron transport. These distinct characteristics make them an attractive material for advancing energy harvesters and other nanoelectronics devices [58].

In a nutshell, the combination of robust in-plane covalent bonding and weak interplanar vdW in 2D materials provides them with high flexibility along with mechanical strength. Their band gap, which spans from 0 eV (GR) to 6 eV (h-BN), electronic properties, and optoelectronic properties make them a promising candidate for various applications ranging from transistors to photodetectors. Additionally, the absence of dangling bonds in these 2D materials not only enables their use in different heterostructure devices but also integrates them with other technologies. Furthermore, nanoscale thickness also ensures high optical transparency of 2D materials. For instance, GR has high light transmittance, as evidenced by its measured white light absorbance ability of approximately 2.3% with minimal reflection of approximately 0.1% [59]. Hence, the exceptional atomic thickness, large surface area, tunable properties, and thermal, electronic, optical, and mechanical properties make 2D materials very attractive for various applications such as transparent, wearable, highly flexible electronic devices, as well as energy and optoelectronic devices, and beyond.

3 | Overview of Synthetic Protocols and Processing of 2D Materials for Energy Harvesting

“Preparation determines the future” is an important quote made by Zhongfan Liu (a renowned nanomaterial scientist), which is becoming an inevitable law for 2D materials.

Hence, it is crucial to prioritize the development of reliable, facile, and feasible techniques to produce 2D materials to make use of their properties and for diverse applications. Therefore, inspired by their intriguing characteristics and potential uses, considerable work has been conducted toward exploring diverse synthetic protocols to generate good-quality and pristine 2D materials with precisely controlled dimensions. In this regard, many synthetic strategies have been established, including top-down (such as mechanical exfoliation, liquid exfoliation, and ion intercalation and exfoliation) and bottom-up (including chemical vapor deposition (CVD) and wet chemical synthesis), as illustrated in Figure 2. Generally, most 2D materials are developed from their layered precursors, distinguished by their weak interplane vdW interaction and robust intraplane covalent/ionic bonding.

Consequently, the top-down technique relies on exfoliation of bulk layers into mono/few layers by utilizing various forces to disrupt the weak interplanar vdW interaction among the stacked layers, whereas the bottom-up strategy involves directly synthesizing 2D materials under specific experimental conditions. In this section, we briefly explore only the state-of-the-art techniques that are currently more frequently used for synthesizing 2D materials with the goal of determining how different synthetic methods influence the characteristics of the resulting 2D materials, their scalability (scalable production of 2D materials is the first step toward bringing these materials from the lab to the market), and ultimately their applicability in energy harvesting technologies. Rather than focusing on the experimental specifics of synthesizing 2D materials, our approach emphasizes establishing a strategic framework to optimize various synthetic techniques for addressing key research challenges in the field.

3.1 | Top-Down Strategies

3.1.1 | Mechanical Exfoliation

As we know, unexfoliated 2D materials have semiconducting properties, where most catalytically active sites remain hidden, and low conductivity limits their catalytic potential. Exfoliation of these bulk materials reveals layered nanosheets with significantly larger surface areas and numerous active sites, enhancing their catalytic efficiency. Mechanical exfoliation is a straightforward approach to producing monolayer nanosheets; however, its low yield rate restricts its practicality for large-scale applications, making it more suitable for mechanistic studies or fabrication of specialized devices. Mechanical exfoliation, also referred to as the scotch tape method, is one of the conventional, simple, and best strategies used for exfoliating stacked bulk crystals to obtain large area and atomically thin 2D nanosheets [60]. Generally, in this process, 2D materials are exfoliated due to greater tape adhesion than the newly generated surface energy by using some mechanical forces to disrupt the weak interplanar vdW interaction. More in detail, in this method, freshly cleaved larger bulk materials (TMDCs or GR) on tape are brought in contact with a target substrate (SiO_2/Si). Typically, a scotch tape is applied on the crystal's external side, as depicted in Figure 2A, followed by gentle pressure to ensure good bulk crystal and substrate contact, and finally, a scotch tape is detached to remove most of the

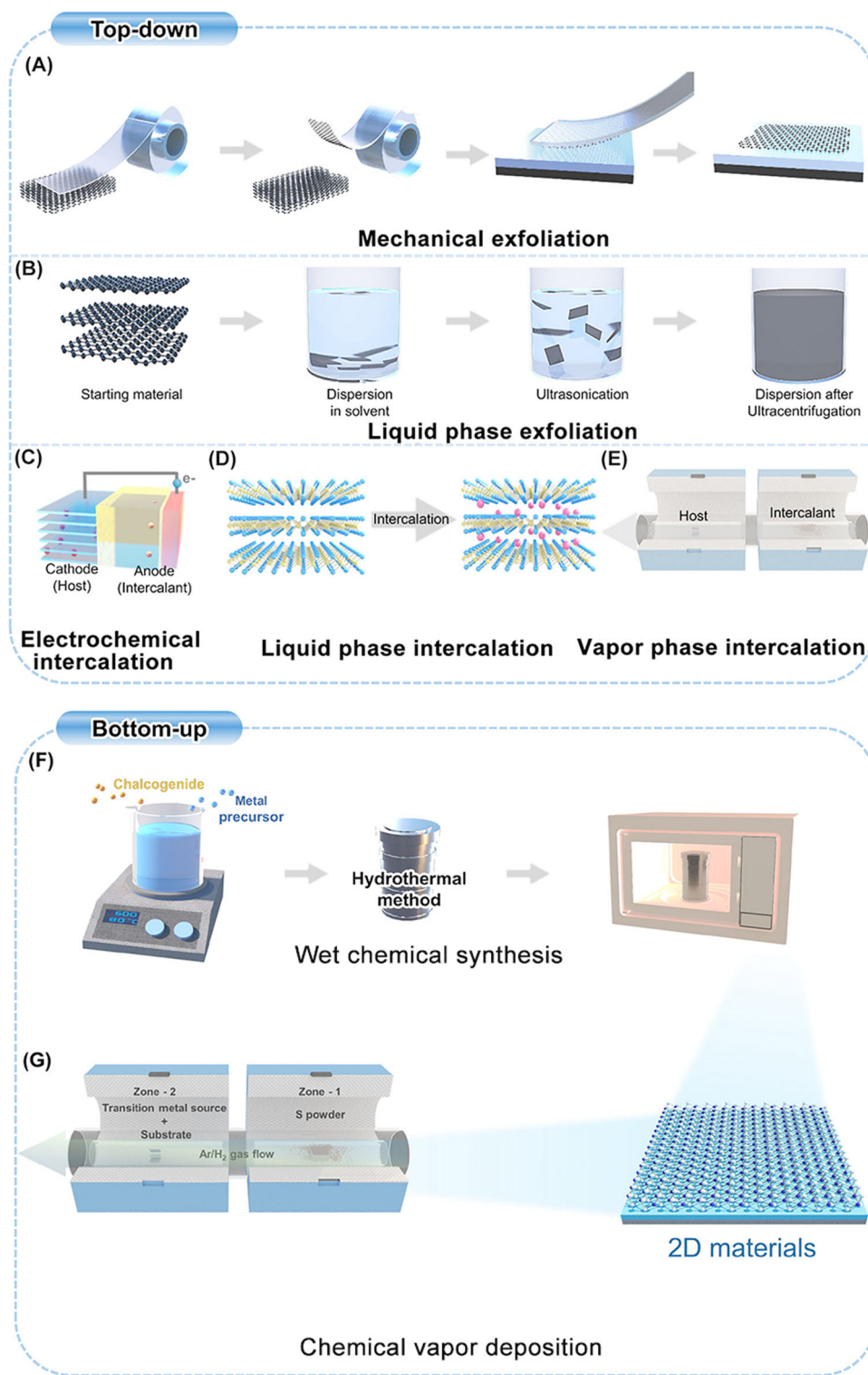


FIGURE 2 | Schematic representation of various synthetic methods utilized for 2D materials, illustrating their distinct approaches to material fabrication. Top-down (including (A) mechanical exfoliation, (B) LPE, (C) electrochemical intercalation, (D) liquid-phase intercalation, and (E) vapor-phase intercalation) and bottom-up (including (F) wet chemical synthesis and (G) CVD).

crystal, leaving only mono/few layers of nanosheets (e.g., TMDCs or GR) on the substrate. Novoselov et al. [61] are pioneers in the use of this technique for successful exfoliation of monolayer GR from bulk layered graphite, and subsequently, this method is extended to other 2D bulk crystals (e.g., h-BN, MoS₂, NbSe₂), owing to low surface energy like GR to exfoliate them into mono/few layers effectively. MoS₂, a representative TMDC, has been

widely studied due to its natural abundance and environmental stability, and in this connection, Mak et al. [62] are pioneers in extending this mechanical exfoliation technique to synthesize monolayer MoS₂, documenting that by controlling the thickness, a transition in the band gap occurs from indirect to direct. Inspired by this study, mechanical exfoliation methods were then widely used to synthesize other types of mono/few-layer

nanosheets like metal oxides, metal–organic frameworks (MOFs), phosphorene, and other 2D TMDCs (TaSe₂, WS₂, TiS₂, WSe₂, etc.) [63–65]. As for device fabrication and other fundamental intrinsic properties, the domain size and thickness are prerequisites, so this technique enables effective synthesis of 2D materials with high quality and large surface area. Therefore, currently, many modified mechanical exfoliation methods exist. For instance, Huang et al. documented a widespread Au-supported mechanical exfoliation method in which the Au layer is initially deposited on a substrate covered with a titanium (Ti) or chromium (Cr) adhesion layer and then freshly cut bulk material on top is pressed onto the Au substrate to ensure optimal substrate/crystal contact. Finally, a Scotch tape is detached to eliminate most of the crystal, leaving only mono/few layers of nanosheets with a large area on the Au substrate [66]. In this approach, the exfoliation strategy was further extended to synthesize other 2D-layered materials (NbOCl₂, PtSe₂, CrSiTe₂, Fe₃GeTe₂, FeSe, NbSe₂, CrOCl, etc.) with superior optical, superconducting, and ferromagnetic properties [67–70]. It is noteworthy that the above-stated methods are only restricted to weak interplane vdW interaction 2D-layered materials and thus do not work for materials with no weak interplane vdW interactions. Taking this into consideration, recently, Jiang et al. [71] presented a novel approach known as the calendaring pretreatment mechanical exfoliation technique in which external force was applied to slide the closely packed layers by weakening interlayer bonding to first form a metastable phase, and then the traditional scotch tape approach was used to fabricate stable mono/few layers 2D materials (e.g., SnO, V₂O₅, etc.) from non-interplane vdW interaction structures. To summarize, the pristine 2D materials, due to the mechanical exfoliation technique, display few defects and very few micrometer-length flakes, which represent the best choice to systematically study their intrinsic properties for use in electronic devices. However, this method has certain limitations (listed below) that restrict its application, especially in energy harvesting.

- The final yield of this method is rather poor, and it cannot satisfy the demands of practical applications that need substantial quantities of materials.
- The insulating substrate is an additional limitation because this method always requires that the exfoliated atomically thick sheets be supported, and may thus leave some adhesive.
- It is difficult to manage the resulting 2D material's thickness, shape, and size.
- This synthetic method is limited only to layered bulk materials. Moreover, its laborious and time-consuming method ultimately impedes its mass production.

Therefore, to exploit this approach properly and meet the practical demands, it is essential to improve both the production rate and yield.

3.1.2 | Liquid-Phase Exfoliation (LPE)

As stated above, the production of 2D materials via the mechanical exfoliation strategy has certain limitations that

impede its practical applications. Therefore, to address these challenges, LPE, as illustrated in Figure 2B, has emerged as a versatile, cost-effective, highly efficient, and scalable method commonly used to exfoliate bulk layered materials to acquire mono- or few-layer nanosheets with reasonable yield (surpassing 1 g L⁻¹) [72, 73]. This exfoliation strategy is primarily classified based on its mechanism as direct exfoliation (including sonication-assisted and shear exfoliation) and chemical exfoliation (involving ion intercalation exfoliation). It should be noted that not all 2D materials can be exfoliated by using any single LPE method, as suitable solvents and appropriate techniques greatly vary among various materials [74, 75]. Generally, via sonication, the multilayer bulk crystals are directly exfoliated in solvents like N-methyl-2-pyrrolidone (NMP) and dimethylformamide (DMF) using the LPE technique. As in the LPE process, the sonication can effectively break the interplanar vdW interaction among layers while preserving the integrity of the covalent bonds within each layer. Moreover, ensuring proper alignment of surface tension among the crystal layers and the solvent is essential for minimizing interfacial energy and optimizing exfoliation efficiency. This exfoliation strategy only works for those 2D materials with weak interlayer forces, whereas for strong interlayer coupling materials, it does not work effectively. The solvent not only stabilizes the exfoliated nanosheets but also plays a crucial role in preventing them from aggregation and restacking. As far as solvents are concerned, water, being a universal solvent, is inappropriate for exfoliation, but instead, organic solvents are used for effective exfoliation. For instance, in one study, the authors described GR synthesis via LPE by using organic solvents (DMF, NMP) to achieve a 1 wt% yield of monolayer with a concentration of 0.01 mg mL⁻¹ [76]. Similarly, Rafi et al. [77] applied the same strategy to successfully exfoliate TMDCs (e.g., MoS₂) by using DMF and NMP as solvents. Furthermore, as an alternative method, it has been found that layered bulk crystals can be exfoliated by subjecting them to sonication in an aqueous solution containing a polymer or a surfactant, which serves to stabilize the nanosheets that have been exfoliated. A variety of different multilayer crystals, including GR, BP, h-BN, MXene, MoS₂, MOF, and metal oxides, can be successfully exfoliated using this LPE method [78, 79]. It is noteworthy to mention here that with LPE via sonication under optimized experimental conditions, the production yield of mono/few layers nanosheets is quite low, which does not meet industrial requirements; therefore, for scale-up fabrication of 2D materials (e.g., GR), Coleman J.N effectively used a shear exfoliation strategy (a scalable alternative to sonication) [80]. As a result of this rotating mixing approach at high speed, they achieved defect-free GR flakes with 300–800 nm lateral dimensions in the presence of an NMP solvent. Typically, in this method, when the mixer is rotated at high speed, it confers substantial shear strength to bulk layered crystals, resulting in the exfoliation of bulk materials. Afterward, this strategy is also effectively extended to BP bulk crystal exfoliation. It is interesting that, being quite simple, this method facilitates large-scale, high-yield, and inexpensive synthesis of mono/few-layer 2D materials, but still, thickness controllability along with morphology is highly desired for practical applications. Additionally, the direct exfoliation process utilizing sonication and shear force in suitable solvents poses challenges in terms of control, leading to a reduction in the monolayer yield, which in turn limits its use as

a scalable and versatile preparation method for 2D materials. Furthermore, ultrasonic energy along with toxic organic solvents, use of a polymer/surfactant, low yield of monolayer sheets, and lateral sizes of sheets inevitably degrade their intrinsic properties and structure, which in turn hinders their optical and electronic applications. However, 2D materials obtained through this method are promising in some fields, like catalysis, printer ink, and SCs. To obtain a high yield of exfoliated monolayer nanosheets along with high lateral sizes, there is an urgent need for further optimization of LPE experimental conditions.

In a critical evaluation summary, liquid exfoliation is a straightforward technique for producing 2D materials with consistent thickness, and yet, controlling the number of layers remains challenging. Although it offers scalability, the process often results in low yields of monolayers and introduces impurities and defects during ultrasonication, limiting its suitability for industrial applications.

3.1.3 | Intercalation-Assisted Exfoliation

As stated above, ion intercalation has emerged as a more promising and controlled synthetic strategy to tackle the issues of thickness dispersion in direct LPE. Typically, this exfoliation strategy involves intercalation/insertion of cations/anions/radicals (Li^+ , Na^+ , K^+ , SO_4^{2-} , $\cdot\text{O}$, $\cdot\text{OH}$, etc.) in weakly bonded layered materials, resulting in an increase in the interlayer spacing and weakening/diminishing of the interplanar vdW between adjacent layers, followed by sonication, to effectively exfoliate the intercalated material to mono/few-layer nanosheets [81]. Notably, in this method, a high yield of monolayer can be achieved by carefully controlled insertion of ions while maintaining the structure and chemical composition of the materials. Generally, so far, various intercalation approaches have been successfully developed, including liquid phase, electrochemical, and vapor phase [82]. Liquid-phase intercalation (Figure 2D) is a typical technique that involves immersing the host 2D material in a liquid solution containing an intercalant (atoms, ions, or molecules), which allows the intercalants to seep into the host material's vdW gaps through the liquid medium. Furthermore, either a solution containing intercalants or the intercalants themselves in their molten state could serve as the liquid source for intercalants [83]. A variety of inorganic and organic molecules, including toluene, isopropyl amine, hydrazine, dimethyl sulfoxide, urea, acetone, *N*, *N*-dimethyl formamide, ethyl alcohol, and chloroform, have been reported to have been effectively inserted into layered materials through liquid-phase intercalation [84]. Importantly, liquid ammonia has been effectively used for the intercalation of alkali/alkaline earth metals into GR, MoS_2 , and BP, because these metals are easily soluble in ammonia [85]. Though the liquid-phase intercalation is a good choice, this method also has some limitations, like in situ quantitative measurements being a challenge, and control of the final product composition is difficult. Another ion intercalation exfoliation technique is the electrochemical intercalation strategy (Figure 2C), which is triggered by external current or system voltage by using conventional electrochemical workstations where intercalants and host materials serve as the anode and the cathode, respectively. In this process, the guest species (intercalant) can act either as a donor (e.g., Li^+ , Na^+ , K^+) or as an acceptor

(e.g., H_2SO_4), where charge transfer takes place between hosts and the intercalants. Therefore, during intercalation, the donor materials donate electrons that effectively permeate into the interlayer gaps of layered materials (serving as a host material). Although this ion intercalation strategy using liquid electrolytes has been proven to be effective in different energy systems, it has some drawbacks that limit its practical applications, like electrolyte reactivity, limited temperature range operation, and dendrite growth. Vapor-phase intercalation (Figure 2E) is another approach of exfoliation in which intercalation is controlled by using heat and has gained widespread attention due to its compatibility with many intercalant and host pairs. For example, this strategy is effectively used to intercalate BP, TMDCs, and GR with Na, K, Cs, Cu, Cd, Sn, Pb, and Bi [86]. However, this exfoliation technique also suffers from limitations like the challenge of in situ determination of the intercalated sample composition, making it difficult to accurately evaluate the transfer of atoms or molecules to the host material.

This ion intercalation technique can be used to effectively synthesize extremely thin 2D sheets from their bulk counterparts like MXenes, h-BN, GR, BP, and TMDCs, and notably, this method is highly scalable, even some TMDCs (MoS_2 and TaS_2) nanosheets' yield exceeding 90%. Importantly, this method can also induce phase transition in TMDCs (MoS_2) via Li intercalation during the exfoliation process, transitioning them from the 2H (semiconductor) phase to the 1T (metallic) phase [87]. As a consequence of this phase, the performance of these materials can be enhanced for some promising applications like field emission transistors (FETs) and energy systems [88]. This method remains a commonly used technique for synthesizing 2D materials because of its controllable and efficient charge doping abilities, and also during this process, the donor/acceptor-type intercalants donate or withdraw electrons from adjoining host layers, resulting in a shift in the Fermi level. Additionally, it can alter the conductivity along with the polarization of 2D materials, making them suitable for optoelectronics applications. For example, $\alpha\text{-MoO}_3$ nanoribbons' optical properties were tuned successfully via intercalation and de-intercalation [89]. This method, due to its ability to modify material properties in terms of tuning the atomically thin vdW, finds application in electronic and memory devices [90].

In a critical evaluation summary, as we know, alkali metals are highly reactive with moisture and oxygen, posing safety risks upon exposure. Although the introduction of chemical reagents can lead to defects and impurities, maintaining strict control over reaction conditions is crucial. This method's advantage lies in its relatively high yield of monolayer 2D nanosheets, which is particularly beneficial for scaling up the production of large-area monolayer 2D materials suitable for industrial applications.

3.2 | Bottom-Up Strategies

3.2.1 | Chemical Vapor Deposition (CVD)

The CVD technique also stands out as one of the most promising bottom-up approaches widely utilized to synthesize wafer-scale ultrathin 2D materials on a substrate due to its high potential for scalable production [91, 92]. Typically, the main goal of this method is to fabricate large-area, tunable thickness,

ultrathin, and low-defect density 2D materials with precise control over the number of layers, morphology, domain size, and quality that are on par with their exfoliated counterparts. In the CVD process, the growth mechanism for 2D materials involves several key steps. Initially, precursor species are transported in the vapor phase, followed by adsorption and decomposition on the substrate surface or within the vapor (Figure 2G). This is followed by crystal nucleation and subsequent growth, which involves adsorption, desorption, diffusion, and surface reactions. The growth typically follows one of the following three modes: island (Volmer-Weber), layer by layer (Frank-van der Merwe), or mixed (Stranski-Krastanov). Various factors like thermodynamics and kinetic surface dynamics, substrate material interactions, and substrate morphology dictate the growth orientation and quality of the resulting 2D materials. Specifically, in this process, a substrate is subjected to reactive precursors at high temperatures and in an inert environment, where the precursor reacts/decomposes on the substrate surface, resulting in large-area solid-phase materials/ultrathin films. Optimal deposition conditions, such as temperature, pressure, and reaction duration, are essential for producing high-quality 2D materials. To improve the yield and expand the domain size of monolayer materials, growth promoters like alkali metal halides/hydroxides are commonly utilized. These additives enhance the evaporation of precursors, stimulate the nucleation of 2D crystals, and promote lateral growth. Furthermore, the choice of substrate, including SiO₂/Si, sapphire, mica, and catalytically active metals (e.g., Au and Cu foils), significantly influences the material's properties. To further improve the homogeneity and decrease the growth temperature, particular CVD approaches like plasma-enhanced CVD and low-pressure CVD have also been effectively used [93, 94]. So far, this technique has been successfully used to synthesize various ultrathin 2D materials, including MXenes, h-BN, TMDCs, GR, metal oxides, alloys, and heterostructures [95–98]. The 2D materials synthesized using the CVD technique have high crystallinity and mechanical flexibility, making them highly suitable for the next generation of electronic devices. For instance, the FETs based on CVD 2D materials have high I_{ON}/I_{OFF} current ratios along with high mobility [99]. Many researchers also attempted to successfully incorporate GR-based electrodes synthesized via CVD into solar cells and organic light-emitting diodes (LEDs) [100, 101]. The exfoliation energy of BP is very high; therefore, preparation of single/few layers directly with a bottom-up strategy is a long-standing challenge. However, BP grown through the CVD method not only changes it from red phosphorus but also yields an atomically thick and small lateral-size film [102]. Furthermore, many macroscopic optoelectronic applications have sizes that are typically in the range of millimeters to centimeters, which is not achievable without a large-scale synthetic technique. As CVD is well known for use in large-scale synthesis, therefore, to tackle this issue, many researchers have successfully incorporated a transparent electrode based on CVD-grown GR into solar cells and also into organic LEDs, and especially TMDCs grown using this technique, characterized by their large band gap, effectively enhance Si-based solar cells' PCE [103]. In one study, CVD-grown MoS₂ PENG was reported through the S vacancy passivation strategy, and excitingly, its current increased threefold, whereas its voltage increased twofold as compared to pristine MoS₂ PENG [104]. This synthetic technique has a remarkable

ability to precisely regulate the thickness, crystallinity, phase structure, and morphology of 2D materials and has become a flexible and scalable approach that offers substantial potential to explore their underlying physics and energy harvesting applications. However, this technique also had certain limitations, like robust scale-up production, uniform control of the film thickness throughout the substrate, requirements for high-temperature and high-vacuum conditions, and the need for a specific substrate for material growth. Therefore, addressing these issues and considering the regularity and crystallinity of 2D films along with the demand for high throughput, cost-effectiveness, and scalability will make this technique a more efficient and reliable option for industrialization.

3.2.2 | Wet Chemical Synthesis

The wet chemical synthetic approach is another frequently used bottom-up technique to produce ultrathin 2D materials. Among various wet chemical synthetic techniques, hydrothermal and solvothermal methods are representative due to their extensive use, where target materials are synthesized from precursors via solution-based chemical reactions at high temperatures and under high pressure (Figure 2F). Since the initial report on the hydrothermal synthesis of tiny quartz crystals, a wide range of nanomaterials have been fabricated in this way, and in the early 21st century, there was a surge in the wet chemical synthesis of 2D materials such as GR and MoS₂. More recently, this method has also been used to produce doped 2D materials, nanocomposites, and heterostructures by introducing different precursors and dopants into the solvent to improve the material properties for particular applications [105, 106]. Significantly, this synthetic strategy facilitates high yield in the liquid phase at a reasonable cost, making it a viable option for industrial production. Additionally, the advantage of this method includes good control of size, surface morphology, and dopants in 2D nanomaterials for various applications like energy harvesting, catalysis, and energy storage. Unfortunately, this synthetic approach also has some drawbacks: for example, it is challenging to generate monolayer nanosheets by the wet chemical technique, due to the reaction parameters, like temperature, time, precursor concentration, and solvent concentration, which can easily alter the synthesis.

In addition to commonly used methods, physical vapor deposition (PVD), molecular beam epitaxy (MBE), and atomic layer deposition (ALD) have emerged as promising techniques for synthesizing a variety of 2D materials [107–109]. In PVD, the bulk precursor material is heated to high temperatures, causing it to vaporize and condense onto a cooler substrate. Beyond thermal evaporation, PVD encompasses other methods like pulsed laser deposition and sputtering, which have shown potential for producing large-scale 2D materials but remain less extensively studied. PVD offers a cost-effective method for synthesizing 2D materials like MoS₂, WSe₂, and alloys in an oxygen-depleted environment, ideal for oxidation-sensitive materials [110]. However, its application is limited to bulk precursors with sufficient vapor pressure and requires precise stoichiometric control during deposition. Additionally, its growth window is constrained for certain 2D materials that are prone to decomposition at high temperatures, posing challenges

for wider material applicability. MBE is a precise thin-film deposition technique widely used for growing high-quality single-crystalline semiconductors. Conducted under ultrahigh vacuum, MBE uses high-purity elemental sources in effusion cells or e-beam evaporators to form molecular beams that react on heated substrates. Despite enabling in situ monitoring and synthesis of complex materials, MBE faces challenges such as small grain sizes in 2D films, high twin boundary density, and high costs due to its sophisticated setup and ultrahigh vacuum requirements, particularly for low-vapor-pressure metals like Mo and W [111]. ALD, a modified form of CVD, introduces vapor-phase precursors sequentially in cycles to enable precise layer-by-layer growth with excellent uniformity across large areas. Its low-temperature operation makes it ideal for integrating 2D materials into device fabrication. However, challenges like random nucleation, limited precursor diffusivity, and steric hindrance at low temperatures hinder monolayer film growth, necessitating post-growth annealing to enhance crystallinity. ALD's reliance on volatile and stable precursors highlights its potential for controlled 2D materials synthesis while underscoring areas for process optimization [112].

In summary, both top-down and bottom-up techniques present distinct advantages and disadvantages in the synthesis of 2D materials. Remarkably, each screening method greatly depends on various factors like material properties, scalability, affordability, shape, size, and thickness control. Therefore, continued investigation into and progress in both methodologies are imperative to propel the synthesis and application of 2D materials throughout diverse domains. Advances in synthesis research are facilitating enhanced growth precision and deeper scientific insight. Site-selective fabrication of microscale 2D crystals directly on pre-patterned substrates is gaining traction as a promising strategy to eliminate the challenges associated with layer transfer. To move beyond traditional trial-and-error approaches and enable scalable, application-oriented synthesis, there is a growing emphasis on data-driven methodologies. Integration of artificial intelligence-guided simulations of growth kinetics with operando characterization techniques and real-time process control is expected to significantly improve synthesis yield, reproducibility, and uniformity. Table 1 highlights the merits and demerits of different synthetic techniques for harnessing energy using 2D materials in energy harvesting applications.

4 | Characteristics of State-of-the-Art 2D Materials for Energy Harvesting (Structural-Property Relationship)

2D materials characterized by their atomic thickness have garnered significant interest for their tunable electronic, thermal, mechanical, and optoelectronic properties, resulting in a surge of interest in their potential application, like energy harvesting, aiming to harness waste or unused ambient energy to electricity. For instance, the well-known 2D material GR, due to its exceptional conductivity, high surface area, good mechanical flexibility, and inherent piezoelectric and thermoelectric properties, has triggered immense interest in terms of use in energy harvesting devices [113, 114]. Additionally, owing to their distinctive characteristics, these materials have been

incorporated into wearable and flexible energy harvesting systems, facilitating the evolution of portable and self-sustaining electronic devices.

In this section, we highlight the characteristics of several notable 2D materials relevant to energy harvesting (Figure 3A–K), aiming to provide readers with a preliminary understanding of representative 2D materials before introducing their energy harvesting mechanisms and performance.

4.1 | GR

Scientists initially believed that nanomaterials would struggle to maintain stability due to thermal fluctuations, leading to rapid degradation. However, GR's remarkable thermal stability and chemical inertness defied these expectations, sparking a surge in research on 2D materials over recent decades. This breakthrough inspired researchers to revitalize traditional techniques, devices, and systems by integrating novel elements, aiming to develop diverse structures and enhance performance.

4.1.1 | Atomic Structure

GR consists of one or a few layers of carbon atoms arranged in a hexagonal lattice, where the atoms form a tightly bonded sp^2 -hybridized structure. This configuration can be visualized as two interlinked sublattices with triangular symmetry. Each carbon atom is bonded to three neighboring atoms through in-plane σ bonds with a bond length of approximately 1.42 Å, whereas the out-of-plane π bonds contribute to GR's remarkable electronic and mechanical properties. GR's exceptional electronic behavior arises from its distinctive band structure. Its valence and conduction bands show a linear relationship near the Fermi level, forming Dirac cones at their intersection. This results in a semimetallic nature with an effectively zero band gap, allowing charge carriers to behave as massless Dirac fermions. Such a structure facilitates ultra-fast electron mobility and minimal resistance to charge flow. Moreover, its electrical properties can be precisely modulated by external factors such as electrostatic gating, chemical doping, and so forth, enabling a seamless transition between electron-dominated (n-type), charge-neutral, and hole-dominated (p-type) states. This intrinsic tunability provides fine control over charge carrier dynamics, optimizing its performance for various energy harvesting applications.

4.1.2 | Configuration

As discussed previously, three of GR's four valence electrons participate in σ bonding, whereas the fourth resides in the delocalized π orbitals, offering a reactive site for additional chemical bonding and enabling the development of novel derivatives. The earliest such modification was the synthesis of hydrogenated graphene achieved by bonding a hydrogen atom to each carbon atom (preferably with alternating sublattice functionalization from opposite sides for enhanced stability). In contrast to GR's semimetallic nature, hydrogenated graphene shows insulating behavior with a band gap of several electron volts. Notably, this

TABLE 1 | Overview of key synthesis techniques for 2D materials tailored for energy harvesting applications.

Method	Efficiency for energy harvesting	Merits	Demerits	Applications
Mechanical exfoliation	Low	Simple, few defects, high purity, good crystal quality, large lateral size, simple.	Low yield. A substrate is required to control the thickness, size, and shape of the crystal, and for non-massive production.	Thermoelectric nanogenerators, photodetectors.
Liquid-phase exfoliation	Low	Scalable, massive production, low cost, insensitive to air and moisture.	Variable thickness, low yield of monolayers, usage of toxic organic solvents, and small lateral size.	Photodetectors, supercapacitors.
Ion intercalation exfoliation	Low	Scalable, solution-based processability, high yield, and large-area monolayer yield.	High temperature and long reaction time, small lateral size with possible defects.	Low-contact resistance field-emission transistors.
Chemical vapor deposition	High	Scalable, high-quality, and large-area crystals, control over thickness, and few defects.	Expensive, requiring a harsh working environment (high temperature, vacuum), use of an insulating substrate.	Photovoltaic, triboelectric, and thermoelectric nanogenerators.
Wet chemical synthesis	High	Low-cost, scalable, high-yield, massive production.	Requiring high temperature and pressure. Sensitivity to solvents and surfactants. Precursor concentration, product quality, and crystallinity are not good.	High-performance energy harvesting devices.

hydrogenation process is fully reversible; for instance, the thermal annealing process can dehydrogenate the material, restoring its pristine electronic structure. Subsequent advancements led to other derivatives such as fluorographene, a 2D analog of Teflon, formed by covalently attaching fluorine atoms to every carbon atom in the lattice. In the realm of energy harvesting, GR is the most investigated nanomaterial owing to its exceptional charge carrier mobility ($200,000\text{ cm}^2\text{ V}^{-1}\text{ S}^{-1}$), lightweight nature, high conductivity (1738 S m^{-2}), chemical inertness, specific surface area ($2630\text{ m}^2\text{ g}^{-1}$), superb optical transparency (97.7%), excellent mechanical strength due to the strong covalent structure (Young's modulus 1000 GPa), and economically feasible production among the range of 2D materials, making it a promising candidate for mechanical energy harvesting. The low contact resistance of approximately $100\text{ }\Omega\text{ }\mu\text{m}$ between GR and metal electrodes enables its integration into electronic devices with a reduced Schottky barrier. Unlike conventional metals, GR stands out for its 2D structure, wide spectrum transparency, ambipolar field-effect transport, and zero band gap, resulting in its classification as a "semimetal" [114, 115]. Additionally, its robust structure provides exceptional transparency across the solar spectrum, positioning it as a strong candidate to replace traditional transparent conductive oxides. GR's unique electronic structure, characterized by its low density of states and band configurations, allows precise tuning of its Fermi level through external electrostatic fields, enabling ambipolar field-effect behavior. Furthermore, its strong ability to form low-resistance Ohmic contacts with metal electrodes and its high optical transparency make monolayer and few-layer GR ideal candidates for use as transparent electrodes in optoelectronic devices, comparable to traditional materials like indium tin oxide (ITO) and fluorine-doped tin oxide (FTO) [116]. It also enhances charge carrier mobility and exciton dissociation, making it highly effective for third-generation PV cells [117]. Its high phonon scattering enhances the material's thermal conductivity at room temperature, a critical requirement for effective thermal energy harvesting [118]. GR also finds its way into dye-sensitized solar cells and fuel cells due to its prominent electrical and thermal conductivity and high surface area, owing to its exceptional mechanical strength and dielectric constant. Graphene oxide (GO), a prominent derivative of GR and a scalable precursor for its synthesis, is produced by the exfoliation of oxidized graphite. Unlike pristine GR, GO (containing oxygen-rich functional groups such as hydroxyl, epoxy, and carboxyl) is hydrophilic, allowing easy dispersion in water for solution-based processing and enabling tunable conductivity. GO forms stable multilayered films through hydrogen bonding, making it highly suitable for water-based applications. For instance, due to its unimpeded permeation of humidity, GO has a high dielectric constant, making it an ideal candidate for piezoelectric energy harvesting, allowing it to produce electricity from atmospheric moisture. By reducing GO to its more conductive form, rGO, its electronic properties can be tailored from an insulating to a semiconducting state, depending on the extent of reduction [119]. All these unique attributes make GR a highly attractive material for advancing energy harvesting technologies. Notably, modification of GR through functionalization is also an effective strategy for tailoring its physical and electronic properties. This approach primarily involves the deliberate integration of heteroatoms such as nitrogen (N) and boron (B) into the carbon framework. The incorporation of these elements induces lattice distortions and

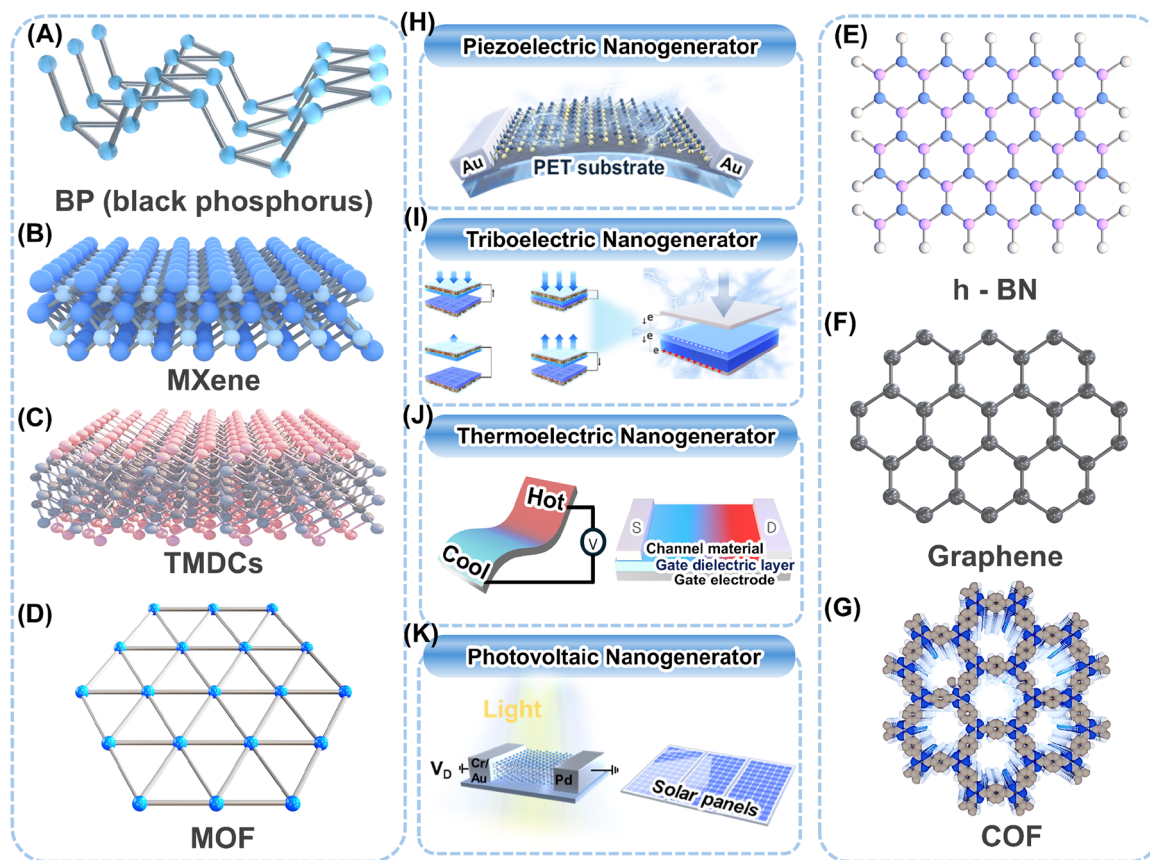


FIGURE 3 | Crystal structures of some common 2D materials and their energy harvesting applications. (A) Black phosphorous [puckered hexagonal lattice, semiconductor, band gap (0.3–2.0 eV), electrical conductivity (100 S m^{-1}), thermal conductivity ($10\text{--}36 \text{ W m}^{-1} \text{ K}^{-1}$), photoluminescence wavelength (918 nm)]. (B) MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) [conductor/semiconductor, band gap (0.95–1.7 eV), electrical conductivity ($2.4 \times 10^4 \text{ S cm}^{-1}$), thermal conductivity ($55.8 \text{ W m}^{-1} \text{ K}^{-1}$), photoluminescence wavelength (503–652 nm)]. (C) TMDCs (MoS_2) [hexagonal lattice, semiconductor, band gap (1.2–1.8 eV), electrical conductivity ($0.2\text{--}1.7 \text{ S m}^{-1}$), thermal conductivity ($52 \text{ W m}^{-1} \text{ K}^{-1}$), photoluminescence wavelength (670 nm)]. (D) 2D MOF [hexagonal lattice having net or network structure, semiconductor, electrical conductivity (10^3 S cm^{-1}), band gap (1.0 eV)]. (E) h-BN [honeycomb lattice, insulator, band gap (5.9 eV), thermal conductivity ($751 \text{ W m}^{-1} \text{ K}^{-1}$), photoluminescence wavelength (203 nm)]. (F) Graphene [hexagonal honeycomb lattice, conductor, band gap (0 eV), a real electrical conductivity (1738 S m^{-2}), thermal conductivity ($3000\text{--}5000 \text{ W m}^{-1} \text{ K}^{-1}$), photoluminescence wavelength (354–729 nm)]. (G) COF [hexagonal layered structure, conductor (electrical conductivity $10\text{--}10^5 \text{ S cm}^{-1}$)/semiconductor (electrical conductivity $10\text{--}10^{10} \text{ S cm}^{-1}$, band gap $< 3.0 \text{ eV}$)]. (H–K) Schematic representation of piezoelectric, triboelectric, thermoelectric, and photovoltaic nanogenerators.

modifies the electronic band structure, consequently altering its conductivity, magnetism, and optical characteristics. By carefully adjusting the type and configuration of dopants, GR can be engineered to have either p-type or n-type behavior. For instance, electron-deficient B and N in pyridinic/pyrrolic forms introduce a p-type doping effect, shifting the Fermi level closer to the Dirac point, whereas graphitic N contributes additional electrons to the π -conjugated network, leading to n-type behavior. These modifications significantly enhance GR's performance in energy harvesting applications. For example, N-doped GR quantum dots (synthesized using a microwave-assisted approach) incorporated as an interlayer into the perovskite/ETL interface have been demonstrated to improve PCE to 21.98%, while also contributing to the long-term operational stability of the device [120]. GR in the form of GO, when used as a conductive layer, significantly improves the efficiency of TENGs by effectively converting mechanical energy from vibrations and human movement into electrical power. The device's (fabricated using the blade-coating technique) enhanced energy harvesting capability enables the activation of 107 LEDs and provides sufficient power for small-

scale electronic devices, including digital watches and other low-power wearables [121]. Furthermore, a recent study introduced a PENG-based sensor for footwear by incorporating GO/rGO into polyvinylidene fluoride (PVDF) using an electrospinning approach, enhancing its β -phase content, which in turn enhances piezoelectric properties. The resulting device demonstrates stable performance over 1500 load cycles, generating a fairly good output voltage. This dual-purpose sensor functions both as an energy harvester and as a wearable sports monitor, paving the way for next-generation self-powered technologies [122].

4.1.3 | Composites' Potential

Advancing this research, researchers are increasingly exploring nanogenerators utilizing composites of 2D materials, aiming to enhance their performance and applicability. These composite-based devices offer key benefits such as cost-effectiveness, mechanical durability, and ease of fabrication, making them highly promising for energy harvesting applications. An

alternative approach to modifying GR's properties involves the development of GR-based composites incorporating polymers, metal oxides, and other functional materials. In such composites, GR serves as a conductive and mechanically robust framework, effectively anchoring nanomaterials and preventing their aggregation. PVDF and its copolymers are widely used for energy harvesting among polymer-based composites due to their flexibility and strong chemical stability. As it is well known, the performance of PENGs in high-temperature environments is often hindered by their thermal sensitivity. To overcome this limitation, GR-based composites have been explored to enhance their efficiency under elevated temperatures. In this context, N-doped GR and MXene were integrated into a PVDF matrix to develop a nanogenerator capable of harvesting energy in high-temperature vibration conditions. Under optimized conditions, the device demonstrates a stable and reproducible energy output. Here, the composite-rich interfacial interactions, superior electrical conductivity, and localized elastic properties collectively boost the piezoelectric response. This device also has great potential for practical energy harvesting applications, such as converting mechanical vibrations from vehicular movement into usable electrical energy, offering innovative solutions for sustainable energy generation in high-temperature environments [123]. Based on composite materials, nanogenerators typically consist of GR/GO/rGO embedded within a polymer matrix and represent a promising strategy for scalable and flexible energy applications. For instance, the dispersion state of GR within polymer composites plays a crucial role in determining their thermoelectric performance at a given filler concentration. Recent advancements have led to the development of stretchable thermoelectric materials utilizing porous GR foam, fabricated using a laser-scribing technique, for self-powered and independent strain and temperature sensing. By incorporating a pre-strain strategy and leveraging the superior thermoelectric properties of poly(3,4-ethylene dioxithiophene)/poly(styrene sulfonate) (PEDOT: PSS), a 3D porous GR foam nanocomposite was engineered to enable precise decoupled sensing. The strong π - π interactions between the GR framework and PEDOT: PSS influence the polymer chain orientation, significantly improving the Seebeck coefficient, which was enhanced nearly fourfold. This nanocomposite-based sensor shows remarkable sensitivity, detecting temperature changes and strain. With a stretchability of 45%, the self-powered platform demonstrates the potential for applications in remote fire detection and real-time monitoring of temperature and strain variations in biomedical settings, such as wound healing. The fundamental design strategies from this study provide a foundation for the development of advanced multimodal sensors with decoupled sensing mechanisms for precise multi-parameter health monitoring [124]. Additionally, flexible sensors hold great promise for wearable energy harvesting applications, and yet, achieving large-area, structurally simple sensors with efficient signal sensing remains a challenge. To address this issue, a scalable rGO-based triboelectric sensor has been fabricated using a straightforward blade-coating technique, where stacked rGO nanosheets are encapsulated and interconnected by polydimethylsiloxane (PDMS), forming a continuous interface between the PDMS dielectric layer and the rGO conductive network, ensuring excellent flexibility and stretchability. During operation, the skin and PDMS function as positive and negative triboelectric layers, due to the difference

in electron affinities. Upon contact, triboelectrification occurs, resulting in electron transfer from the skin to the PDMS surface. When the skin was separated, the potential difference between the triboelectric layers induced electron flow within the rGO/PDMS conductive layer, generating an electron current until electrostatic equilibrium was reached. Upon recontact, electrostatic induction drives electron movement in the opposite direction, enabling the sensor to operate in single-electrode mode with high reliability, achieving a peak voltage of 129 V and a power density of 0.5 W m^{-2} for self-powered sensing applications [125].

4.2 | TMDCs

TMDCs have also garnered significant attention in energy harvesting applications, as they are semiconductor analogs of GR, coupled with outstanding chemical, electrical, and optical properties. These 2D materials represent a fascinating material library, extending from metals (NbS_2 , VSe_2), semimetals (WTe_2 , TiSe_2), and semiconductors (MoS_2 , WS_2), to insulators (HfS_2) [126, 127].

4.2.1 | Atomic Structure

TMDCs are represented by the general formula MX_2 (where M represents the transition metal belonging to groups 4–10 and X represents chalcogens such as S, Se, or Te), and have diverse chemical and physical properties owing to their unique structural configurations. The groups 4–7 TMDCs primarily crystallize in a layered configuration, whereas those belonging to 8–10 groups often adopt a non-layered configuration, leading to variations in their electronic and mechanical characteristics. In layered TMDCs, individual sheets, typically 6–7 Å thick, consist of a central metal atom layer sandwiched between two chalcogen layers in a hexagonal manner. Strong covalent bonding within these layers ensures structural stability, whereas weak vdW interactions between adjacent layers facilitate easy exfoliation. This layered architecture minimizes dangling bonds at the surface, enhancing environmental stability. These 2D TMDCs in monolayer form often show a ripple-like structure to maintain thermodynamic equilibrium. The oxidation states of metal and chalcogen atoms are typically +4 and −2, respectively, with metal–metal bond lengths ranging from 3.15 to 4.03 Å, which is approximately 15%–25% longer than those in elemental transition metals, reflecting reduced d-orbital overlap. The metal coordination in TMDCs, either trigonal prismatic or octahedral (often distorted into a trigonal-antiprismatic form), is dictated by the metal–chalcogen combination and plays a crucial role in shaping their electronic, optical, and mechanical properties, making them highly suitable for applications in flexible energy harvesting. Notably, the bulk TMDCs have diverse polymorphs and stacking variations due to individual MX_2 monolayers' ability to adopt distinct structural arrangements. The most common polymorphs, denoted as 1T, 2H, and 3R, correspond to trigonal, hexagonal, and rhombohedral symmetries, respectively, with the numerical prefix indicating the number of X–M–X layers within a unit cell. The 2H phase, for instance, exists in three polytypic stacking sequences,

whereas a single TMDC shows multiple polymorphs depending on its synthetic conditions. For example, natural MoS₂ predominantly crystallizes in the 2H phase with an AbA BaB (capital letters represent chalcogens, whereas lowercase letters represent metal atoms) stacking order, whereas its synthetic counterpart often adopts the 3R phase with an AbA CaC BcB sequence. The group 4 TMDCs, such as TiS₂, typically stabilize in the 1T phase, characterized by an AbC AbC stacking arrangement and octahedral metal coordination. In the monolayer form, TMDCs can only adopt trigonal prismatic (D_{3h}) or octahedral (D_{3d}) configurations, which significantly influence their electronic characteristics. The electronic properties of TMDCs are governed by both the metal coordination environment and d-electron occupancy. In both 1H and 1T phases, non-bonding d orbitals are situated within the energy gap between bonding (σ) and antibonding (σ^*) states of M–X bonds, contributing to varied electronic properties across different TMDCs. As d-band occupancy increases from group 4 to group 10 elements, the electronic character shifts from metallic (e.g., 2H-NbSe₂ and 1T-ReS₂) to semiconducting (e.g., 1T-HfS₂, 2H-MoS₂, and 1T-PtS₂). Although the metal atoms primarily determine the electronic structure, chalcogen composition also plays a role, with increasing atomic number leading to d-band broadening and a gradual reduction in the band gap, as illustrated by the decrease in band gap from 1.3 eV in 2H-MoS₂ to 1.0 eV in 2H-MoTe₂ [128].

4.2.2 | Configuration

TMDCs' atomically thin nature, tunability in the band gap, high spin-orbit coupling, strong excitonic effect, natural abundance, and unique electrical and mechanical characteristics make them intriguing for fundamental research and energy harvesting applications. Maximizing the functional potential of TMDCs necessitates an in-depth understanding and precise control of their structural imperfections. Recent advancements have focused exclusively on characterizing, manipulating, and engineering defects in TMDCs through doping, functionalization, and heterostructure assembly. Among these strategies, doping is particularly effective in tailoring the properties of TMDCs, exerting a more pronounced influence on nanosheets than on their bulk counterparts. Various doping techniques, including electrostatic, substitutional, intercalation, and surface charge transfer doping, have been used to modulate TMDCs' properties, and emerging approaches such as reversible solid-state, remote modulation, and thickness modulation doping are also gaining traction. For instance, a recent study using the spin-coating technique investigated surface engineering of a MoS₂-based TENG by integrating an additional transition layer, such as polyethylene terephthalate (PET), alongside Kapton to enhance charge storage. The surface of MoS₂ (serving as an active layer) is functionalized with various aromatic carboxylic acids to improve the efficiency of vertical TENG. This configuration can effectively alter the work function (WF) and surface charge density of MoS₂-based TENG, leading to enhanced output performance. Here, the strong electron-withdrawing nature of carboxylate functionality facilitates a shift in MoS₂ thin films toward a more tribo-negative state in the triboelectric series, thereby optimizing their energy harvesting capability [129]. However, discrepancies between theoretically predicted and

experimentally realized device efficiencies often stem from intrinsic defects, such as vacancies, substitutions, adatoms, and grain boundaries, which significantly impact optical, electronic, and optoelectronic behaviors. For instance, sulfur vacancies in MoS₂ introduce electron donor states within the band gap, underscoring the necessity for controlled defect engineering to enhance energy harvesting applications. Additionally, alloying provides additional flexibility in tailoring band gaps and electronic properties, although achieving such tunability in TMDCs is challenging due to their dimensional constraints. A self-powered photosensitive vertical TENG was successfully developed by a spin-coating strategy using a ternary TMDC alloy (Mo_{0.5}W_{0.5}S₂), which functions as a triboelectric layer and a photo-absorbing material. This approach effectively addresses the limitations of conventional binary TMDCs by leveraging their enhanced optical properties. Due to its efficient visible-light absorption, the device achieves superior triboelectric output and enhanced energy harvesting performance under illumination [130]. Another promising research avenue is the development of heterostructures, which enhance the electronic and optical performance of TMDCs. These structures can be assembled in vertical or lateral configurations, with vertical stacking forming vdW heterostructures. This architecture minimizes lattice mismatch and mechanical strain by preventing direct atomic contact between layers. Additionally, the band structure of monolayer TMDCs is governed by interactions between transition metal and chalcogen orbitals, with vdW heterostructures further modifying band alignment through interlayer coupling, unlocking novel energy harvesting applications (see Section 6 for more details). Furthermore, these heterostructures have advantageous characteristics, including optimized band gaps, superior light-harvesting efficiency, and tailored electronic properties. These attributes make them highly promising for advanced photonic devices and next-generation solar energy technologies. For instance, a novel mixed-dimensional heterostructure composed of single-walled carbon nanotubes and MoS₂ has recently been shown to facilitate remarkably long-lived charge-separated states [131]. However, the long-term stability of certain TMDCs in solar cells is compromised by prolonged exposure to light and environmental factors, and scalable, cost-effective production remains a challenge. To overcome these limitations, researchers are exploring interface engineering, heterostructures, doping, and novel device architectures to enhance efficiency and durability. Despite advancements in band gap engineering, large-scale integration of 2D TMDCs into PV devices is still in its early stages. Significantly, excellent photoluminescence is produced during the exfoliation process of multilayers due to the expansion of band gaps in single-layer materials through the interaction of s–p orbitals. This transformation results in the conversion of indirect band gap into direct band gap materials. The electronic properties of TMDCs vary significantly with the number of layers. For instance, multilayer MoS₂ has an indirect band gap of 1.29 eV, which shifts to a larger direct band gap of 1.80 eV in its monolayer form. This direct band gap, aligned with visible and near-infrared wavelengths, makes semiconducting TMDCs ideal candidates for advanced photonics and optoelectronic applications in these spectral regions [132, 133]. In piezoelectric materials, the large band gap is crucial for preventing current leakage; photoluminescence offers a more effective method of harvesting solar energy, and due to this

band gap tuning ability, theoretically, it can absorb extra incident sunlight compared to traditional Si or GaAs materials of the same thickness [134–136]. Notably, certain TMDCs, such as TiS_2 and WT_2 , have a zero-band gap structure, akin to GR. Hence, for nanodevice applications, modulating the band gap in TMDCs is advantageous, as it allows for adjustable electronic properties tailored to specific requirements. For this purpose, the electronic band structure of TMDCs can be dynamically tuned using techniques such as strain engineering, electric field modulation, and alloying. These materials also have ambipolar field-effect characteristics, with MoS_2 demonstrating exceptional carrier mobility exceeding $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, highlighting its potential for use in high-performance electronic devices [137]. Moreover, some TMDCs like MoS_2 have very attractive features, including exceptional electron and hole states and a noticeable Seebeck coefficient, which signifies significant potential for thermoelectricity applications [138, 139]. Because of their large band gap, TMDC-based devices can also operate at high frequencies, which is crucial for applications involving data processing and transmission. These 2D thermoelectric modules have a multitude of adjustable characteristics through functionalization with diverse polymers, rendering them outstanding contenders for flexible and transparent electronics applications [140].

4.2.3 | Composites' Potential

The exceptional properties of TMDC materials open avenues for a wide range of applications, extending beyond their functionalities to their composite forms. The integration of nanoscale building blocks within TMDC-based composites enables the development of novel functionalities and superior performance characteristics. Traditional energy sources, such as batteries, remain prevalent in low-voltage applications like mobile devices. However, they present several limitations, including reliance on hazardous materials, limited durability, and a relatively short operational lifespan. These challenges have driven the exploration of alternative energy harvesting strategies, particularly through the development of electroactive material-based composites. Such composites enable sustainable energy generation by harnessing mechanical motion, offering an environmentally friendly approach to powering electronic devices. The integration of electroactive materials (TMDCs) into composite structures enhances their energy conversion efficiency while ensuring a reduced environmental footprint. Additionally, these composite-based devices are gaining attention due to their cost-effectiveness, sustainability, nontoxic nature, and widespread availability, making them a promising alternative for next-generation energy harvesting technologies with broad societal implications. These composites have emerged as promising materials, capitalizing on synergistic interactions to enhance efficiency across diverse harvesting platforms. For instance, the incorporation of TMDCs into PVDF-based PENGs significantly improves their mechanical, electrical, and thermal properties. The high surface area of TMDC nanosheets facilitates strong interfacial interactions with PVDF dipoles, resulting in an amplified piezoelectric response. Specifically, MoSe_2 with its atomically thin structure and high carrier mobility acts as a low-resistance pathway for charge transfer, thereby enhancing polarization and dipole alignment, which are critical

factors in optimizing piezoelectric performance. Under mechanical stress, these 2D TMDC nanosheets serve as nucleation sites for β -phase formation in PVDF, reinforcing dipole orientation and charge redistribution. This leads to greater polarization, enhanced charge accumulation, and higher energy conversion efficiency. Consequently, the resulting composite-based device (fabricated using the doctor blade method) can effectively harvest energy from various mechanical stimuli. The generated energy was utilized to power light emitting diodes (LEDs) and small-scale electronic devices, demonstrating the potential of TMDC composites for use in sustainable energy harvesting applications [141]. TENGs represent an emerging class of miniature energy harvesting devices; however, those constructed from conventional polymers often have limited friction-induced electrical output, restricting their practical applications. To address this issue, a TENG device incorporating a nanocomposite fiber membrane composed of MoS_2 (serving as a filler) and PVDF-hexafluoropropylene (HFP) was engineered using a common electrostatic spinning approach. The resulting device (MoS_2 -based composite acts as a dielectric layer) demonstrated stable and enhanced electrical output, effectively converting mechanical energy into usable electrical power [142]. Moreover, MoS_2 has also emerged as an effective reinforcing agent in composite materials, enhancing their mechanical performance without compromising structural integrity. Recent studies have demonstrated its ability to significantly improve toughness and stretchability while maintaining stiffness and modulus. When incorporated into hybrid composites, MoS_2 contributes to superior mechanical and electrical properties. This is due to the porous architecture and the 2D morphology of MoS_2 , which facilitates efficient fluid flow, making it particularly suitable for electromechanical applications based on piezoelectric mechanisms [143].

4.3 | MXene

MXenes (2D transition-metal carbide, nitride, or carbonitride) are a recent addition to the 2D material family; owing to many distinctive properties, they hold great promise for the harvesting of a variety of ambient energy sources as follows: (I) as they typically show electronic conductivity, thus, during electrical energy transformation, the captured energy loss during current flow is minimal; (II) due to their 2D nature, they have a high surface-to-volume ratio; (III) compared to other counterparts, the mechanical stability of these materials is exceptional, making them highly durable during energy harvesting; and (IV) the abundant surface chemistry endows these materials with not only hydrophilicity but also a unique charge distribution [51–53].

4.3.1 | Atomic Structure

MXenes represent a diverse class of 2D transition-metal carbides and nitrides, characterized by a layered structure where transition-metal (M) atoms form a honeycomb-like lattice interspersed with carbon and/or nitrogen (X) atoms occupying octahedral sites. These materials are derived from their parent MAX phases through selective etching of the A-layer elements

(e.g., Al, Si, Ga), resulting in loosely stacked layers that can be further exfoliated into single-layer sheets. The general formula for MXene is $M_{n+1}X_nT_x$, where M denotes an early transition metal, X represents carbon or N, T_x signifies surface terminations, and n varies from 1 to 4. Over the past decade, MXenes have been synthesized in more than 30 distinct stoichiometric configurations, with theoretical studies predicting the stability of many more. They can be categorized into six major types, three of which are defined by their transition-metal site occupancy.

- a. **Random solid-solution MXenes (s-MXenes):** Characterized by a random distribution of two or more transition metals, such as $Ti_{2-y}V_yCT_x$.
- b. **Out-of-plane ordered MXenes (o-MXenes):** Composed of alternating transition metal layers, such as $Mo_2Ti_2C_3T_x$.
- c. **In-plane ordered MXenes (i-MXenes):** Featuring an ordered arrangement of different metals within the same atomic layer, such as $(Mo_{2/3}Y_{1/3})_2CT_x$.

Beyond these classifications, MXenes with uniform halide or chalcogenide surface functionalization, referred to as iso stoichiometric MXenes (*t*-MXenes), have also been synthesized, such as $Ti_3C_2Cl_2$ and Ti_3C_2Se , demonstrating unique physico-chemical characteristics distinct from conventionally synthesized MXenes with mixed surface functionalization. Additionally, high-entropy MXenes (*h*-MXenes) have recently emerged, incorporating multiple transition metals into a single phase, such as $TiVNbMoC_3T_x$ and $TiVCrMoC_3T_x$, showing enhanced thermal stability and tunable electronic properties [144].

4.3.2 | Configuration

Due to their thermodynamic instability, MXenes are highly susceptible to oxidative degradation, primarily driven by exposure to oxygen and water. Among different MXene forms, such as films, composites, and colloidal dispersion, the latter shows the greatest instability due to its high surface area. Various strategies have been explored to mitigate oxidation, including inert gas protection, ultralow-temperature storage (e.g., -80°C and -20°C) [145], reductive capping with sodium L-ascorbate, storage in nonaqueous polar solvents (e.g., DMF), and preservation in saturated salt aqueous solutions (e.g., NaCl, LiCl, $CaCl_2$). Of these, only ultralow-temperature storage and saturated solutions have demonstrated the ability to extend the shelf-life of $Ti_3C_2T_x$ MXene [146]. However, these methods present challenges: deep refrigeration requires high energy consumption, and specialized solvent storage complicates handling and scalability. Recent studies suggest that adjusting the precursor composition, specifically by introducing excess Al during synthesis, significantly enhances the crystallinity and carbon stoichiometry of Ti_3AlC_2 MAX-phase precursors. The resulting MXene nanosheets have remarkable electrical conductivity and prolonged stability in aqueous environments, with a shelf-life exceeding 10 months, even at room temperature [147]. This synthesis approach preserves the intrinsic properties of $Ti_3C_2T_x$ MXene while improving its environmental resilience, making it highly promising for diverse applications.

Additionally, in energy harvesting technologies, MXene-based films demonstrate superior stability, further underscoring their potential for long-term operational reliability. The surface terminations in MXenes significantly configure their properties, and computational studies have identified four preferred sites in the MXene lattice the top site (positioned directly above a transition-metal atom), a hexagonal close-packed site (located within the hollow formed by surface metal atoms with an underlying X layer), a bridge site (situated above a metal-metal bond), and a face-centered cubic (fcc) site (present within a hollow region of surface metal atoms). These structural variations and surface modifications enable MXenes to have a wide range of tunable properties, making them highly promising for energy harvesting applications. Consequently, due to these advantages, MXene-based nanostructures have been used for ambient energy capture with promising outcomes. Numerous energy sources, like PV, piezoelectric systems, triboelectric systems, and thermoelectric systems and so forth, have been effectively captured with distinctive energy mechanisms by using MXene structures. Furthermore, owing to their superior electrical and mechanical characteristics, MXenes can be incorporated into TENGs, positioning them as a viable material for future TENG development. Therefore, MXene's distinctive composition, high compatibility with a wide range of devices, and good tunable properties render it an appealing choice for improving energy harvesting technologies. MXenes have surface terminations ($-O$, $-F$, $-OH$, $-Cl$), enabling tailored functionalization that enhances their stability and electronic properties. Their surface characteristics can be modified via noncovalent interactions (vdW, hydrogen bonding, electrostatic attraction) or covalent modifications, including small-molecule attachment, polymerization, and heteroatom doping. Owing to their inherent hydrophilic nature, MXenes can be seamlessly integrated into a variety of solution-based fabrication techniques, such as spinning, coating, vacuum-assisted filtration, and spray drying. These methods enable the efficient processing of MXenes into diverse macroscopic architectures, expanding their applicability across multiple advanced energy systems. For instance, as a transport layer, when MXene ($Ti_3C_2T_x$) nanosheets are integrated into self-assembled monolayer-based interfaces, they will efficiently improve charge transport, enhance the anode WF, and mitigate interfacial trap states, leading to reduced non-radiative recombination in perovskite solar cells. The device fabricated via spin-coating shows increased PCE and also improved film quality and moisture resistance, highlighting the use of MXenes as a promising solution for stable, high-performance PVs [148]. The tunable surface chemistry and structural adaptability also make MXenes highly effective for charge-trapping in TENGs. Leveraging these properties, a polyvinyl chloride/MXene (conductive electrode)-based TENG has been developed using an electrospinning technique. This configuration improves charge retention and confers superior environmental stability, long-term durability, and multifunctional sensing capabilities, highlighting its potential for use in next-generation energy harvesting applications [149]. A novel approach to modifying MXene has recently been developed, leveraging its controlled oxidation to enhance the performance of TENGs while simultaneously preventing further degradation. As MXene has rich surface terminations, it effectively attracts and traps electrons, allowing mobile electrons to cross its nanosheets. Through an in situ oxidation

reaction, MXene can be transformed into TiO_2 , a high-permittivity material with a relative permittivity of approximately 80, playing a critical role in dielectric constant regulation and polarization enhancement. Consequently, in the MXene/ TiO_2 hybrid system, the layered MXene structure effectively disperses TiO_2 nanoparticles, forming additional conductive pathways for efficient electron transport throughout the hybrid material. As a result, this architecture enables optimal charge trapping and storage, maximizing triboelectric charge retention. As the interfacial electric field in TENGs can cause surface charges to dissipate into the air or migrate into the tribo-material, charge regulation through effective charge storage is just as crucial as charge generation. In this regard, an MXene/ TiO_2 hybrid film was fabricated using a suction filtration technique, incorporating partially oxidized MXene as an intermediate layer between the negative tribo-material and the bottom electrode in a contact-separation mode. This configuration led to significant performance improvements due to the synergistic effects of electron trapping, charge blocking, and polarization, ensuring effective charge retention and minimizing diffusion losses [150].

4.3.3 | Composites' Potential

In the field dedicated to the enhancement of mechanical, electrical, and functional properties, the development of stable and adaptable composite materials has gained considerable attention in recent years. MXenes, owing to their 2D morphology, layered configuration, and inherent flexibility, have emerged as promising building blocks for multifunctional composite synthesis. Generally, they are incorporated into energy harvesting nanogenerators as electrode/active layers by using various fabrication techniques such as spray-coating, vacuum filtration, chemical cross-linking, inkjet printing, and dip-coating. Their unique properties have driven extensive research into MXene-based hybrid materials, leading to the fabrication of novel composites incorporating polymers, metal oxides, and carbon nanotubes. The highly tunable surface chemistry of MXene not only improves its compatibility within polymer matrices but also ensures uniform dispersion, thereby maximizing interfacial interactions in composite materials. Polymer-based composites offer advantages such as flexibility and lightweight properties, making them suitable for energy harvesting applications. By integrating MXene nanosheets into a polymer matrix, a synergistic balance is achieved because the polymer provides mechanical stability and environmental resistance. At the same time, MXene contributes exceptional electrical conductivity and mechanical strength. This combination ensures long-term functionality while mitigating environmental degradation. For instance, by utilizing a printing technique, a composite of poly(vinylidene fluoride-co-trifluoro ethylene) was fabricated with MXene (used as a template), where the polymer aligns perpendicular to the basal plane of MXene nanosheets. The strong electrostatic interactions within this structure resulted in a significant piezoelectric response. These nanoscale interactions at polymer-2D material interfaces provide new research avenues for designing high-performance piezoelectric composites with diverse applications [151]. MXenes also serve as efficient hole transport layers (HTLs) in PSCs, leveraging their high conductivity and tunable surface chemistry to enhance hole extraction, minimize recombination losses, and

boost device efficiency. Their intrinsic chemical stability also acts as a protective barrier, mitigating environmental degradation and extending PSC's long-term stability. A novel approach integrates $\text{Ti}_3\text{C}_2\text{T}_x$ monolayer nanoplates into a phenyl ethyl ammonium iodide solution, forming a modified 2D perovskite layer atop a 3D perovskite. This 3D/2D composite, achieved via in situ thermal annealing, enhances carrier transport and interface properties. Hydrogen bonding between $\text{Ti}_3\text{C}_2\text{T}_x$ functional groups and phenylethylamine (PEA^+) stabilizes the perovskite matrix, improving their carrier lifetime and defect passivation. The resulting optimized energy band alignment and superior hole mobility lead to an impressive PCE with 95.5% retention over a prolonged period of time in ambient conditions without encapsulation, highlighting the MXene-based composite's potential for use in durable and high-performance PSCs [152]. Moreover, developing flexible, eco-friendly, and high-performance silk-based composite membranes for energy harvesting remains a significant challenge. To address this issue, a crosslinked silk-based single-electrode TENG was fabricated by integrating MXene into a polyvinyl alcohol, silk fibroin, and a silk sericin matrix, forming a uniform film. This crosslinking-based device demonstrated outstanding electrical performance and power density, highlighting the MXene-based composite's role in triboelectric performance [153]. Nowadays, the integration of PENGs with photo-piezo catalysis presents a promising approach for simultaneous energy harvesting and environmental remediation. Addressing the dual challenge of clean energy generation and pollution control, this hybrid approach aligns with the principles of a circular economy by minimizing waste, optimizing resource use, and reducing emissions. For instance, recently, a nanofiber composite incorporating zinc oxide (ZnO) nanoparticles and MXene nanosheets (serving as electrodes) was fabricated via electrospinning, demonstrating a substantial enhancement in energy conversion efficiency [154].

4.4 | h-BN

h-BN, analogous to GR, consists of a 2D honeycomb lattice formed by alternating B and N atoms. BN, with a 1:1 boron-to-nitrogen ratio, has insulating characteristics that distinguish it from other semiconductors in the same group III-V. In its hexagonal phase (h-BN), this layered inorganic crystal demonstrates exceptional thermal stability and thermal conductivity, making it a valuable 2D material.

4.4.1 | Atomic Structure

BN generally exists in several stable crystalline forms, categorized based on their hybridization. These include sp^3 -hybridized tetrahedral cubic (c-BN) and wurtzite structures, along with sp^2 -hybridized hexagonal (h-BN) and rhombohedral phases. The cubic phase has a zincblende-type crystal structure, where B and N atoms form an interpenetrating fcc lattice, closely resembling the atomic arrangement in diamond. Each B and N atom in c-BN undergoes sp^3 hybridization, resulting in a B-N-B/N-B-N characteristic tetrahedral bond angle of 109° . Notably, the densely packed lattice configuration in c-BN contributes to its exceptional hardness, making it one of the hardest

known materials. The wurtzite phase (metastable polymorph) consists of tetrahedral B and N atoms arranged in a close-packed structure with an AA–BB–AA stacking sequence. This phase shows a B–N bond length of approximately 157 pm, but has the same characteristic tetrahedral bond angle (109.5°). The rhombohedral phase shares structural similarities with h-BN (AA′) but adopts an ABC stacking order [155]. In this configuration, B and N atoms remain aligned vertically; however, the layers are shifted, causing the first and third N atoms to align with the fourth and sixth B atoms in the layer below. In contrast, h-BN (referred to as white GR) adopts a layered honeycomb hexagonal structure ($P6_3/mmc$ space group, D_{6h} point group, $a = b = 2.5 \text{ \AA}$, $c = 6.6 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$ unit cell parameters) with sp^2 hybridization, forming a 2D lattice. Strong in-plane B–N σ bonds contribute to high thermal conductivity, chemical stability, and mechanical strength, and the π orbitals remain unoccupied, rendering h-BN an insulator with a wide band gap ($\sim 5.955 \text{ eV}$). Due to the electronegativity difference between B and N, ionic bonding characteristics emerge with the h-BN layers. This leads to an AA′ stacking configuration, aligning N atoms directly above B atoms in adjacent layers. Additionally, sometimes, AB (Bernal) stacking has been observed in h-BN films, though it represents a higher energy metastable state [156].

4.4.2 | Configuration

As discussed, h-BN represents the most thermodynamically stable polymorph of BN, featuring an alternating B–N arrangement. In this configuration, the adjacent layers are held together by weak vdW interactions, leading to an interlayer spacing of $\sim 3.3 \text{ \AA}$. Like GR, h-BN also shows various geometric configurations, such as zig-zag and armchair edge configurations, depending on its crystallographic orientation. In monolayer form, it can assemble into various nanostructured morphologies, including nanotubes, nanoflakes, nanoribbons, and nanoscrolls, further expanding its potential applications in advanced material design for energy harvesting. It has remarkable thermal and chemical stability (due to strong B–N bonds and the absence of dangling bonds), making it highly suitable for use in energy-related technologies across various functional applications [157]. For instance, the exceptional thermal conductivity and electrical insulating properties of hybrid h-BN make it an ideal material for efficient heat dissipation and enhanced durability in PV devices [158]. Additionally, the tunable band gap of their hybrid structures enables more effective light absorption across a broader spectral range, paving the way for the development of next-generation high-performance solar cells. It demonstrates tensile strength (around 100 GPa) and an elastic modulus (approximately 0.8 TPa) comparable to those of GR [159]. Unlike GR's brittle fracture behavior, single-layer h-BN has remarkable fracture toughness and plasticity, attributed to its asymmetric lattice structure, which facilitates stable crack propagation. Moreover, h-BN has excellent thermal conductivity (up to $33 \text{ W m}^{-1} \text{ K}^{-1}$), comparable to that of stainless steel, and a low thermal expansion coefficient (between 2.0 and $6.5 \times 10^{-6}/^\circ\text{C}$), nearly matching that of silica glass, making it ideal as a protective layer and passivation buffer in 2D devices, while primarily serving as an insulating barrier or substrate in nanodevices [160].

Furthermore, defects in h-BN combined with other 2D materials enhance thermoelectric properties by increasing phonon transport, and yet, still fall short of practical use. Being an electrically insulating and mechanically robust material with a wide band gap and high thermal stability makes h-BN an ideal candidate for modifying and enhancing the properties of other materials, like GR, TMDCs, and polymers, through heterostructure. Additionally, its intrinsic properties can be tailored or improved via doping or alloying, expanding its applicability in advanced functional materials. Such advances in h-BN heterostructures and doping strategies, like incorporating chemical species, show potential for improving thermoelectric efficiency by tuning electronic properties and enhancing charge carrier mobility, paving the way for future applications [161].

4.4.3 | Composites' Potential

The exceptional mechanical strength and thermal insulation properties of h-BN make it an ideal candidate for enhancing the dielectric performance of composite materials. Advanced composite synthesis techniques (e.g., 3D printing, cross-linking, etc.) have been used to integrate it into energy harvesting systems. The key fabrication methods include free-standing composite films, nanosheet alignment, 3D printing with polymer, and in situ polymerization, each contributing to the effective utilization of h-BN in a multifunctional composite structure. Due to its non-centrosymmetric structure and the ionic nature of B–N bonds, h-BN has exceptional capabilities for use in energy-related applications. The mechanical and functional performance of h-BN composites is strongly influenced by factors such as volume fraction, dispersion, and alignment of 2D reinforcements. Additive manufacturing techniques, particularly direct ink writing, facilitate the fabrication of well-aligned, defect-free carboxymethyl cellulose-h-BN composites, where h-BN acts as a rheology modifier to reinforce the polymer. A sustainable 3D printing approach, using aqueous-based inks composed of an eco-friendly cellulose polymer and citric acid as a green linker, yields a high-strength h-BN composite with enhanced mechanical, thermal, and flexoelectric properties [162]. Beyond structural applications, h-BN-based composites are gaining prominence in TENGs for self-powered sensing and energy generation. For instance, PDMS@h-BN composite films, engineered via contact–separation mode TENG with a simple two-electrode configuration, demonstrated superior dielectric properties and enhanced power output for portable electronics. Notably, these advancements have enabled the development of a self-powered handwriting recognition system, integrating TENG technology with machine learning algorithms for real-time handwriting sensing, marking a significant step toward next-generation intelligent electronics [163]. Recently, polymers (such as PVDF and its copolymers) have gained significant attention in mechanical energy harvesting due to their exceptional properties (lightweight, durable, environmentally sustainable). Among the various polymorph phases of PVDF, the β phase is the most electroactive phase. Therefore, to enhance the β -phase content in the PVDF matrix, the incorporation of 2D nanomaterials as a filler is an effective strategy. As h-BN nanostructures show intrinsic piezoelectric behavior; therefore, to leverage the advantages of both materials, free-standing standing PVDF/h-BN composite films were fabricated using the

solvent casting method, demonstrating enhanced PENG performance. The enhanced device performance is attributed to the piezoelectric filler (h-BN), driven by interfacial Maxwell–Wagner–Sillars polarization at the h-BN/PVDF interface [164].

4.5 | BP

BP is a well-studied monoelemental 2D inorganic layered material and the most thermodynamically stable form of phosphorus. It can be synthesized from white or red phosphorus under high-pressure and high-temperature conditions. BP remains the most thermodynamically stable allotrope of phosphorus among others (white, red, and violet phosphorus). Monolayer BP has a thickness of 0.7–0.85 nm and can be produced through mechanical or liquid exfoliation. Its unique puckered honeycomb structure with broken inversion symmetry leads to notable in-plane anisotropy in properties like carrier transport, and an ambipolar field effect driven by its low crystal symmetry [165].

4.5.1 | Atomic Structure

BP has an orthorhombic crystalline structure composed of puckered double layers, where each half consists of parallel zig-zag chains of covalently bonded phosphorus atoms. Each phosphorus atom undergoes sp^3 hybridization, forming two in-plane covalent bonds with adjacent atoms and a third bond linking it to a phosphorus atom in the adjacent half-layer, resulting in a quadrangular pyramidal arrangement of P_6 rings in a chair conformation [22]. The P–P bond length within the zig-zag chains is approximately 2.224 Å, nearly equivalent to the interlayer distance of 2.244 Å, whereas the closest spacing between adjacent double layers is 3.314 Å with an average bond angle of 102°. Notably, the lattice constants and angles are influenced by the number of layers. The weak vdW interactions between stacked puckered layers facilitate exfoliation into single- or few-layer BP. Under increasing pressure, BP undergoes structural transformations, shifting to a rhombohedral phase at 5 GPa and further transitioning to a simple cubic structure at 10 GPa, with no additional changes observed up to 60 GPa [166].

4.5.2 | Configuration

BP shows significant instability under ambient conditions due to its unique chemical structure. Each surface phosphorus atom forms three covalent P–P single bonds while retaining a lone pair of electrons, which contributes to its high carrier mobility and bipolar transport characteristics. However, these lone-pair electrons make BP highly susceptible to oxidation and degradation in air. Light-induced oxidation is the primary degradation mechanism, where photon exposure generates electron–hole pairs, facilitating electron transfer from BP to surrounding oxygen molecules. This results in the formation of superoxide radicals, which react with phosphorus to form P–O and P=O bonds. Water molecules then interact with these oxidized sites, leading to the detachment of phosphorus atoms via hydrogen

bonding, further accelerating BP deterioration. Consequently, their prolonged exposure results in layer-by-layer etching, forming phosphoric acid, and deteriorating BP's electronic properties [167]. Therefore, to address the stability issues of BP, various passivation strategies can be used for practical applications, such as covalent and noncovalent functionalization, surface coordination, physical encapsulation, and edge passivation. For instance, encapsulating BP with other 2D-layered materials like GR or h-BN, forms vdW heterostructures, enhancing its stability without lattice-matching constraints [168]. This encapsulation strategy for enhancing the stability of BP involves applying a protective passivation layer to shield it from environmental degradation. In optoelectronic devices, this is achieved by using either a polymethyl methacrylate (PMMA) coating, which minimizes optical interference, or through h-BN encapsulation, which prevents light absorption in the communication band. Both methods effectively improve BP's durability, and in turn, the device displays phenomenal performance [169]. BP (also referred to as phosphorene) has recently emerged as a rival to GR due to its various advantages, including piezoelectricity, a greater on/off current ratio, a tunable band gap, robust light–matter interaction, and extraordinary carrier mobility. As a result, it has generated significant research interest in energy harvesting applications. Remarkably, the BP-based devices are efficient, flexible, and lightweight, revealing great potential for sensors and nanogenerators [170]. Moreover, in its intrinsic form, BP displays a direct band gap (2 eV), but by increasing the thickness, it can be decreased to as low as 0.3 eV, and due to this, it is effectively utilized in solar cells, displaying its efficacy as an electron transport layer and a HTL; also, theoretically, it can display good solar PCE of about 20% [171]. This tunable band gap positions BP uniquely between zero-band gap GR and wide-band gap TMDCs, allowing absorption across the visible to infrared light ranges. Additionally, BP has been investigated as a possible option for mechanical energy harvesting through the piezoelectric effect, facilitated by its non-centrosymmetric lattice structure, and due to thermal anisotropy, it is also a viable option for thermoelectric applications [172]. Its in-plane anisotropy and high carrier mobility, exceeding $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ in phosphorene-based FETs, make it ideal for optoelectronic applications [173].

4.5.3 | Composites' Potential

2D BP has gained attention as a promising alternative to GR due to its tunable band gap, high on/off current ratio, and piezoelectric and PV properties. However, its rapid degradation under ambient conditions significantly limits its practical applications. In addition to various stabilization strategies, BP-based composites have emerged as multifunctional materials offering enhanced performance. Theoretical and experimental studies suggest that the interaction between BP's P_z orbitals and GO's π orbitals, along with P–C bond formation, facilitates charge transfer. Leveraging this synergy, a solvothermal method was developed to synthesize BP–GO composites, which were incorporated into perovskite grain boundaries using an advanced solvent posttreatment technique. These BP–GO composites effectively passivate perovskite defects, suppress nonradiative recombination, and hinder the diffusion of active ions, leading to improved lattice stability, passivation capabilities, and charge

transport efficiency [174]. Furthermore, integration of GO with BP also creates an ideal hybrid platform for exploring piezoelectric-driven charge transfer mechanisms. A novel surface-enhanced Raman scattering chip strategy was recently established by incorporating a BP-GO composite semiconductor with a PVDF film to examine the influence of the piezoelectric effect on chemical enhancement. Here, BP nanosheets were encapsulated within multilayer GO, effectively mitigating BP's inherent instability. As a result, this piezoelectrically activated substrate demonstrates a remarkable increase in photocurrent, confirming the substantial improvement in charge transfer efficiency within the BP-GO composite system [175]. Though currently BP has emerged as a promising material for enhancing both thermal conductivity and photothermal conversion efficiency, attributed to its distinctive orthorhombic pleated honeycomb structure, challenges remain in developing phase-change composites with optimized solar energy harvesting efficiency. It is a well-known fact that, unlike chemically inert nanofillers such as carbon nanotubes and GR, BP nanosheets are prone to oxidation under ambient conditions. Additionally, achieving hierarchical alignment of BP nanosheets using conventional 3D thermal transport frameworks remains a challenge. To overcome these limitations, a refined exfoliation method, followed by a freeze-casting strategy, was used to produce ultrathin BP nanosheets with minimal oxidation. These nanosheets were then integrated into a polyethylene glycol (PEG) matrix, resulting in a 3D BP/PEG composite that demonstrates enhanced solar energy harvesting capabilities [176]. Table 2 summarizes the fundamental characteristics of representative 2D materials. The incorporated data are the combination of both experimental measurements and theoretical predictions.

4.6 | 2D Conjugated MOF (2D MOF)

2D metal-organic frameworks (MOFs) have garnered significant interest as a novel class of multifunctional materials, owing to their well-defined structural tunability, inherent porosity, superior charge transport properties, and remarkable electrical conductivity. These distinctive attributes position 2D MOF as a highly promising candidate for advancing next-generation electronic and energy-related applications.

4.6.1 | Atomic Structure

2D MOF has a layered structure like graphite, where metal nodes and organic ligands coordinate in a planar arrangement to form extended 2D lattices. These layers are stacked along the z-axis through π - π interactions, creating well-defined nanochannels. Typically, the organic ligands used in 2D MOF are highly planar and conjugated, incorporating multifunctional groups (e.g., $-\text{NH}_2$, $-\text{OH}$, $-\text{SH}$), whereas the coordination centers are transition-metal ions (e.g., Fe, Cu, Co, Ni, Pt) with unpaired electrons. The formation of a square-planar coordination geometry facilitates strong $d\pi$ - $p\pi$ orbital hybridization, enabling in-plane charge delocalization. Additionally, the rigidity and extensive π conjugation of the ligands contribute to structural stability and efficient electronic π delocalization across the 2D framework.

TABLE 2 | Physical, mechanical, and electrical/thermal properties of typical 2D materials.

2D materials	Physical properties		Mechanical properties		Electrical/thermal properties		
	Density (g cm^{-3})	Specific surface area ($\text{m}^2 \text{g}^{-1}$)	Young's modulus (GPa)	Tensile strength (GPa)	Electrical conductivity (S cm^{-1})	Dielectric constant	Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
Graphene	2.267	2630	1000	130	10^6	-1.8 (in plane) -3 (out of plane)	1500-2500
MoS ₂	5.06	5.28-27.8	270 ± 100	23	10^4	15.5-15.9 (in plane) 6.2-6.9 (out of plane)	23.2-155
MXene	2.6	49.5-61.4	133-517	14-70	5000-15,000	—	55.8
h-BN	2.1	198.41-964.3	~860	~70	10^{-13} - 10^{-6}	6.82-6.93 (in plane) 3.29-3.76 (out of plane)	340
BP	2.69 (orthorhombic)	~2630	58 (zig-zag) to 27 (armchair)	8.42 (zig-zag)	40-60	6.1	110 (zig-zag) 36 (armchair)

4.6.2 | Configuration

Although most MOFs are electrically insulating with low charge mobility, recent advancements in 2D MOFs have introduced a new class of multifunctional materials with remarkable electrical conductivity and tunable chemical structure. These 2D MOFs have layered architectures like other 2D materials, allowing their exfoliation into ultrathin nanosheets. Due to their unique structural characteristics, including large surface area, abundant active sites, and open channels, they are highly suitable for energy-related applications. The strong in-plane covalent bonding within these frameworks facilitates electron delocalization, enhancing their electrical properties and making them ideal for electronic devices. Additionally, their high specific surface area promotes efficient charge transfer and redox reactions, whereas their solution processability enables easy integration into thin films and functional devices, making them a promising candidate for next-generation thermoelectric materials. The topology and pore structures of 2D MOFs are largely determined by the symmetry of their conjugated ligands. For instance, trigonal ligands based on benzene or triphenylene form honeycomb lattices with hexagonal channels [177], phthalocyanine- and porphyrin-based ligands generate tetragonal or rhombic lattices, and D_{6h} symmetric ligands such as perthiolated coronene facilitate the formation of Kagome lattices. This ability to predict and control framework architectures by ligand design enables the development of functional MOFs tailored for specific applications [178, 179]. Despite their promising electronic properties, most reported 2D MOFs show metallic behavior, limiting their use in semiconductor applications. To address this, researchers recently synthesized rhombic 2D MOF with D_{2h} symmetry. The resulting p-type semiconductor has an indirect band gap, high electrical conductivity, and exceptional charge carrier mobility [180]. Moreover, post-synthetic modifications offer additional functionalization routes, such as converting 2D MOFs into 3D structures via solvent-assisted linker exchange or tuning interlayer distances through ligand replacement reactions, further configuring their versatility in material design.

4.6.3 | Composites' Potential

The integration of 2D MOFs into hybrid composite materials has emerged as a promising approach for advancing energy-related applications. The incorporation of 2D MOF with various functional materials has been proven to be an effective approach for enhancing their properties and introducing novel functionalities for practical applications. To date, a wide range of 2D MOF-based composites have been synthesized by integrating 2D MOFs with diverse functional materials, such as GO and polymers. This synergistic integration not only preserves the intrinsic advantages of the individual constituents but also generates emergent physical and chemical properties. Moreover, the inherent sensitivity of 2D MOF, like other 2D materials, to moisture presents a significant challenge when used in energy harvesting devices. Therefore, the integration of 2D MOF into composite materials has also emerged as a viable strategy for effectively enhancing their structural integrity and environmental resilience, thereby facilitating the development of more robust and reliable devices. A key example is the

incorporation of 2D MOFs as an active filler into polymer-based composite materials for TENGs. Due to their high aspect ratio and superior charge-trapping capabilities, 2D MOFs significantly improve the energy output of TENGs. Notably, a composite material composed of 2D MOFs embedded in silk fibroin was fabricated using a sonication-assisted method, where the composite functioned as a triboelectric-positive layer. This integration successfully enhanced the power density of TENGs, enabling efficient energy harvesting from human motion [181]. Beyond TENGs, 2D MOFs are also useful in salinity gradient power conversion, an innovative approach to harvesting energy from ion transport in electrolyte solutions. In this scenario, conventional 2D membranes face challenges related to restricted ion flux due to narrow interlayer spacing and elongated transport pathways. To overcome these limitations, the researcher developed a 2D MOF-based nanocomposite membrane based on silk nanofibers using a vacuum-assisted approach. This incorporation of space-charge nanofibers enhanced membrane stability while enabling multiple one-dimensional ion pathways, significantly improving ion transport efficiency. By mixing artificial seawater with river water, this membrane achieved a remarkable power density [182]. Furthermore, existing membranes suffer from low charge carrier generation due to disordered microstructures and inefficient charge transport mechanisms. Addressing this challenge, a highly conductive 2D MOF-cellulose nanofiber (serving as a reinforcing mechanical crosslinker) composite membrane was synthesized using a vacuum-assisted technique for photoelectric energy conversion. This system capitalizes on extended π - d conjugation within the ab plane and π - π stacking interactions, creating optimized charge carrier pathways that maximize both generation and ion transport. These advancements highlight the immense potential of 2D MOF-based composites in next-generation energy harvesting technologies [183].

In summary, to rationally evaluate and select suitable 2D materials for specific energy harvesting applications, it is important to know the interplay between the atomic structure and functional properties. Figure 4 shows the comparative electrical conductivity, mechanical strength, thermal conductivity, band gap tunability, surface functionalization, and environmental stability of typical 2D materials, highlighting the high conductivity and mechanical strength of GR, the rich surface chemistry of MXene and 2D MOF, and the tunable electronic properties of TMDCs and BP.

4.7 | Beyond Conventional 2D Materials

Apart from the aforementioned 2D materials, additional ones, such as perovskites, covalent organic frameworks, and layered double hydroxides, among others, demonstrate effective energy harvesting capabilities [184–187]. Group IV monoelemental 2D materials, such as silicene, germanene, and stanene, have also gained significant attention due to their GR-like honeycomb structures, which give rise to unique physical properties. Unlike GR, these materials can only be synthesized in situ under ultrahigh vacuum conditions, not through exfoliation techniques. Their compatibility with existing semiconductor technologies offers a substantial advantage for developing energy-efficient

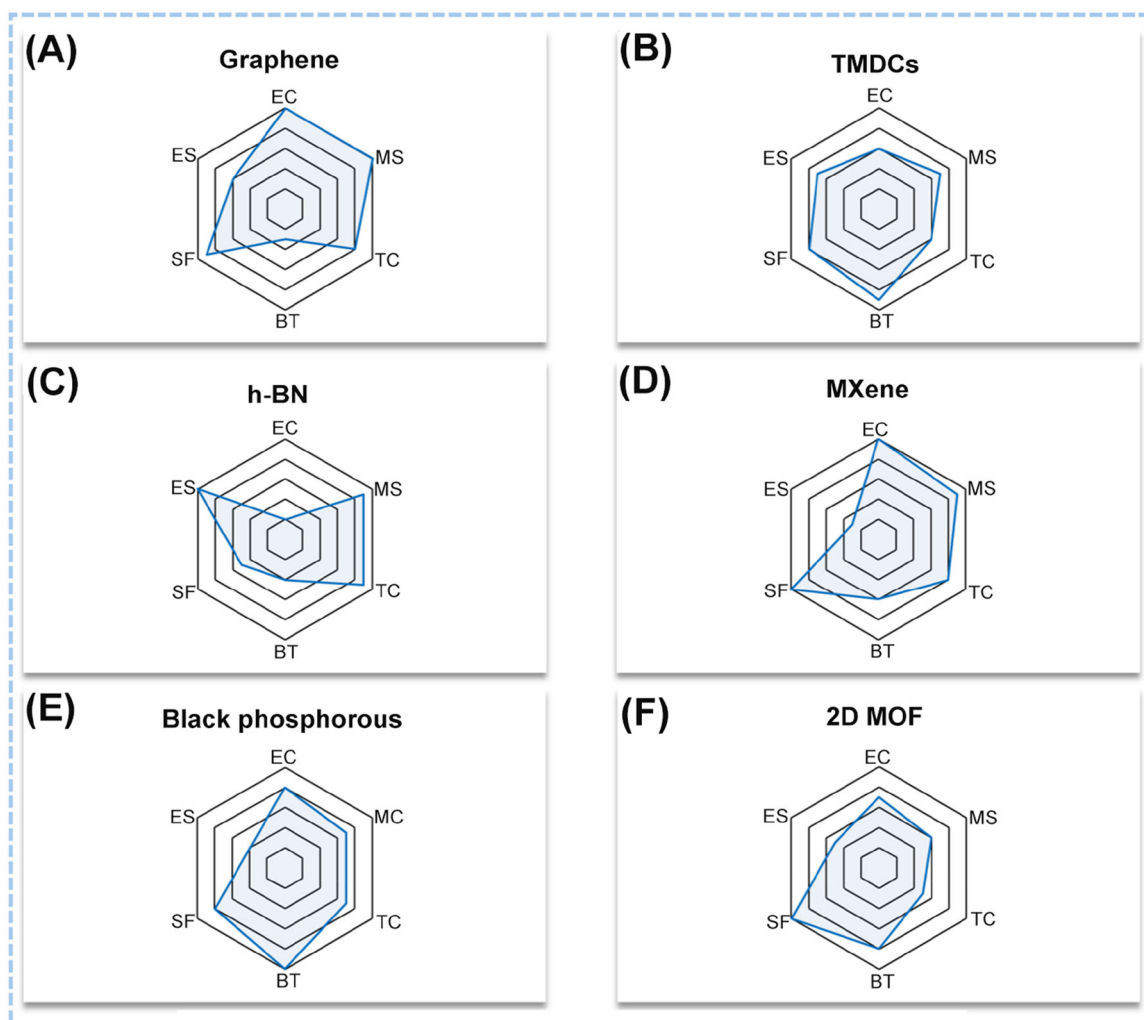


FIGURE 4 | Comparison of the structure–property relationship: Comparative radar chart of (A) graphene, (B) TMDCs, (C) h-BN, (D) MXene, (E) black phosphorous, and (F) 2D MOF, highlighting their diverse properties for energy harvesting applications, where EC, MC, TC, BT, SF, and ES represent electrical conductivity, mechanical strength, thermal conductivity, band gap tunability, surface functionalization, and environmental stability, respectively.

nanodevices, positioning them as key players in advancing next-generation electronic materials [188]. Moreover, newly developed elemental 2D materials like antimonene, tellurene, bismuthine, and arsenene have garnered significant interest for their potential in optoelectronics and energy applications. Their unique attributes, including large surface areas, high carrier mobility, tunable band gaps, and structural anisotropy, make them highly promising for advanced energy conversion technologies, driven by their exceptional electronic properties [189]. Significant advances have been made in exploring elemental 2D materials, uncovering their vast potential in energy harvesting technology due to their exceptional properties. However, challenges remain in scalable synthesis, stability, and bridging the gap between theoretical predictions and experimental results. Further, the integration of 2D materials into devices presents significant challenges, primarily due to issues in compatibility with traditional interfaces, such as electrodes and dielectrics, which result in contact resistance and performance losses. High-quality synthetic approaches coupled with innovative transfer techniques are essential to overcome these hurdles. Researchers are increasingly focusing on developing heterostructure strategies and utilizing advanced characterization techniques to boost performance and explore new energy applications, paving the

way for innovative uses of these 2D elemental materials beyond traditional electrochemical systems [190].

In conclusion, 2D materials are particularly appealing due to their superior in-plane flexibility compared to traditional materials, making them well-suited for integration into flexible energy harvesting devices.

5 | Interface Engineering in 2D Materials for Energy Harvesting

In materials science, the unique capabilities of 2D-layered materials have spurred a wave of innovation, enabling their integration into a wide range of multifunctional devices. Central to harnessing these capabilities is the discipline of interface engineering, which focuses on precisely controlling the interactions at the boundaries between different materials to enhance functionality. By optimizing these interfaces, researchers can significantly improve the performance of optoelectronic applications. Due to their atomically thin structure, 2D materials are highly sensitive to external conditions, making the quality of the interfaces crucial for achieving high efficiency in energy harvesting

devices. The advanced field of interface engineering in 2D materials is highlighted by ongoing progress in optimizing device interfaces. This is achieved by fine-tuning the interactions at different layers and component interfaces to improve charge transport, minimize energy losses, and improve device stability. Key developments include strategies like passivation defects, enhancing contact quality, tuning dielectric properties, and introducing charge transfer within 2D heterostructures. These innovations are crucial for improving the device's performance and functionality in PVs, thermoelectric, and piezoelectric applications. Regarding the use of interfaces in heterostructures created by layering 2D materials, Zhao et al. explored the photoresponsive behavior of a 2D MnSe/WSe₂ heterojunction, demonstrating impressive device performance. This heterostructure achieved a current rectification ratio of 6557, with a wide spectral response from 300 to 2200 nm, an exceptional responsivity of 156 A/W, detectivity of 2.21×10^{12} jones, and external quantum efficiency of 36,400%. Furthermore, the type-II band alignment and inherent potential within the heterojunction effectively separated photoexcited electron-hole pairs, highlighting the potential of MnSe-based heterostructures in advanced optoelectronic applications [191]. In another study, the authors enhanced the photoelectric conversion efficiency of dye-sensitized solar cells by modifying the device interface with localized surface plasmon resonance techniques. Their finding indicates that this interfacial engineering significantly improved current density and incident photon-to-current efficiency compared to those without Ag nanoparticles. This study highlights the importance of an optimal quantity of Ag nanoparticles in boosting photon-electric conversion efficiency. Moreover, it also identifies the ideal ratio of metal nanoparticles emerging as an effective strategy for improving the dye-sensitized solar cell's performance. This study offers valuable insights into the mechanism and optimization approaches for PV devices [192]. Among various 2D materials, MXene stands out as a promising candidate for wearable electronics due to its remarkable physicochemical properties. However, pure MXene alone does not meet all the criteria needed for effective energy harvesting applications. Recently, the field has witnessed a surge in multi-interface engineering techniques aimed at modifying the properties of MXene and its composites to enhance their functionality and adaptability [193]. Several strategies have been investigated to optimize the MXene interface, such as regulating terminations groups, modifying through polymerization (by incorporating external functional groups such as -COOH, -SO₃H, and -NH₂), introducing heteroatoms (like N, P, and S), and manipulating defects, all of these playing a crucial role in influencing the active sites and electronic properties of MXene, facilitating property optimization [194]. Thanks to advancements in interfacial engineering, high-performance MXene has been developed for various energy harvesting devices, including solar cells, TENGs, and thermoelectric nanogenerators [195].

6 | Moiré Superlattices and Twistronics in 2D Materials for Flexible Energy Harvesting

Some of the most intriguing and complex phases of matter emerge from strong electron correlations or topologically nontrivial electronic structures. Moiré-engineered systems have recently become a pivotal platform for exploring the

interplay between these phenomena, leveraging the diverse library of 2D vdW materials. A defining feature of 2D materials is their tunable electronic landscape, which can be engineered by vertically stacking atomically thin crystals. This stacking facilitates significant overlap of surface electronic wavefunctions. Moiré superlattices in 2D vdW materials have emerged as a groundbreaking platform for investigating exotic electronic and optical properties. These properties can be precisely tailored by selecting constituent atomic layers and manipulating their twist angles. Moiré superlattices not only inherit the intrinsic characteristics of their building blocks but also show novel emergent phenomena. The dangling bond-free surfaces of 2D flakes facilitate stacking heterostructures or twisted Moiré patterns, enabling atomic interface design, energy band engineering, and crystal symmetry modulation. By stacking single-layer materials, where atoms primarily reside at the surface, interlayer interactions define new systems within vdW heterostructures. However, when the crystalline orientation of one layer is rotated relative to another, unique θ -dependent Moiré superstructures emerge. These structures result in varying atomic interactions between the two layers, forming the foundation of the field known as twistronics [196]. By adjusting the rotation angle θ , new interlayer coupling phenomena arise. Furthermore, when two flakes with similar or identical lattice structures are slightly misaligned, periodic patterns called Moiré superlattices are generated (Figure 5A), offering opportunities to explore and manipulate material properties at the atomic level [197, 199]. These structures, characterized by superlattice potentials with much larger periodicities than those of the parent lattices, allow unprecedented control over band structures, symmetry breaking, quantum geometry, and Bloch wavefunction topology. Consequently, they unlock diverse special properties such as electrical, thermal, mechanical, optical, and optoelectronic properties, which diverge significantly from those of the individual layers. Nowadays, twisted heterostructures with Moiré patterns have become a central topic in 2D materials research because the conventional vdW heterostructures rarely have interlayer alignment. Moiré superlattices formed by stacking crystals with varying crystalline symmetries provide exceptional potential for engineering interfacial symmetry breaking, enabling unique bulk photogalvanic effects. For instance, combining WSe₂ with BP or SiP, which possess distinct rotational and mirror symmetries, creates stripe Moiré superlattices featuring a single mirror line (Figure 5B) [200]. This symmetry reduction induces a Berry curvature dipole, facilitating a nonlinear quantum response, such as spontaneous quantum-mechanical shift currents, which are absent in the individual materials due to symmetry constraints. Although linear photocurrent responses in isolated monolayers arise from contact-induced mechanisms, nonlinear bulk photogalvanic effects emerge when light illuminates the Moiré superlattice device center (Figure 5C). This discovery opens avenues for significant nonlinear optoelectronic phenomena, particularly in the terahertz (THz) range, with promising applications in systems like twisted bilayer GR and aligned WSe₂/WTe₂ heterostructures. Moiré superlattices with robust twist-proximity effects in low-dimensional systems have the potential for use in the development of nanoscale thermoelectric devices. In this context, vdW heterojunctions combining metallic MX₂ and wide-band gap semiconducting

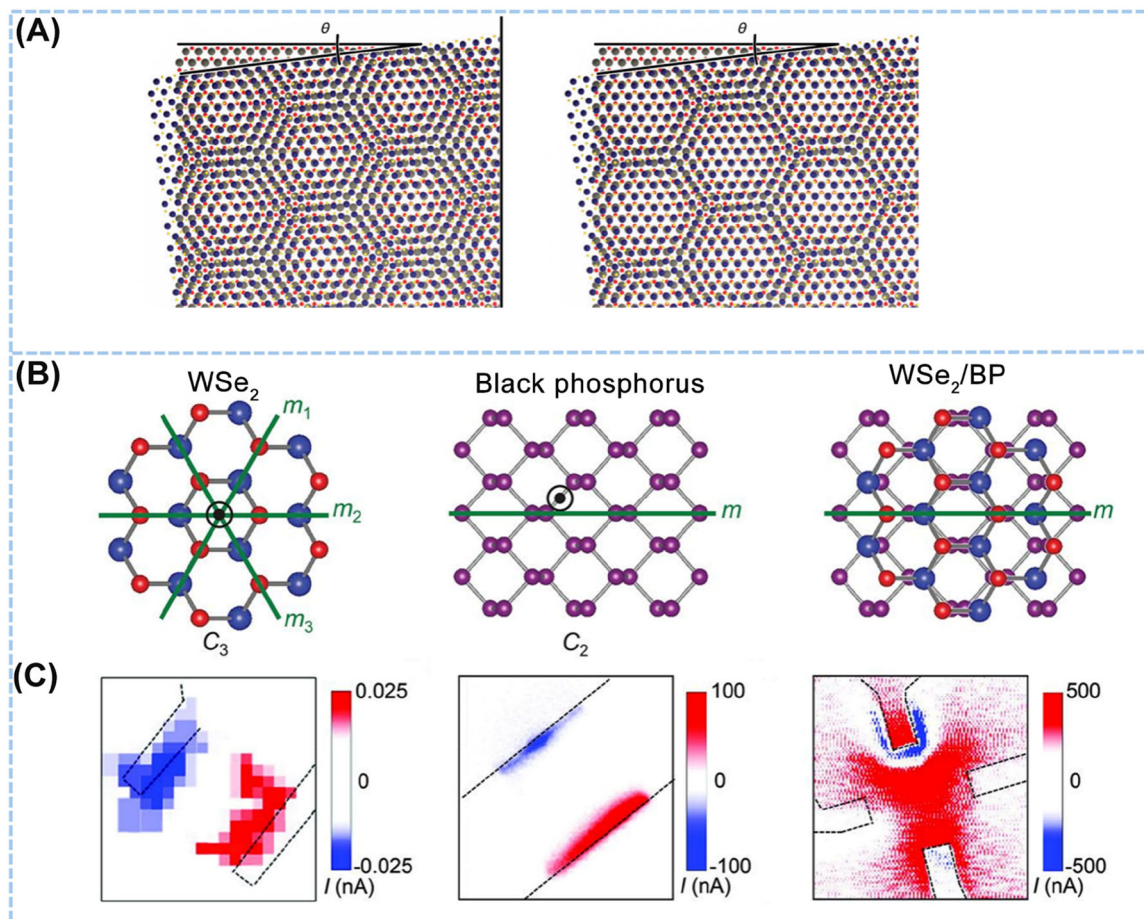


FIGURE 5 | Moiré superlattices and Twistronics in 2D materials. (A) Schematic depiction of Moiré patterns generated by twisting two-dimensional crystalline layers relative to each other by an angle θ . Reproduced with permission: Copyright 2022, Nature Publishing Group [197]. (B) and (C) Schematic presentations and photocurrent maps illustrating monolayer WSe_2 , BP, and the WSe_2/BP Moiré superlattice. The green lines indicate mirror planes, whereas the circled dots denote rotational symmetry axes. Reproduced with permission: Copyright 2021, Science (AAAS) [198].

magnetic MX_3 materials with high-quality interfaces provide an ideal platform due to their efficient spin and charge transport alongside spin-charge interconversion. Singh et al. [200] demonstrated significant thermoelectric performance in $\text{CrI}_3/\text{1T-MoS}_2$ Moiré superlattices, attributed to microscopic band structure modulation induced by local twist effects. This modulation causes electron localization and macroscale thermoelectric enhancement, yielding a Seebeck coefficient of $\sim 413 \mu\text{V K}^{-1}$ and a figure of merit of ~ 4.3 at 200 K for 30° twist angles. These findings present a novel strategy for designing high-efficiency thermoelectric devices through twist-induced band engineering in Moiré systems. In summary, Moiré materials have emerged as leading platforms for investigating the intricate interplay of electronic correlations, topology, and superconductivity in condensed matter physics, largely due to their remarkable tunability. Advancing this field will require a comprehensive strategy that encompasses scalable and precise fabrication methods, innovative characterization and control techniques, and robust theoretical models to interpret and predict novel quantum phenomena. In addition to enabling fundamental breakthroughs, these engineered quantum systems offer significant potential for next-generation electronic technologies, such as ultra-sensitive quantum sensors and devices for quantum information storage and processing.

7 | Bio-Inspired Design of 2D Materials for Flexible Energy Harvesting

Bio-inspired materials have revolutionized material design, drawing from the extraordinary structural and functional features of nature. Recent advancements in bio-inspired materials have drawn from nature's ingenuity, resulting in innovations like nacre-inspired damage-tolerant structures and lotus leaf-inspired self-cleaning surfaces. Additionally, technologies such as insect compound eye-inspired antifogging coatings, gecko-inspired adhesives, and desert beetle-inspired water harvesting systems exemplify the potential of biomimicry. Other breakthroughs include opal and butterfly wing-inspired photonic materials, insect-trapping plant-inspired ultrafast water transport systems, and fish-gill-inspired oil-water separators [201]. These developments highlight how nature is a blueprint for pioneering sustainable materials and advanced technologies. Therefore, a comprehensive understanding of the relationship between structure and properties is crucial for effectively designing and developing bio-inspired materials. 2D materials with unique properties at interfacial regions have facilitated the development of nature-inspired structures. The iridescent inner layers of seashells, such as those of abalones and pearl oysters, have inspired the development of photonic 2D nanostructures

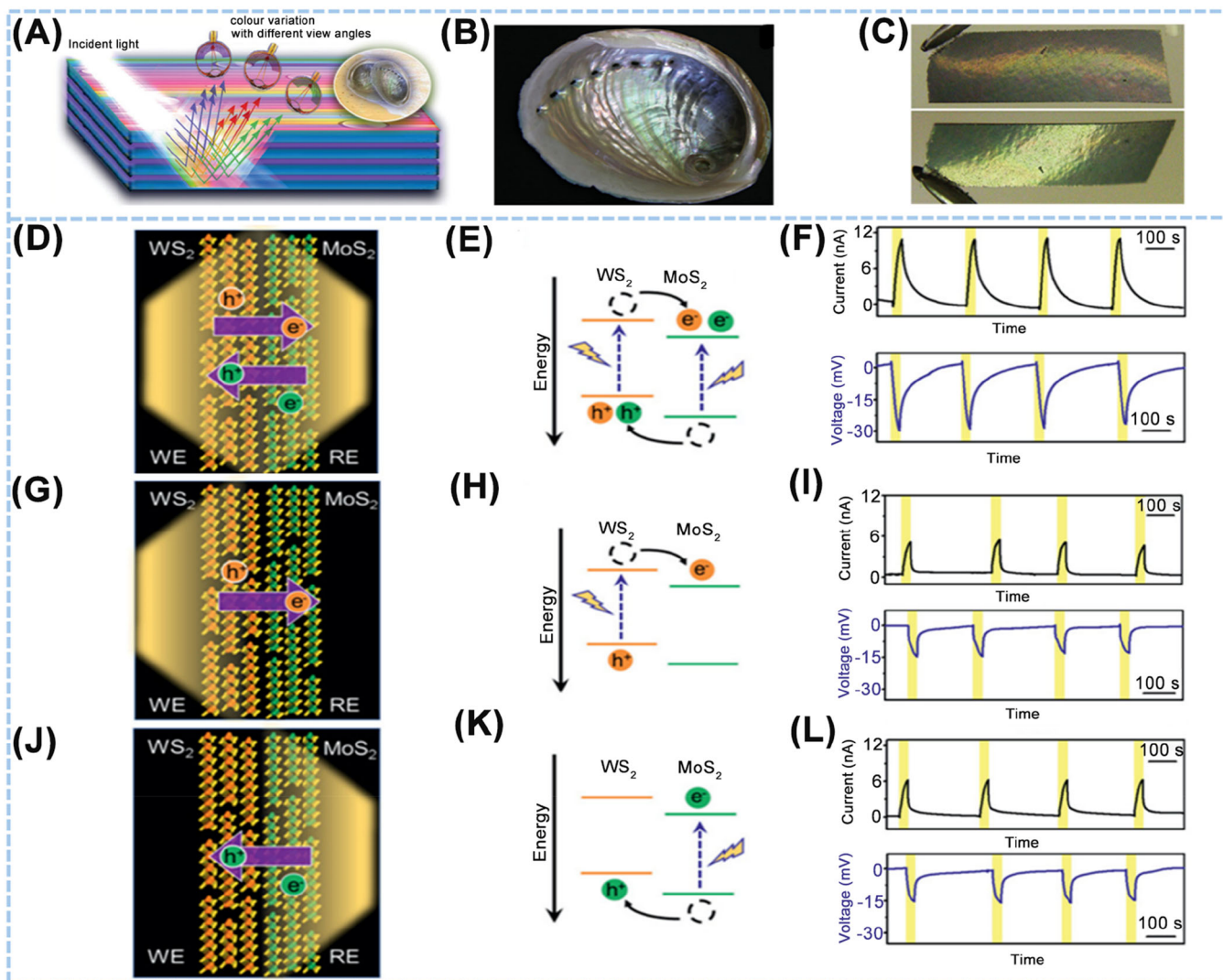


FIGURE 6 | Bio-inspired design of 2D materials for flexible energy harvesting. (A) Diagrammatic representation illustrating the angle varying optical variations in seashell structures. (B) Image showcasing the iridescent inner surface of an abalone shell. (C) Seashell-inspired heterostructure engineered from graphene and 2D TiO_2 . Reproduced with permission: Copyright 2019, Wiley-VCH [202]. (D, G, J) Illustration of the WS_2/MoS_2 heterostructure and schematic representation of electron and hole transfer dynamics across the energy band applied from both sides (D) or a single side (G and J). (E, H, K) In WS_2/MoS_2 heterostructures forming type II semiconductor junctions, dual-side illumination (E) causes electron transfer from WS_2 to MoS_2 and hole transfer from MoS_2 to WS_2 , facilitating charge separation, generating ionic photocurrents and photovoltages. Single-side illumination (H and K) localizes charge carrier generation, with electrons (H) or holes (K) selectively transferring. (F, I, L) The polarity of the ionic response remains constant, but the magnitudes are halved compared to dual-side illumination. Reproduced with permission: Copyright 2021, Wiley-VCH [203].

(Figure 6A,B). Mimicking the optical properties of these natural systems, researchers fabricated 2D heterostructures by layering GR and TiO_2 nanosheets [204]. These bio-inspired materials show superior interlayer charge transfer, rapid electron mobility, and exceptional light responsiveness. Such nacre-inspired nanomaterials hold significant promise for applications such as photoelectrodes, showcasing the seamless integration of natural inspiration with advanced 2D material science (Figure 6C). Similarly, bio-inspired 2D nanofluidic membranes have emerged as an advanced platform for efficient ion transport systems, mirroring the sophisticated process observed in biological organisms [205]. In recent years, osmotic energy harvesting via 2D nanofluidic channels has garnered significant global interest for its potential to address the limitations of traditional ion exchange membranes. For instance, 2D MXene

membranes, equipped with positively and negatively charged nanofluidic channels, achieve a high output power density of 4.6 W m^{-2} by mixing artificial seawater and freshwater, approaching the commercial benchmark of 5 W m^{-2} [202]. Moreover, heterogeneous 2D membranes, composed of multiple 2D materials, show promise for photoelectric conversion. A notable example is the WS_2/MoS_2 heterostructure, where sequential stacking of layers under 1 nm in spacing enables directional cation transport from WS_2 to MoS_2 under visible light irradiation (Figure 6D–L). This process is driven by charge separation within the heterostructures, resulting in a maximum ionic photo-response of 45 mV and $21 \mu\text{A cm}^{-2}$ under solar intensity (1 kW m^{-2}), showing superior performance compared to other 2D membranes [203]. This ability to generate photon-ion power at the angstrom scale offers exciting prospects for

biomimetic energy conversion, light-controlled ionic circuits, and artificial photosynthesis, marking a significant advancement in energy harvesting and nanofluidic technologies. Similarly, taking inspiration from biological membranes, a sub-nanoscale ion channel termed a double angstrom-scale confinement (DAC) ion-permselective membrane 2D $\text{Ti}_{0.87}\text{O}_2$ was developed for osmotic power generation. These membranes feature an interlayer spacing of ~ 2.2 Å and Ti atomic vacancies, ensuring high ion selectivity and permeability through strong interactions with Na^+ ions. As a result, DAC achieved an outstanding power density of 17.8 W m^{-2} using 0.5/0.01 M NaCl solutions and up to 114.2 W m^{-2} with a 500-fold salinity gradient, surpassing all previously reported macroscopic membranes [206]. In summary, future advancements in bio-inspired 2D materials depend on interdisciplinary collaboration across biology, material science, and engineering. This synergy promises innovative and efficient energy systems aligned with global sustainability objectives.

8 | Defect-Driven Dynamics in 2D Materials for Energy Harvesting

Crystallographic phenomena that disrupt the periodicity and symmetry of lattice structures are referred to as defects. 2D materials, defined by their significant aspect ratios along with tunable surface attributes, have shown exceptional promise in advancing energy harvesting technologies. To tailor the physical and chemical attributes of these materials for specific applications, extensive research has been devoted to defect engineering, which has become a pivotal strategy for enhancing reaction kinetics. Strategically introduced defects can adjust carrier concentrations and modify the energy band structure, thereby improving electrical conductivity in energy-related systems [207]. The inevitability of defects in crystalline materials is supported by the second law of thermodynamics. These defects, a ubiquitous phenomenon in crystallography, manifest in various forms and are categorized into four primary types based on the extent of atomic disorder.

- a. **Point defects:** These are localized disruptions in the lattice structure, typically spanning one to several lattice constants, and they include features such as vacancies and heteroatoms (Figure 7A).
- b. **Line defects:** These result from misaligned crystal planes, with examples including edge and screw dislocations (Figure 7B).
- c. **Planar defects:** These emerge at interfaces that divide a perfect three-dimensional crystal into smaller domains. Within these domains, atomic arrangements remain periodic, but significant misalignments occur at the interfaces, as seen in stacking faults, grain boundaries, and twin boundaries (Figure 7C).
- d. **Volume defects:** These involve the removal of multiple adjacent atoms, leading to the formation of vacancy clusters such as voids and cracks (Figure 7D).

Intrinsic defects are frequently observed in real crystals, stemming from conditions during crystal formation and the thermal

motion of atoms. The distinctive structural features and extensive surface areas of 2D materials create an ideal setting for the development and impact of defects. In particular, ultrathin 2D materials possess higher surface energy and more coordination-unsaturated sites than their bulk counterparts, facilitating the emergence of vacancies, lattice dislocations, and distortions [214]. The influence of various defects is profound, significantly altering the properties of 2D materials and driving robust research into defect engineering. This field focuses on refining the intrinsic characteristics of 2D materials, including electronic, optical, magnetic, and mechanical properties. A wide range of techniques has been developed to implement defect engineering in 2D materials. These include electron beam irradiation, plasma treatments, in situ oxidation, ozone exposure, ball milling, chemical reduction, laser treatments, and CVD. By applying these approaches, researchers aim to enhance the functional performance of 2D materials for diverse applications [215]. Each of these methods used for defect engineering in 2D materials has distinct benefits and drawbacks. For example, mechanical ball milling enables scalable production with adjustable defect concentration but it lacks precision and is energy-intensive. Plasma etching efficiently generates defects by manipulating plasma power and exposure time, though it risks structural damage and demands expertise. Chemical reduction facilitates defect formation through redox reactions, offering in situ engineering options but limited defect control. CVD is ideal for high-quality 2D films and heteroatom doping but requires careful parameter management. Furthermore, phase transition and LPE provide alternative strategies, though both face challenges like complexity and lattice disorder. Now, in the subsequent section, we will discuss the key impacts of defect engineering on 2D materials in the context of energy harvesting applications. Because defects in 2D materials play a pivotal role in optimizing electronic structures, enhancing active sites, and lowering energy barriers, harnessing electrical energy from mechanical energy sources that are otherwise wasted in nature, has gained significant attention as a sustainable solution for powering low-energy portable devices and enabling self-powered electronics. In this regard, extensive research is dedicated to developing high-performance PENGs based on piezoelectric nanostructures to create components for transparent, flexible, and self-powered electronic devices. A notable advancement in this field involves the fabrication of high-performance PNG using a CVD-grown MoS_2 nanosheet with a point defect, where S vacancy passivation was achieved through S treatment of the pristine MoS_2 surface [216]. This S vacancy site in the MoS_2 lattice has a strong tendency to form covalent bonds with S functional groups, enabling the capture of free electrons and subsequent chemisorption of S atoms at these vacancies. This treatment reduces the charge-carrier density on the monolayer MoS_2 surface, effectively mitigating the screening effect of piezoelectric polarization charges caused by free carriers. Consequently, the output peak current and voltage of the S-treated monolayer MoS_2 nanosheet PNG improved significantly, increasing by more than threefold (100 pA) and twofold (22 mV), respectively. Furthermore, the S treatment process enhanced the maximum power output by nearly tenfold (Figure 7E). Similarly, TENGs offer a promising approach for directly transforming broadband mechanical energy into electrical energy, providing a sustainable and environmentally friendly energy solution. Therefore, the design of

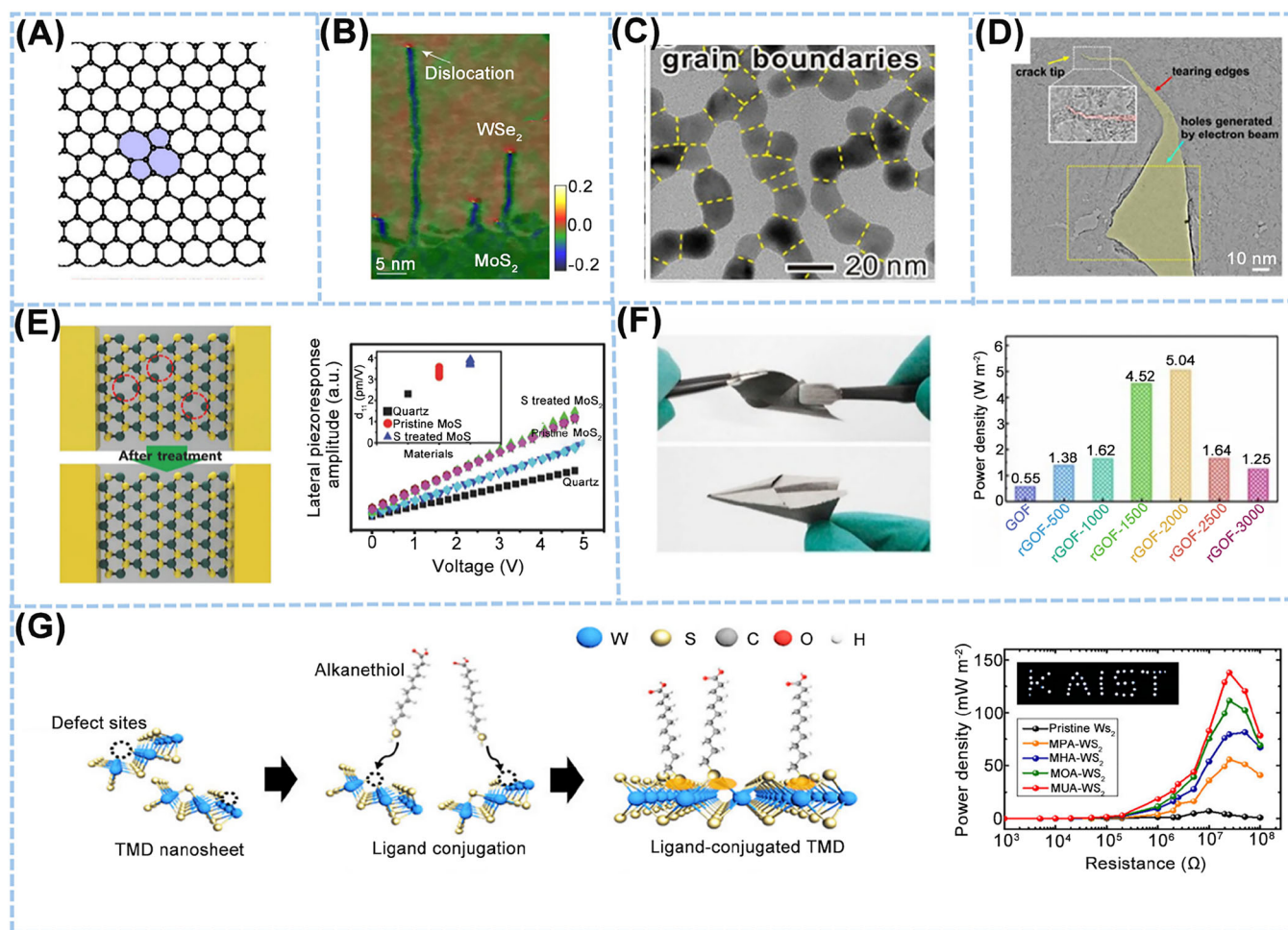


FIGURE 7 | Defect-driven dynamics in 2D materials for energy harvesting. (A) Depiction of topological vacancy defects, such as Stone–Wales defects, formed in graphene through alterations in the bonding configurations around vacancies during defect reconstruction. Reproduced with permission: Copyright 2022, American Chemical Society [208]. (B) Atomic-scale annular dark-field scanning transmission electron microscopy image revealing periodic dislocation at the MoS₂/WS₂ interface. Reproduced with permission: Copyright 2017, Springer Nature [209]. (C) Transmission electron microscope (TEM) image highlighting the existence of a high concentration of grain boundaries. Reproduced with permission: Copyright 2022, American Chemical Society [210]. (D) TEM image of MoS₂ revealing crack propagation initiated by focused e-beam irradiation, showcasing lattice-directed edges and detailed crack tip morphology in high resolution. Reproduced with permission: Copyright 2016, American Chemical Society [211]. (E) Illustration of the intrinsic sulfur vacancies in pristine MoS₂ and their passivation following sulfur treatment, alongside their piezoelectric measurements, highlighting that the pristine MoS₂ showed superior piezoelectric performance compared to its counterparts. Reproduced with permission: Copyright 2018, Wiley-VCH [212]. (F) Optical image demonstrating the folding adaptability of rGOF into various shapes, alongside a graph depicting the maximum power density of rGOF-2000-TENG reaching 5.04 W m⁻², approximately nine times and four times higher than those of GOF-TENG and rGOF-3000-TENG, respectively. Reproduced with permission: Copyright 2022, Elsevier [213]. (G) Schematic representation of defect engineering of WS₂ through thiolation with different alkane chains, alongside a graph depicting the output power density of pristine WS₂ and functionalized TENG device, successfully illuminating 45 LEDs. Reproduced with permission: Copyright 2021, American Chemical Society [213].

triboelectric materials with customized WFs is essential for maximizing the output performance of TENGs. Despite their potential, controlling the WF of triboelectric materials remains a significant challenge due to constraints in the doping rates of particles or ions. To address this issue, Zhou et al. [212] introduced a defect strategy to modulate the WF of GR-based triboelectric materials. This approach involved generating varying levels of oxygen and carbon defects by annealing rGO at temperatures ranging from 500°C to 3000°C. This study demonstrated that rGO annealed at 2000°C, characterized by fewer oxygen defects and potentially more carbon defects, showed the lowest functionality. This optimized WF effectively promoted electron loss in the positive triboelectric materials, significantly

enhancing the TENG's output performance. Specifically, the output voltage and current reached 190 V and 14 μA, respectively, with a power density of 5.04 W m⁻² (Figure 7F). The defect sites in TMDCs nanosheets, like S vacancies present in the basal planes or edges, significantly influence their electrical and chemical properties, impacting carrier mobility and charge density. Therefore, precise control of the defect sites with thiol chemistry is also crucial for fabricating electrical and energy devices with tailored functionalities. In this context, researchers have introduced an innovative method for defect engineering in TMDCs, aimed at enhancing the performance of TMDCs-based TENGs. They explored the relationship between ligand-conjugated TMDCs and their triboelectric charging behavior to

provide fundamental insights into energy harvesting devices [213]. By chemically exfoliating WS₂ nanosheets using thiolation (thiol-containing ligands), they effectively modulated S vacancy sites, leading to significant performance improvements in the TENG device. The resulting device achieves an output voltage of 12.2 V and a power density of 138 mW m⁻². Additionally, this device, besides high chemical stability, also maintains consistent performance over 10,000 contact-separation cycles (Figure 7G).

9 | Utilizing 2D Materials for Energy Harvesting Applications

As long as the ambient energy resources are persistently available, an energy harvester can continuously generate electrical energy. Unlike fossil fuels or nuclear power, these devices are environmentally friendly regarding pollution production. Of these benefits, energy harvesters have significant capability to serve as an energy source for the IoT era. In this section, we will discuss the representative kinds of energy harvesters, namely, PV, thermoelectric, piezoelectric, triboelectric, and RF energy-based harvesters. As we know, all forms of motion in our environment, including natural sources like vibrations, wind, river currents, and even bodily movements such as walking, speaking, and typing, represent valuable sources of kinetic energy that can be harnessed and converted into electricity by using these energy harvesting phenomena. Table 3 summarizes different energy harvesting methods, their corresponding operational mechanism, and specific properties based on 2D materials.

9.1 | PV Energy Harvesting

9.1.1 | Mechanism and Performance

Among all the energy sources, solar energy is deemed the utmost abundant exploitable renewable energy resource, owing to its unparalleled abilities, like safe, infinite, and eco-friendly. Solar cells (or PV devices) have been significantly explored and utilized to produce electricity from sunlight via the PV effect [226]. The widespread availability of light makes PV technology a promising solution for energy harvesting, enabling its versatile deployment across various environments. Usually, the PV phenomenon ensues within the p-n junction/Schottky junction. More typically, in a PV device, the incoming sunlight creates electron-hole pairs (photogenerated carriers) with the help of absorbed photons in the active material; subsequently, these are then separated and transported to the outer electrodes to provide an electrical output (Figure 8A). Importantly, the development of affordable semiconducting materials that allow for simple, scalable, and cost-effective technical processes is crucial for realizing economically viable solar energy conversion on a worldwide scale. 2D materials have emerged as a promising class for enhancing the performance of PV energy harvesting systems due to their exceptional electronic, optical, and mechanical properties. These materials have high carrier mobility, tunable band gaps, and strong light-matter interactions, making them suitable for next-generation solar cells. The

TABLE 3 | Summary of different energy harvesting methods, their corresponding operational mechanism, and specific properties based on 2D materials.

Energy harvesting methods	Mechanism	Specific properties based on 2D material	2D materials	References
Photovoltaic harvester	Sunlight creates electron-hole pairs, which are then separated and transported to the outer electrodes to provide an electric output.	Increase light absorption, improve charge transport, flexibility, and tunable band gap.	Graphene, MXene, BP, TMDCs, 2D perovskite	[217, 218]
Thermoelectric harvester	The temperature gradient within the material produces electricity.	Boost thermal and electrical conductivity along with flexibility.	Graphene, BP, TMDCs, MXene	[219, 220]
Piezoelectric harvester	Mechanical stress/strain produces an electrostatic potential in the polarization axis of the piezoelectric substances.	High piezoelectric coefficient, superior charge carrier mobility, improved mechanical flexibility.	h-BN, BP, TMDCs, 2D perovskite	[12, 221]
Triboelectric harvester	A generated electrical output was acquired by combining the triboelectric charging and the electrostatic induction technique.	Improved triboelectric properties, flexibility, durability, and increased surface charge density.	Graphene, MXene, BP, TMDCs, 2D, MOF	[222, 223]
Radiofrequency harvester	Capture radiofrequency signals from the radio environment and convert them into electricity to power low-voltage electronic devices.	Improve antenna performance, lightweight, and improved rectification.	MXene, TMDCs	[224, 225]

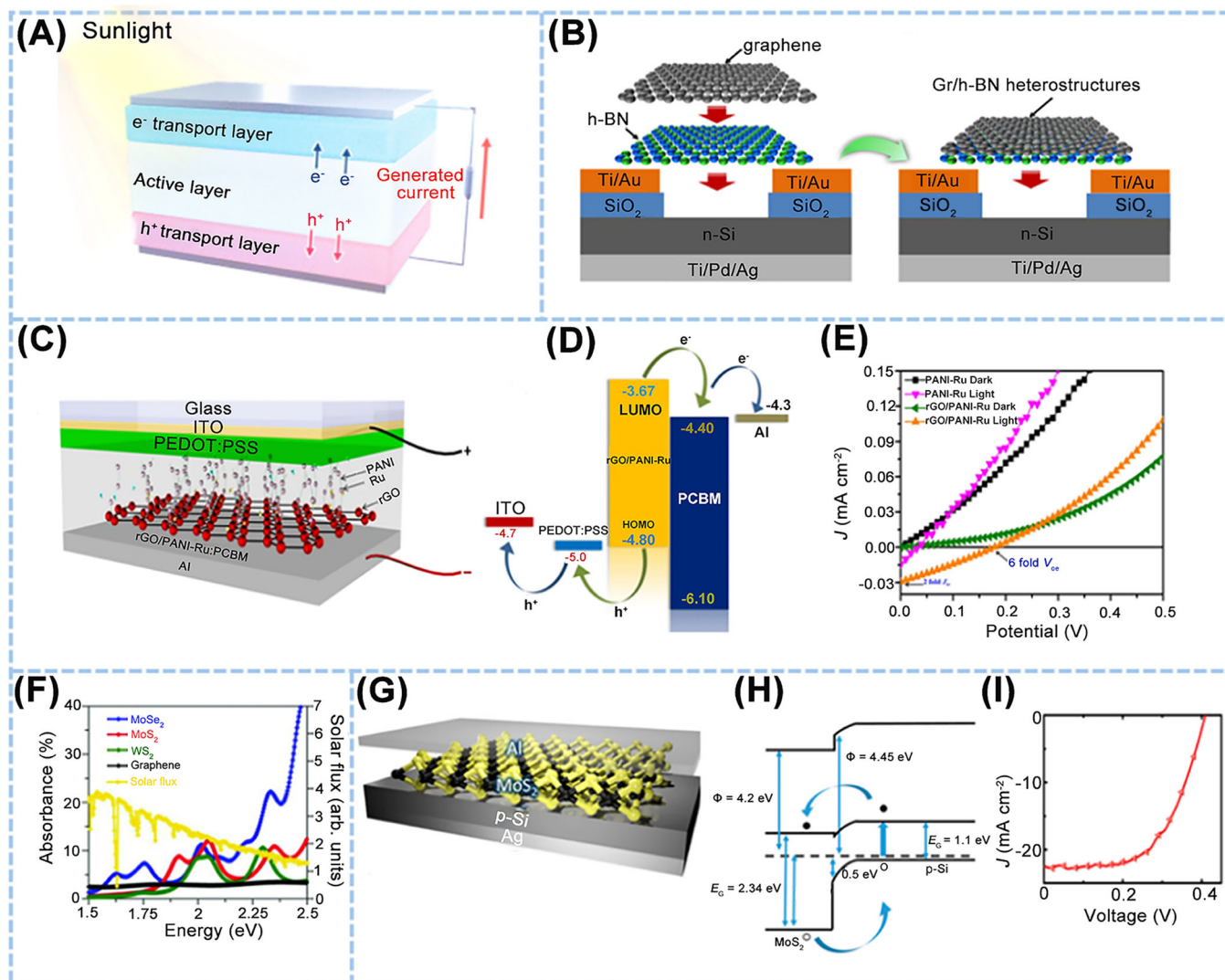


FIGURE 8 | 2D materials for photovoltaic energy harvesting. (A) Schematic representation of the working mechanism of photovoltaic energy harvesting. (B) Illustration of one-and two-step fabrication of solar cells. Reproduced with permission: Copyright 2016, Elsevier [227]. (C) Schematic illustration depicting the fabrication of a PSC device incorporating rGO/PANI-Ru as the electron donor. (D) Energy-level diagram illustrating the configuration of rGO/PANI-Ru within PSC. (E) The current density-voltage (J - V) representation is a characteristic of the PSC device. Reproduced under the terms of the CC-BY-NC-ND license: Copyright 2017, The authors, published by Springer Nature [228]. (F) Absorbance of monolayers of MoSe_2 , MoS_2 , WS_2 , and graphene, overlapping with the incident AM1.5 G solar flux. Reproduced with permission: Copyright 2013, American Chemical Society [229]. (G) The device structure illustrating a heterojunction solar cell composed of MoS_2 /p-Si. MoS_2 has solar radiation ability, whereas p-Si supports and transports charge carriers; as a result, the MoS_2 /p-Si heterojunction separates and collects electron-hole pairs, producing electrical current. (H) Band diagram of the heterojunction between MoS_2 /p-Si, demonstrating their corresponding energy levels. (I) The J - V characteristics which show the electrical behavior of the MoS_2 /p-Si heterojunction solar cell under variable voltage and current conditions, offering insights into its performance and efficiency. Reproduced with permission: Copyright 2014, American Chemical Society [230].

mechanism of 2D materials in PV energy harvesting is primarily driven by their ability to manipulate light absorption and electron transport at the atomic level. The specific mechanism of 2D materials for PV energy harvesting is as follows:

- i. **Photon absorption and exciton generation:** In 2D materials, incident photons are effectively absorbed owing to the direct band gap nature, and then this absorption creates electron-hole pairs (excitons), crucial for producing electrical current.
- ii. **Charge separation and transport:** The atomically thin nature of 2D materials allows for a high surface-to-

volume ratio, promoting rapid charge separation while suppressing recombination losses. In addition, heterostructures with stacked layers of different 2D materials (e.g., MoS_2 /GR) improve the mobility and stability of charge carriers.

- iii. **Interfacial engineering:** The layered nature of 2D materials provides a good platform for the manipulation of interfacial properties, which is a very promising way to establish an efficient charge transport path. By tailoring these interfaces, it becomes possible to minimize energy losses and enhance the overall efficiency of the PV cell.

The specific mechanical design needs to be engineered to achieve the required flexibility, as hard Si and GaAs wafers lack transparency and flexibility. Furthermore, despite the advanced development of silicon-based PV technology, its restricted solar spectrum absorption leads to a high material cost per unit of output power. Another issue that the current solar cell business faces is its reliance on specific materials like Pt/In (ITO electrode), which are scarce, costly, and environmentally hazardous. Conversely, the emerging class of 2D materials has traits such as semi-transparency, ultra-thinness, copiousness, durability, high absorbance per unit thickness, and sustainability. These characteristics render them viable for advanced PV applications of the future, which can be seamlessly integrated onto a variety of surfaces, including the human body. Additionally, a diverse array of 2D materials with different band gaps offers numerous opportunities for the development of high-efficiency solar cells through band gap engineering. Thus far, a great deal of research has focused on examining the potential of a multitude of monolayer/few-layer semiconducting crystals, which are presently exemplified by GR, TMDCs, and BP for use in next-generation ultrathin PV systems [231, 232]. Remarkably, 2D stacked materials with a band gap in the visible region can be generally utilized as active materials in PVs; hence, they have attracted significant interest as potential candidates for next-generation solar cells, thanks to their distinctive electrical and optical properties. GR, a semimetal with remarkable optical transparency ($\sim 97.7\%$ transmittance in the visible region), high electrical conductivity ($\sim 10^4 \Omega^{-1} \text{cm}^{-1}$), carrier mobility ($2 \times 10^5 \text{cm}^2 \text{V}^{-1} \text{s}^{-1}$), ON/OFF ratio (5.5–44), thermal conductance ($5 \times 10^3 \text{W m}^{-1} \text{K}^{-1}$), fracture strain (27%–38%), and flexibility, has emerged as a promising material for its potential application in conductive/counter electrodes in flexible solar cells and this is particularly significant, as conventional electrodes ITO and FTO are expensive and also brittle, making them unsuitable for such purposes [233]. Furthermore, although ITO transparency is restricted to only visible light, GR offers high transmittance along with low transparency degradation, making it a flexible electrode option for solar cells [234]. GR is also positioned in energy-efficient applications, offering high-performance, lightweight, efficient, cost-effective, and flexible energy devices, to address global energy resource constraints [235]. Also, it has been incorporated into numerous layers in the solar cells, enhancing their efficiency and performance [236]. Remarkably, interface engineering is also a promising strategy to enhance the efficiency and performance of 2D materials, as shown in Figure 8B. An interface modification is established with the addition of h-BN, resulting in reductions in the solar cell resistance due to h-BN's higher electrical conductivity [237]. Following the pioneering work on GR, a new class of extremely thin, layered, semiconducting 2D materials has emerged. These 2D materials have distinct crystal structures and an intrinsic band gap, making them promising contenders for a wide array of applications in optoelectronics and next-generation electronics [238]. Vinoth et al. [228] investigated the integration of rGO into a polymer solar cell (PSC) alongside a pyridyl benzimidazole-based assembly incorporating a Ru complex (PANI-Ru), resulting in notable device performance stemming from the exceptional electrical capabilities of rGO. To evaluate the PV performance, a bulk PSC device was fabricated (Figure 8C), where PANI-Ru and rGO/PANI-Ru act as electron donors (p-type) and fullerene act as an acceptor (n-type),

whereas Figure 8D represents its highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) levels acquired from a cyclic voltammogram, showing -4.8 and -3.67eV for the HOMO/LUMO level of rGO/PANI-Ru, suggesting the easy transfer of electrons from the donor to PCBM. Additionally (Figure 8E), the hybrid nanocomposite current density–voltage (J - V) shows that under dark conditions, both devices (PANI-Ru and rGO/PANI-Ru) do not have any short-circuit current density.

Notably, beyond GR, other 2D materials like TMDCs, MXene, h-BN, and BP have also been effectively utilized in solar cells, especially in organic solar cells, serving as an interfacial layer for the electron/hole transport layer (ETL/HTL) with additional protective properties. Theoretical investigation revealed that monolayer TMDCs (with thickness $\leq 1 \text{nm}$) can produce electric currents of up to 4.5mA cm^{-2} and could effectively capture as much solar radiation as either silicon (50 nm thickness) or 15 nm thickness GaAs-based solar cells. Notably, MoS_2 displays outstanding properties like carrier mobility ($10\text{--}200 \text{cm}^2 \text{V}^{-1} \text{s}^{-1}$), ON/OFF ratio ($10^6\text{--}10^8$), thermal conductance ($34\text{--}52 \text{W m}^{-1} \text{K}^{-1}$), and fracture strain (25%–33%), making it a suitable candidate for PV applications. Figure 8F displays the calculated absorbance of a monolayer of WS_2 , MoS_2 , MoSe_2 , and GR [229]. For instance, a monolayer MoS_2 (direct band gap $\sim 1.9 \text{eV}$) can form a p–Si Schottky junction (Figure 8G–I), showing a PCE of 5.23% and an open-circuit voltage (V_{oc}) of 0.41 V [230]. Phosphorene and its derivatives also have good electrical characteristics, high charge carrier mobility ($1000 \text{cm}^2 \text{V}^{-1} \text{s}^{-1}$), ON/OFF ratio ($10^3\text{--}10^5$), thermal conductance ($10\text{--}36 \text{W m}^{-1} \text{K}^{-1}$), fracture strain (24%–32%), and superb optical transparency, making them a viable option for PV energy harvesting. Therefore, they serve both as a light absorber and as a charge transfer layer in solar cells, resulting in approved light absorption ability. The phosphorene-based solar cells have shown a PCE of $\sim 8.25\%$ and notably, theoretically, PCE of up to 20% has been calculated, making it also a viable candidate for solar cells [239]. $\text{Ti}_3\text{C}_2\text{T}_x$ is a promising member of the MXenes family that has exceptional potential as an electrode designed for solar cells due to its distinct physicochemical characteristics, including carrier mobility ($\sim 0.7 \text{cm}^2 \text{V}^{-1} \text{s}^{-1}$), ON/OFF ratio ($10^3\text{--}10^6$), thermal conductance ($10\text{--}472 \text{W m}^{-1} \text{K}^{-1}$), and sufficient flexibility. To boost the PV performance of MAPbI_3 -based perovskite cells, Ma et al. used $\text{Ti}_3\text{C}_2\text{T}_x$ as an additive in the absorbent layer, resulting in grain size enhancement, which in turn not only increases the perovskite cell light absorbing ability but also boosts the carrier charge transfer due to the inherent properties of $\text{Ti}_3\text{C}_2\text{T}_x$, that is, high electrical conductivity and mobility. Interestingly, the PCE of the device based on $\text{Ti}_3\text{C}_2\text{T}_x$ as an additive increased from 15.5% to 17.4%, paving the way for the investigation of other MXenes in solar cell technology [13]. Remarkably, MXene's flexible WFs allow them to be directly used in solar cells as ETL/HTL materials. In this connection, Yang et al. [240] used $\text{Ti}_3\text{C}_2\text{T}$ as ETL in $\text{CH}_3\text{NH}_3\text{PbI}_3$ -based perovskite solar cells. They reported that UV–ozone treatment of the $\text{Ti}_3\text{C}_2\text{T}$ remarkably enhances its PCE up to 17.17%, demonstrating the significant potential of MXenes for PSC use as well as their potential for use in solar technology applications. The subject of energy harvesting could advance with continuous research and development of 2D materials in solar cell technologies, paving the way toward the development of flexible, lightweight, and efficient energy-generating systems for a range of wearable and portable applications. In PV energy harvesting, 2D materials have revolutionized the field by initiating new

TABLE 4 | Summary of recent advancements in 2D material-based photovoltaic energy harvesters.

Materials	Substrate	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)	References
Graphene	FTO	0.96	23.00	68.43	15.18	[241]
MXene	$\text{Al}_2\text{O}_3/\text{Si}$	0.45	29.50	59.00	7.89	[242]
MoSe_2	SiO_2/Si	0.24	4.56	25	1.9	[243]
Graphene	Si wafer	0.55	29.48	64.41	10.57	[244]
WS_2	SiO_2/Si	0.46	—	39.00	5.0	[245]
MoS_2	Si wafer	0.4	23.99	54	5.06	[246]
BP	Glass/ITO	0.71	23.44	0.70	11.6	[247]
MXene	FTO	0.21	0.15	25.9	0.34	[248]
MOF	Glass/ITO	0.69	10.8	69.0	5.2	[247]
Graphene	FTO	1.12	19.85	56.7	12.6	[249]
Graphene QDs	FTO	0.935	22.88	60.17	12.87	[250]
WS_2	FTO	1.02	34.35	87.27	30	[251]
MXene	ITO	0.89	26.04	74.67	17.31	[252]
MXene	Glass/ITO	1.28	20.74	66.51	17.69	[253]
MoS_2	Ti mesh	0.81	27.26	0.735	16.39	[254]
ZrSe_2	Glass/ITO	0.93	18.14	71.35	11.96	[255]
MOF	ITO	1.17	24.71	74.95	21.68	[256]
BP-GO	Glass	1.13	24.90	78.63	22.11	[174]
MoS_2	FTO	0.76	9.13	0.74	5.14	[257]
MXene	ITO	0.56	29.86	71.12	11.92	[195]
2D LDH	ITO	0.70	12.40	0.62	5.4	[258]
MoS_2	FTO	0.72	9.21	0.50	4.18	[259]
2D Perovskite	ITO	1.08	23.00	76.52	19.03	[260]
2D Perovskite	ITO	1.10	21.37	79.26	18.56	[261]
2D Perovskite	ITO	1.17	22.18	81.12	21.07	[262]

Abbreviations: BP, black phosphorous; FF, fill factor; J_{sc} , short-circuit current; LDH, layered double hydroxide; MOF, metal-organic framework; PCE, power conversion efficiency; QDs, quantum dots; V_{oc} , open-circuit voltage.

mechanisms for improving light absorption, charge carrier generation, and overall device stability. The high efficiency realized by integrating materials like TMDCs, GR, and 2D perovskites highlights their potential to surpass conventional silicon-based PV devices. The recent advancements in 2D material-based PV energy harvesters are summarized in Table 4.

9.1.2 | Strengths and Challenges

2D materials, like GR and MXene, due to their exceptional strengths, have substantial potential for PV energy harvesting. Their good electrical conductivity aids efficient charge transport, whereas their high surface-to-volume ratio increases light absorption and furnishes sufficient sites for charge separation. The tunable electrical and optical properties offered by these materials can be modified to enhance specific applications. Moreover, their mechanical flexibility and chemical inertness make them ideal candidates for durable solar cell applications. However, challenges persist, like low PCE in several materials, suboptimal charge transfer at interfaces, environmental stability issues, and industrial-scale manufacturing. To address these

limitations, various strategies such as surface and interface engineering, heterostructures, heteroatom doping, and innovative encapsulation are being adopted. Ongoing advancements in scalable production techniques are paving the way for 2D materials to become key components in creating more stable, efficient, and flexible solar energy harvesting solutions.

9.2 | Thermoelectric Energy Harvesting

9.2.1 | Mechanism and Performance

As we know, there is a lot of thermal energy in our surroundings. Hence, thermoelectric devices can harvest and convert heat from the environment, including geothermal energy, waste heat from power plants, and heat from automobile exhaust, into electrical energy. Moreover, for years, researchers have been fascinated by the prospect of harnessing the heat generated by human bodies to generate power. Interestingly, an average adult male at rest emits approximately 100–128 W of heat energy; therefore, harnessing body heat energy presents a viable step toward eco-friendly and long-lasting wearable power

solutions. Currently, the preferred technique for harvesting body heat energy is through a thermoelectric nanogenerator, which creates electricity from the temperature difference between the human body and its surroundings. Thus, thermoelectric materials play a crucial role in developing sustainable and energy-efficient technology by transforming thermal energy from temperature gradients into electrical power or vice versa [230, 263, 264]. Figure 9A shows the schematic operational mechanism of thermoelectric energy harvesting. Typically, in

the design and enhancement of thermoelectric materials and systems, two essential concepts, that is, Peltier and Seebeck, play a crucial role. The phenomenon where heat is either released or absorbed at a junction when an electric current passes through it is referred to as the Peltier effect, whereas the voltage differential generated by a temperature difference between two materials is referred to as the Seebeck effect. In essence, the dimensionless figure of merit (ZT) factors in electrical conductivity (σ), the Seebeck coefficient (S), thermal

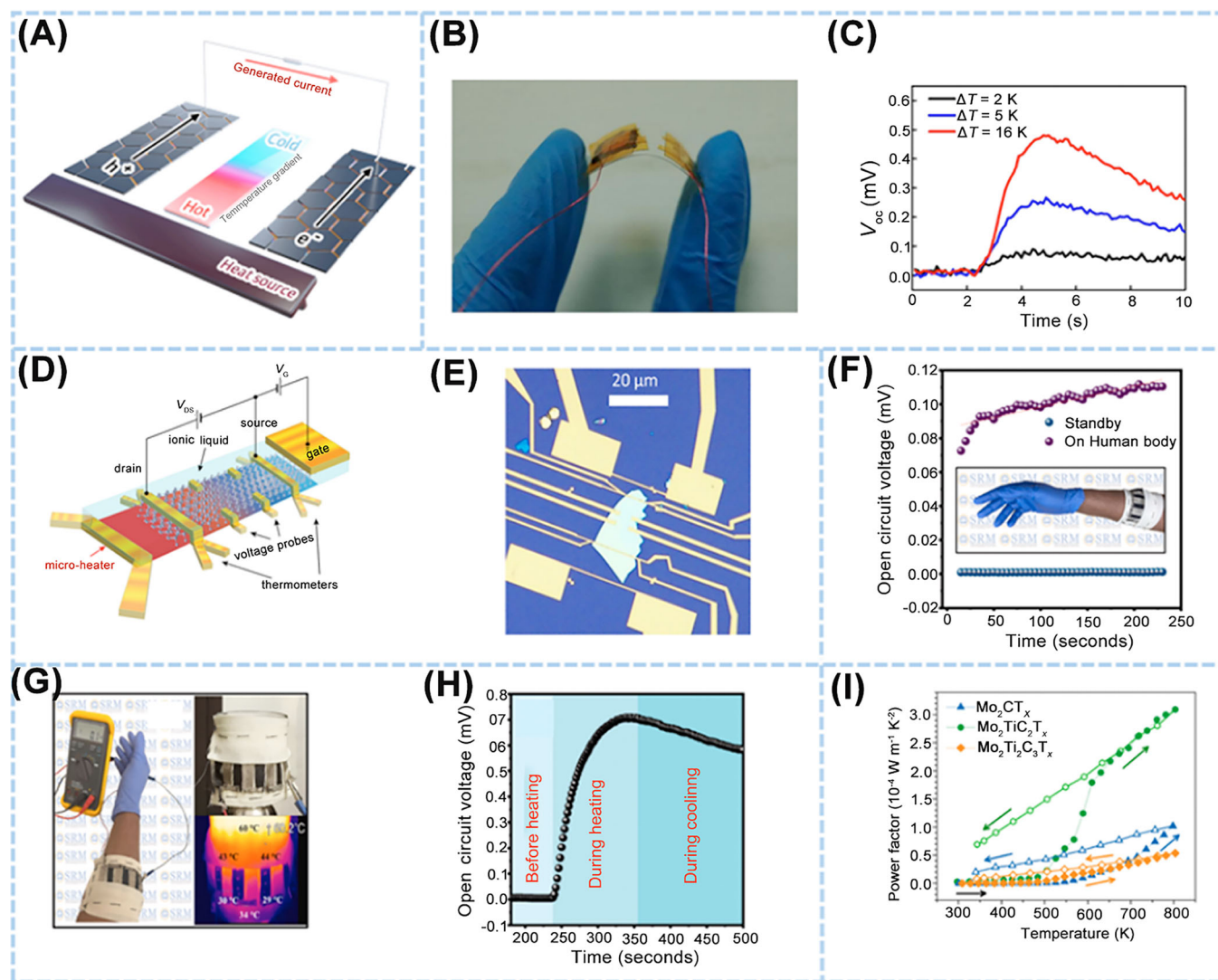


FIGURE 9 | 2D materials for thermoelectric energy harvesting. (A) A scheme illustrating the operational mechanism of thermoelectric energy harvesting, elucidating how heat energy is converted into electrical energy through this phenomenon. (B) Image showcasing the flexible nanocomposite thermoelectric nanogenerator by incorporating the $\text{MoS}_2/\text{graphene}$ nanocomposite, displaying a noticeable power density and output voltage. (C) The graphical presentation of output voltage generated by the nanocomposite thermoelectric nanogenerators across various temperature differentials, which provides insight into device performance under variable thermal environments, demonstrating their efficiency in harvesting energy from temperature gradients. Reproduced with permission: Copyright 2017, IOP Publishing Ltd [265]. (D) Schematic representation of a BP-based device for thermoelectric measurements. (E) Optical microscope image of BP flakes with a thickness of 40 nm. Reproduced with permission: Copyright 2016, American Chemical Society [266]. (F) Graphical depiction of V_{oc} determination by exposing the human arm to the fabricated WTEG. (G) Representation depicting the output measurement from a human arm on the left, and from the glass on the right, highlighting the relative investigation of energy harvesting proficiencies from different surfaces. (H) Graph of V_{oc} on the glass surface showing that before the warm water introduction, the device showed no V_{oc} generation, but with the addition of warm water, V_{oc} steadily increased to a peak of 0.7 mV, and subsequently, upon water cooling, V_{oc} decreased to 0.6 mV. Reproduced with permission: Copyright 2024, Elsevier [267]. (I) The graphical presentation of thermoelectric properties of Mo-based MXenes (including Mo_2CT_x , $\text{Mo}_2\text{TiC}_2\text{T}_x$, and $\text{Mo}_2\text{Ti}_2\text{C}_3\text{T}_x$) shows that the $\text{Mo}_2\text{TiC}_2\text{T}_x$ sample showed outstanding performance, characterized by a notably high thermoelectric power factor. Reproduced with permission: Copyright 2017, American Chemical Society [268].

conductivity (k), and absolute temperature (T), which together describe how well a thermoelectric material converts heat into electrical energy [269, 270]. It is noteworthy that, despite having lower ZT values than traditional bulk thermoelectric materials, 2D materials have garnered significant interest as effective thermoelectric materials due to their outstanding mechanical and electrical properties, particularly for flexible and wearable thermoelectric devices. Moreover, unlike conventional bulk thermoelectric materials, 2D materials have a quantized electron density of states due to quantum confinement effects, which significantly contributes to the improvement of the Seebeck coefficient. This characteristic enables better thermoelectric performance by enhancing energy conversion efficiency. 2D-layered materials can be exfoliated into ultra-thin sheets, with dimensions approaching the mean free path of phonons, allowing for independent tuning of either electrical or thermal conductivity. Additionally, the band gap of these materials can be precisely controlled through “thickness engineering.” These features collectively enhance their potential for achieving a high thermoelectric ZT. The known Seebeck coefficient value and thermal conductivity value for GR range from approximately 40 to 180 $\mu\text{V K}^{-1}$ at temperatures of 250 to 300 K and from 2000 to 5000 $\text{W m}^{-1} \text{K}^{-1}$ at 300 K, respectively. GR’s superior thermal conductivity and minimal Seebeck coefficient preclude its direct use as a thermoelectric material, although studies have indicated that adding GR or rGO to other thermoelectric materials can increase their power factor ($PF = \sigma S^2$) [271]. Xie and colleagues [265, 272, 273] documented a thermoelectric flexible nanogenerator incorporating the MoS_2/GR nanocomposite (Figure 9B,C), displaying a power density of 8.8 nW cm^{-2} and an output voltage of -0.73 mV under a temperature differential of -35 K , surpassing those without GR. Phosphorene, another 2D material, has also garnered significant attention in thermoelectric energy harvesting owing to its notable characteristics, including a considerable Seebeck coefficient ($335 \mu\text{V K}^{-1}$), robust carrier mobility, and adjustable band gap. Moreover, its inherent in-plane anisotropy contributes to the formation of a distinct electrical conducting pathway perpendicular to the direction of heat conduction. This phenomenon results in a substantial ratio of electrical conductivity to thermal conductivity (σ/k), which in turn leads to a high ZT value [274, 275]. In a recent study by Saito et al. [266], they introduced a thermoelectric harvester utilizing phosphorene, featuring a double-layer electric transistor with gate tuning (Figure 9D,E); by adjusting the gate voltage, they managed to achieve an increase in the Seebeck coefficient to $510 \mu\text{V K}^{-1}$, contrasting with the $340 \mu\text{V K}^{-1}$ observed in the bulk analog. In another study, Suresh Prasanna et al. developed a flexible thermoelectric device by depositing MoS_2 layers onto conductive carbon cloth. To assess its practical performance, they measured the open-circuit voltage ($V_{\text{oc}} = 0.11 \text{ mV}$) in real time during contact with a human arm (Figure 9F) and also on a curved glass surface, as illustrated in Figure 9G. Moreover, before the introduction of warm water, the device showed no V_{oc} generation, but with the addition of warm water, V_{oc} steadily increased to a peak of 0.7 mV , and subsequently, upon water cooling, the V_{oc} decreased to 0.6 mV (Figure 9H) [267]. Kim et al. conducted a temperature-based study that achieved remarkable outcomes while investigating the thermoelectric properties of Mo-based MXenes (including Mo_2CT_x , $\text{Mo}_2\text{TiC}_2\text{T}_x$, and $\text{Mo}_2\text{Ti}_2\text{C}_3\text{T}_x$). Among them, the $\text{Mo}_2\text{TiC}_2\text{T}_x$ sample showed

outstanding performance, characterized by a notable Seebeck coefficient ($-47.3 \mu\text{V K}^{-1}$) and a high thermoelectric power factor ($PF = S^2\sigma$) ($3.09 \times 10^{-4} \text{ W m}^{-1} \text{K}^{-2}$). These findings surpassed those of their MAX phases, but even rivaled those of traditional thermoelectric materials (Figure 9I) [268]. The recent advancements in 2D material-based thermoelectric energy harvesters are summarized in Table 5.

9.2.2 | Strengths and Challenges

Unlike conventional bulk thermoelectric materials, 2D materials display a discrete electron density of states due to quantum size effects, which helps to improve the Seebeck coefficient. As in bulk materials, charge carriers are typically incorporated during the material growth phase, whereas in 2D materials, carrier concentration can be easily adjusted through ion intercalation or electric gating. Additionally, these layered 2D materials can be exfoliated down to just a few layers, with a thickness comparable to the mean free path of phonons, allowing for precise control of both electrical and thermal conductivity. The band gap of these materials can be effectively tuned through thickness manipulation techniques. These combined attributes make 2D materials highly advantageous for achieving superior thermoelectric performance, reflected in a higher ZT value. However, achieving a high ZT value, which improves heat-to-electricity conversion efficiency, remains a major challenge for 2D thermoelectric materials. As their performance is tied to field-effect-transistor devices, enhancing electrical ohmic contact is essential for accurate measurement of their intrinsic properties. Additionally, integrating these materials into practical thermoelectric devices requires overcoming challenges related to their environmental sensitivity and microscopic scale. Future research should focus on designing new materials and contacts, while also clarifying phonon and electron transport mechanisms to optimize ZT performance. Additionally, it is essential to use precise nanofabrication techniques during both the synthesis of 2D materials and the fabrication of thermoelectric devices based on them. In terms of evaluating the thermoelectric properties of these materials, there is a pressing need for the development of standardized, commercially available characterization methods, which are currently insufficient and should be the main focus in future research.

9.3 | Piezoelectric Energy Harvesting

9.3.1 | Mechanism and Performance

A typical method of converting mechanical energy into electricity is piezoelectric energy harvesting, and because of its ease of production and variable configuration, piezoelectric energy is a viable option for a power source in small electronics. Generally, materials with polarization domains or non-centrosymmetric structures show piezoelectricity, which permits the reversible conversion of electrical signals into mechanical signals and vice versa. Therefore, the utilization of PENGs for mechanical energy harvesting has garnered significant interest due to their ability to efficiently convert mechanical energy into electricity, making them versatile tools for various integrated applications [298, 299]. Briefly, a

TABLE 5 | Summary of recent advancements in 2D material-based thermoelectric energy harvesters.

Materials	PF ($\mu\text{W m}^{-1} \text{K}^{-2}$)	Seebeck coefficient ($\mu\text{V K}^{-1}$)	Electrical conductivity (S cm^{-1})	ZT	References
GO/SWCNT	5.3×10^{-2}	29	6.5×10^{-1}	—	[276]
MoS ₂ /Sb ₂ Te ₃	4.9×10^{-3}	530	176	2.08	[277]
WS ₂ /CNT/PANI	94.6	78	153	0.17	[278]
MXene/Cu ₂ Se	1.3×10^{-2}	250	—	1.77	[279]
SWCNT/MXene	203.29	39.64	1293	6.20×10^{-3}	[280]
Te-BP	9.4	250	—	1.1	[281]
h-BN/SWCNT	116.1	50.6	—	—	[282]
MOF/CNT	24.86	40.7	150	8.77×10^{-3}	[283]
PAg@BP	107.2	24.4	1785.73	—	[284]
GO/Cu ₂ Se	881	88	11	—	[285]
Re/MoS ₂	8	−100	1800	—	[286]
SiO ₂ /MoS ₂	253	39.2	1646	1.23×10^{-3}	[287]
MnO ₂ /MoS ₂	5.4×10^{-3}	−21	28.2	—	[267]
Bi ₂ Te ₃ /MXene	1.2×10^3	159	5.6×10^2	1.1	[288]
MXene/Ag ₂ Se	869	−129.5	591.2	2.7×10^{-1}	[289]
Cu/WS ₂	1.6×10^{-2}	118.7	67.6	0.68	[290]
Nb/WSe ₂ /MoS ₂	4.2×10^{-2}	135	282	0.63	[291]
MoSe ₂ /Bi ₂ Te ₃	3×10^{-1}	−232	558	1.24	[292]
MXene/SrTiNbO ₃	3050	−165	2.5×10^2	0.9	[293]
MXene/TiS ₂	220	−80.4	322.12	—	[294]
MoS ₂ /SWCNT	107.99	47.10	561.6	—	[295]
MXene/ Polyurethane	0.086	−8.3	1.2×10^5 (S m^{-1})	1.7×10^{-3}	[296]
MoS ₂ /CF	0.55	12.8	4242 (S m^{-1})	—	[297]

Abbreviations: BP, black phosphorous; CF, carbon fabric; PF, power factor; ZT, figure of merit.

standard piezoelectric energy harvesting module typically consists of two current collectors positioned above and below, sandwiching a central layer of piezoelectric material. Whenever an external force compresses the generator, it reduces the piezoelectric material's dipole moment, thereby modifying the charge density of its poles, resulting in an imbalance of charges on both current collectors. Finally, an external circuit plays a crucial role in transferring electrons to balance the charge, thereby generating an output current referred to as piezoelectricity (Figure 10A). However, upon removal of the external force, the piezoelectric material's deformed structure reverts to its original state, thereby generating a reverse current signal. Normally, the density of polarization charge (produced from the electrical moment) is directly correlated with the mechanical stress applied, as described by the equation ($\rho = dX$), where ρ represents the polarization charge density, d denotes the piezoelectric coefficient, and X signifies the applied stress. Notably, this stress or strain originates from various sources that are ubiquitous in our surroundings, including vibrations, body movements, acoustic vibrations, and airflow; therefore, a piezoelectric effect generally represents a phenomenon that couples mechanical and electrical states. Mostly, the majority of the previously reported PENGs

have relied on conventional piezoelectric materials like ZnO (first coined by Wang and Song [304] in 2006), PVDF, PZT (lead zirconate titanate), and others [305]. Following a significant breakthrough, researchers are directing increased attention toward the investigation of piezoelectric materials at the nanoscale. Therefore, among other materials, 2D piezoelectric materials offer several advantages over their bulk counterparts. For example, 2D materials have greater flexibility and can endure significant strain, thanks to their atomic-scale thickness and exceptional mechanical properties. This capability to resist high deformation broadens the potential applications of devices operating in complex and high-strain environments, such as human activities [306, 307]. Therefore, 2D materials have demonstrated potential in PENG mechanical energy harvesting. Indeed, for a material to demonstrate intrinsic piezoelectricity, it must possess both a band gap and a non-centrosymmetric crystal structure; therefore, surface engineering emerges as a potent strategy for introducing asymmetry and, consequently, enhancing its piezoelectric properties [308]. An illustration of this concept is GR, which does not have inherent piezoelectric characteristics because of its centrosymmetric structure; however, the adsorption of external atoms can disrupt inversion symmetry, thereby

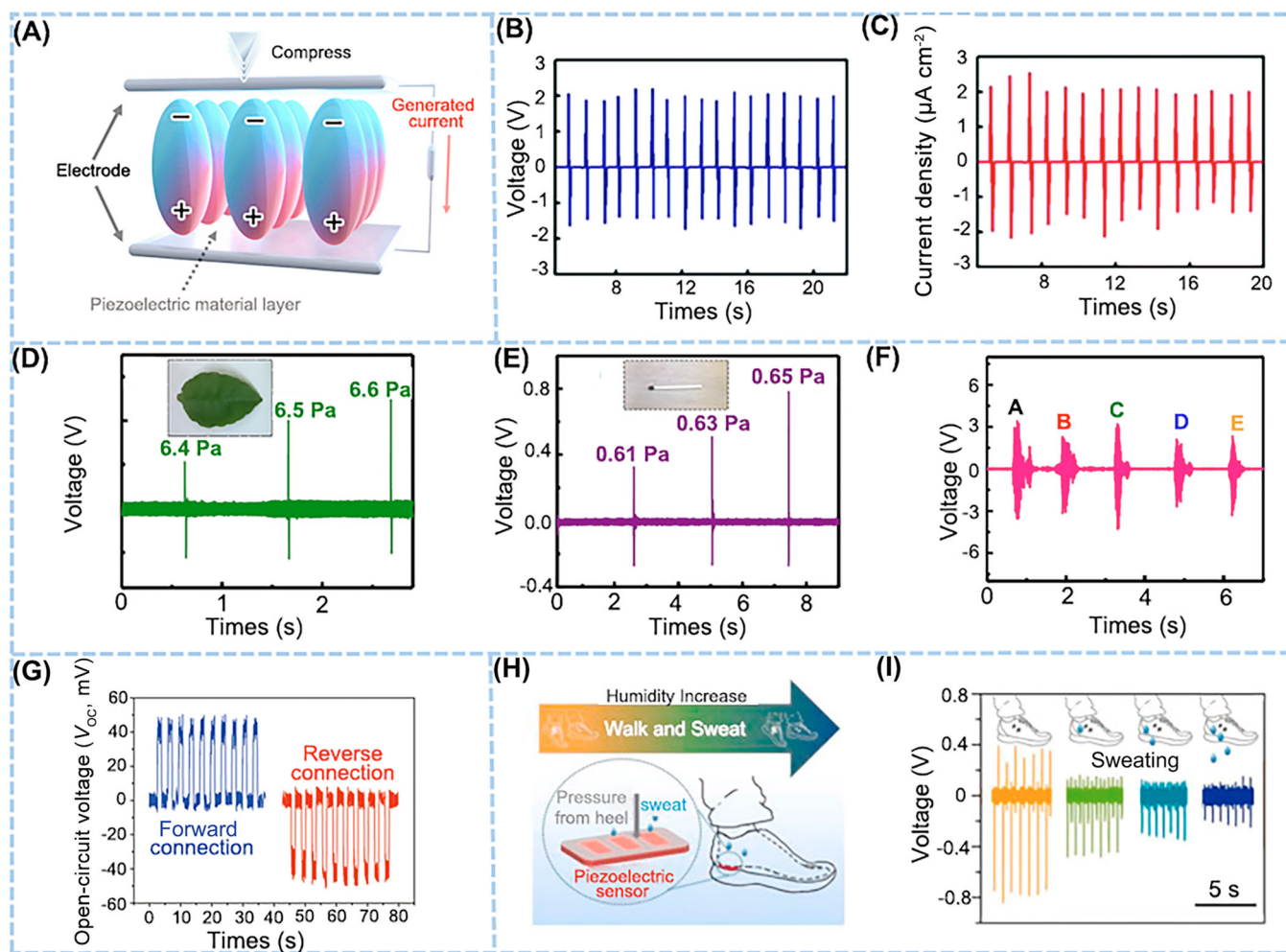


FIGURE 10 | 2D materials for piezoelectric energy harvesting. (A) A Scheme illustrating the operational mechanism of piezoelectric energy harvesting, elucidating how mechanical energy is converted into electrical energy through this phenomenon. (B and C) Graphical presentation of voltage (B) and current density (C) of a PZT-based piezoelectric nanogenerator, with graphene, which serves as the electrode. When subjected to a force of 0.9 kg, this device showed an output voltage of approximately 2 V and a current density of around $2.2 \mu\text{A cm}^{-2}$. Reproduced with permission: Copyright 2012, The Royal Society of Chemistry [300]. (D–F) Illustration of piezoelectric output voltage responses of leaves (D), matchsticks (E), and finger touch (F) for various applied stresses when dropped from varying heights. Reproduced with permission: Copyright 2017, John Wiley & Sons [301]. (G) Graphical presentation of an open-circuit voltage under stress for a monolayer h-BN nanoflower in both forward and reverse connections. Reproduced with permission: Copyright 2019, American Chemical Society [302]. (H) Diagram depicting the piezoelectric sensors with the porous electrode that can be affixed beneath the heel to identify foot sweat signals. (I) Diagram showing a voltage signal reduction under pressure during sweating, as during prolonged walking, more sweat is produced, leading to a consistent step frequency (~ 1.5 steps/s) and pressure resulting in a decrease in the sensor's output voltage. Reproduced with permission: Copyright 2021, Elsevier [303].

effectively inducing piezoelectricity. Following this, GR-based materials are being utilized as electrodes in conjunction with piezoelectric materials to boost the piezoelectric efficacy of wearable PENGs, especially textile-based PENGs [309]. For instance, Kwon et al. presented a flexible PENG using PZT, with GR, which serves as the electrode. Remarkably, when subjected to a force of 0.9 kg, this device showed an output voltage of approximately 2 V, a current density of around $2.2 \mu\text{A cm}^{-2}$, and a notable power density of about 88 mW cm^{-3} (Figure 10B,C) [300]. Both theoretical predictions and experimental studies indicate that numerous 2D materials, such as TMDCs (e.g., MoS_2) and the monochalcogenide family (e.g., SnS), have significant intrinsic piezoelectric properties due to their noncentrosymmetric lattice structures. Maity et al. presented a PENG comprising PVDF nanofiber webs and MoS_2 . Their PENG demonstrates the capability to

detect lightweight objects, such as leaves or matchsticks, dropped from varying heights, as well as the voltage produced by finger touch, suggesting its significant potential in industry for applications in automated frequency detection (Figure 10D–F). They found that MoS_2 nanosheets contribute to enhancing the overall crystallinity and piezoelectric-phase composition in PVDF nanofiber webs, rendering them suitable for constructing PENGs [301]. Apart from TMDCs, K. Park showcased the presence of piezoelectricity in h-BN by synthesizing them through a mechanochemical exfoliation process, followed by transferring them onto a plastic substrate decorated with an electrode, allowing for assessing the energy harvesting potential under external stress. When subjected to mechanical bending, this monolayer h-BN generated alternating piezoelectric outputs of approximately 50 mV and 30 pA with a piezoelectric voltage coefficient (g_{11}) of approximately $2.35 \times 10^{-3} \text{ V m N}^{-1}$ (Figure 10G) [302]. Wang

et al. have recently developed an interesting piezoelectric film pressure sensor with (PVDF-TrFE)/MXene nanofiber that can effectively detect body motion, making it viable for healthcare applications. Moreover, MXene's hydrophilicity nature facilitates the film's ability to detect changes in humidity, and similarly, these sensors can also harness energy, making them suitable for self-powered pressure-sensing applications. As it is known that excessive sweating of the feet poses health risks, piezoelectric sensors with porous electrodes can be affixed beneath the heel to identify foot sweat signals (Figure 10H). However, when an individual walks for a longer period, more sweat is produced, leading to a consistent step frequency (~1.5 steps/s) and pressure resulting in a decrease in the sensor's output voltage (Figure 10I) [303]. The recent advancements in 2D material-based piezoelectric energy harvesters are summarized in Table 6.

9.3.2 | Strengths and Challenges

Traditional piezoelectric energy harvesters often lack versatility for modern wearable applications. Therefore, advances in PENGs leverage the unique properties of 2D materials, which can withstand high strain and offer adjustable flexibility. Their ability to convert mechanical signals into electrical energy makes them ideal for compact nanogenerators, achieving power outputs of up to milliwatts per square meter, highlighting their potential for

next-generation energy harvesting technologies. Despite progress in 2D material-based PENGs, challenges remain in achieving consistent response due to factors like in-plane piezoelectricity, orientation-dependent behavior, and limited control over material alignment during synthesis. Experimental validation is limited by difficulties in preparation and scalability, and the lack of a suitable growth substrate further complicates large-scale production. Even minor thickness variations can disrupt piezoelectric properties, and environmental factors can significantly affect their behavior due to the material's sensitivity. Integrating theoretical and experimental analysis of the interfacial properties between 2D materials and electrodes can provide valuable insights into critical phenomena, including how different metals and electrode configurations influence charge transfer and transport at the interface. This knowledge is essential for the rational design and advancement of future PENGs.

9.4 | Triboelectric Energy Harvesting

9.4.1 | Mechanism and Performance

TENGs represent an emerging technology rooted in the well-known contact electrification effect, designed to efficiently harness ambient mechanical energy (such as ocean waves, wind, and human motion) for various applications such as

TABLE 6 | Summary of recent advancements in 2D material-based piezoelectric energy harvesters.

Materials	V _{oc} (V)	Load (Ω)	Power output	I _{sc} (μA cm ⁻²)	References
MoS ₂	0.03	8 M	62.72 pW	—	[310, 311]
SnO ₂ /PVDF	42	30 M	4900 W m ⁻³	6.25	[312]
SnS	0.15	1 G	—	0.002	[313]
GR/PVDF/BaTiO ₃	10	1 M	5.8 μW	0.31	[314]
PVDF-BZT-BCT-GO	4	—	2.5 μW	1.6	[315]
TOCN/MoS ₂	43	80 M	1.29 μW	0.16	[316]
CNF/MXene@ZnO	17.15	—	4.64 μW	997 nA	[317]
MXene/PVDF	3.15	—	—	134 nA	[318]
PVDF-WSe ₂	116	—	48.5 μW	—	[319]
h-BN/PVDF	50	—	2 μW	250 nA	[164]
N-graphene/MXene	9.1	3 M	3.85 μW	1.51	[123]
Graphene/PVDF	26	10 M	16.52 μW	1.13	[320]
V ₂ CT _x /PVDF	124	1 M	21.62 μW	2.2	[321]
CNF/MXene/ZnO	17.15	300 kPa	4.64 μW	997 nA	[317]
OWPM/MXene/ZnO	4.71	45 N	—	234.17 nA	[322]
MoS ₂ /PVDF/Cellulose	20	10 kPa	56.25 μW	03	[323]
2D MoO ₃ /PVDF	50	30 M	1512 W m ⁻³	—	[324]
WS ₂ /PVDF/HFP	20	1 kgf	294.55 pm/V	1.06	[325]
Ti ₃ CNT _x /PVDF	14.6	9 N	2.2 μW	1.48	[326]
Perovskite/PVDF	135	12 N	15.7 μW	0.4	[327]
BaTiO ₃ /polylactic acid	11.97	24 N	—	560 nA	[328]
MXene/PDMS	80	1 M	13.5 W m ⁻²	14	[329]
Li-MoS ₂	13.5	3 kgf	—	30	[330]

Abbreviations: CNF, cellulose nanofiber; HFP, hexafluoropropene; I_{sc}, short-circuit current; OWPM, oxidized wood porous material; V_{oc}, open-circuit voltage.

self-powered systems, biomedical equipment, and portable electronics, owing to their adaptability, cost-effectiveness, high-power density, and simple production. In contrast to electromagnetic generators and PENGs, TENGs excel at efficiently converting mechanical energy under low force and low frequencies, while also facilitating sustainable and cost-effective production processes [331]. Generally, TENGs operate through four primary mechanisms, including (i) contact-separation, (ii) single electrode, (iii) lateral sliding, and (iv) free-standing modes. Among these, the contact-separation mode is the most widely implemented, due to its significantly low-frequency mechanical contact in the environment. Typically, TENGs exploit the phenomenon wherein electric charges shift from one material to another upon contact between two substances. The process of TENGs involves generating electrical energy from

mechanical energy through triboelectrification and electrostatic induction. When external mechanical energy causes two dissimilar materials to come into contact, triboelectrification induces electrostatic charges on the surface of each material. As the materials separate due to additional mechanical energy, an electrical potential is created between the electrostatic charges, resulting in prompt charges on conductive materials to migrate from one side to the other [332]. It is noteworthy that this charge transfer is not immediately reversible, resulting in an accumulation of excess charges on one side and a deficiency on the other. Consequently, alternating current (AC) can flow based on the polarity of the charges, whether positive or negative (Figure 11A) [335, 336]. Notably, improving energy harvesting efficiency remains a crucial challenge across all types of energy harvesters, including TENGs, and since their inception

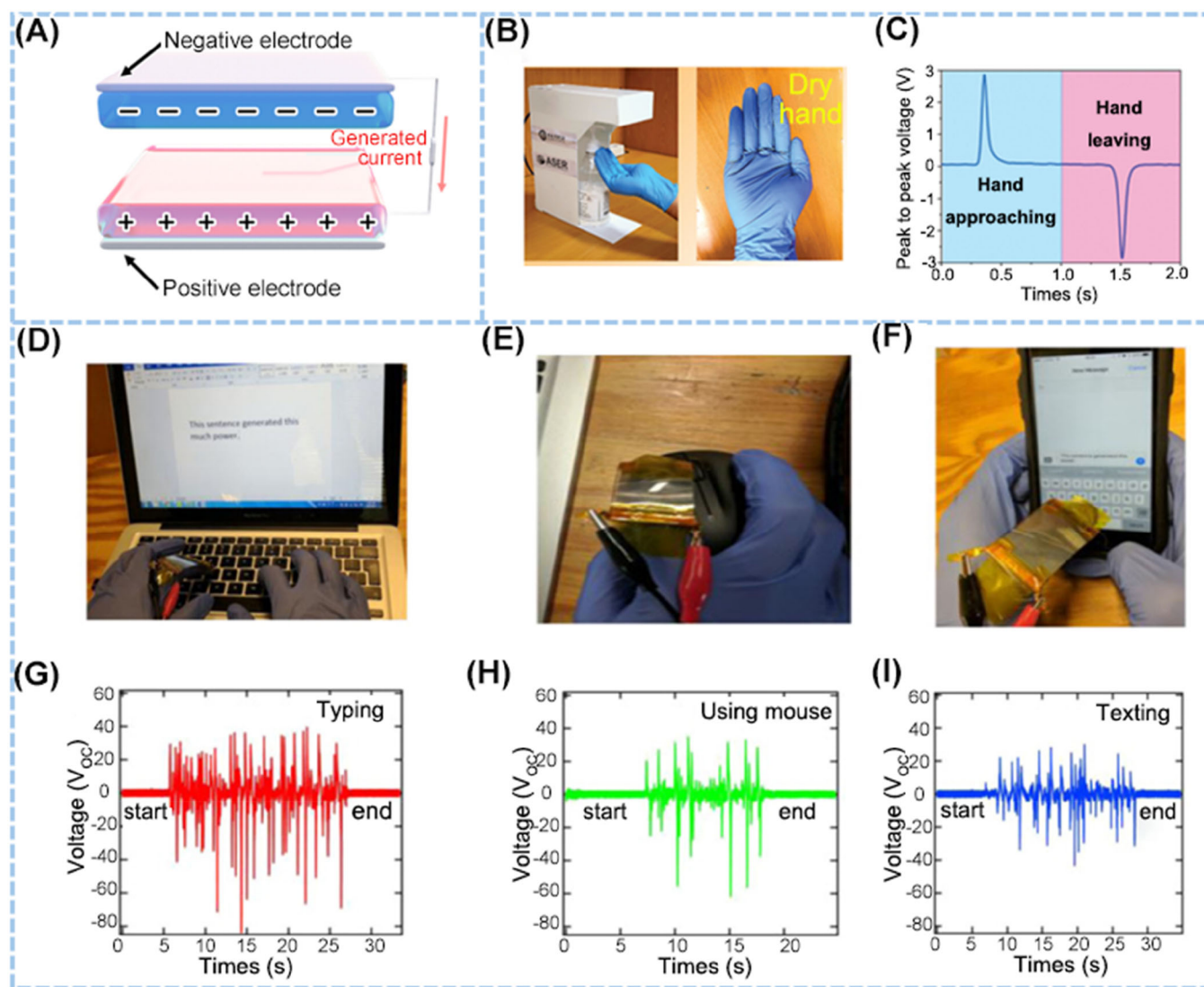


FIGURE 11 | 2D materials for triboelectric energy harvesting. (A) A Scheme illustrating the operational mechanism of triboelectric energy harvesting, elucidating how mechanical friction energy is converted into electrical energy through this phenomenon. (B) Presentation of a self-powered triboelectric sensor based on a MoS₂/LIG touchless hand sanitizer and the resulting innovative device, working as a self-powered triboelectric sensor, which was used for touchless hand sanitizer to combat the spread of COVID-19. (C) Graphical presentation of the peak-to-peak voltage signal waveform depicting the hand's proximity to and removal from the sanitizer. Reproduced with permission: Copyright 2022, John Wiley & Sons [332]. (D) Presentation of a flexible MXene harvesting mechanical energy during (D) typing, (E) mouse-clicking, (F) smartphone texting, and V_{oc} generated during (G) typing, (H) mouse-clicking, and (I) texting. Reproduced with permission: Copyright 2018, Elsevier [333].

in 2012, significant endeavors have been dedicated to enhancing their efficiency. The electrostatic force of attraction (F_e) and charge displacement (D_e) are the two key parameters that significantly impact the efficiency of TENGs. For instance, the product of F_e and D_e determines the energy transferred from mechanical energy to electrical energy. Virtually all everyday materials, including silk, plastics, wood, polymers, and metals, have triboelectric properties, making them suitable candidates for TENG applications. A wide range of materials is utilized in TENG fabrication, with the polymer being the most extensively studied. Fluorine-containing polymers like polytetrafluoroethylene, PVDF, and fluorinated ethylene propylene are commonly chosen as negative triboelectric materials due to fluorine's strong electron affinity, whereas polymers with electron-donating groups, such as nylon, wool, and silk, serve as positive triboelectric materials. These polymers offer advantages like flexibility, scalability, stretchability, and ease of processing, but they are limited by their low melting points and weak mechanical properties, restricting their use at high temperatures. In contrast, 2D materials excel in terms of mechanical strength, and even small additions of these materials to a polymer matrix can substantially enhance its durability. Therefore, recently, 2D materials have attracted significant interest due to their unique properties, which set them apart from traditional bulk materials. These materials possess robust in-plane mechanical stability, attributed to strong covalent bonds, while showing minimal interlayer interactions, allowing for easy exfoliation from bulk forms. Moreover, their distinctive electronic characteristics, such as variable charging polarities, offer an excellent selection criterion for choosing triboelectric materials. Therefore, 2D materials' incorporation into TENGs is widely recognized for the ability to significantly boost their conversion efficiency and enhance the flexibility and durability. Specifically, integrating 2D materials into the triboelectric layer enhances the effective surface area for charge capture and storage within the 2D material layer. This addition minimizes charge recombination between the electrode and the triboelectric layer, ultimately boosting the triboelectric charge density and improving the nanogenerator's output performance. Nowadays, researchers are exploring various applications of these 2D materials in TENGs, including their electrode applications, as a charge-trapping additive, and triboelectric layers [337]. For example, T. Mondal introduced natural rubber (NR) as an eco-friendly material for developing TENGs. By incorporating GR, both the dielectric constant and triboelectric properties of NR were enhanced, significantly improving performance. This material was also pressure-sensitive, allowing it to detect human motions like finger or neck bending, making it highly suitable for real-time motion monitoring applications [338]. Remarkably, the TENG device's large lateral size and atomic thickness resulted in a high surface area, which improved the effective contact area among its active layers. MoS₂, for instance, offers efficient charge-trapping sites because of its particular surface area along with quantum confinement characteristics [339, 340]. Researchers are nowadays prioritizing the development of flexible, portable, and self-sufficient power supplies to meet the growing need for adaptable, stretchable, wearable electronics, sensors, and electronic skin. Due to their capability to bend, fold, stretch, and twist, power sources, like TENGs, demonstrate versatility when equipped with appropriate active layers, electrode material, and suitable substrate

design. In this regard, a recent development showcased the use of MoS₂-incorporated laser-induced graphene (LIG) as a charge retention layer in a contactless TENG using Siloxene/Ecoflex. The introduction of MoS₂/LIG resulted in a fourfold enhancement in surface potential, and as a charge-trapping layer, MoS₂/LIG showed superior performance compared to LIG and MoS₂. Moreover, combining Silicone/Ecoflex with a MoS₂ and LIG layer resulted in an output voltage of 32 V with a current density of 7 $\mu\text{A m}^{-2}$, resulting in this innovative device, working as a self-powered triboelectric sensor, which was used for touchless hand sanitizer to combat the spread of COVID-19 (Figure 11B). Additionally, the signal waveform depicting the hand's proximity to and removal from the sanitizer was observed (Figure 11C) [332]. MXenes are highly regarded as excellent materials for triboelectric layers and conductive electrodes in TENGs, owing to their exceptional electrical conductivity, hydrophilic nature, and rich variety of surface functional groups such as $-\text{OH}$, $-\text{O}$, and $-\text{F}$, making them an ideal alternative to the electronegative friction material in TENGs. In this regard, Dong et al. highlighted the efficiency of TENGs in converting the surplus mechanical energy generated from human motion into electrical power. Their study showcased that a flexible MXene (Ti₃C₂T_x) TENG can harness electrical energy from muscle movements, such as texting, indicating the seamless integration of TENGs with wearable electronics (Figure 11D–I) [333]. However, the use of flexible MXene-based TENGs for athlete health monitoring and training support remains largely unexplored. Additionally, MXenes are prone to rapid oxidation and degradation in air and water, primarily forming TiO₂, which compromises their performance. To make these nanogenerators viable for practical applications, enhancing the stability of MXenes and their composites is crucial to ensure consistent output and long-term reliability. Further research should focus on improving their durability. In summary, these findings show that due to their unique properties and potential applications in different energy harvesting devices, 2D materials are also attractive for triboelectric energy harvesting. The recent advancements in 2D material-based triboelectric energy harvesters are summarized in Table 7.

9.4.2 | Strengths and Challenges

2D materials are emerging as leading candidates for electrode and triboelectric materials in TENGs due to their outstanding electrical conductivity. 2D-TENGs are known for their excellent output performance, flexibility, and durability, making them ideal for energy harvesting and self-powered sensors. However, several challenges hinder the advancement of 2D-TENGs. There is a need to explore novel ultrathin 2D materials, as current research is limited. Additionally, the engineering of the crystal phase is vital, as it affects the material's properties. Although theoretical studies on the triboelectric properties of 2D materials are abundant, there is a pressing need for more experimental investigation to effectively expand their potential in flexible and wearable electronic applications. Improving the operational stability of 2D TENGs is also crucial, as some materials are environmentally sensitive, which can negatively affect performance. A promising strategy to address these issues involves integrating 2D materials with more robust materials to ensure stable output under extreme conditions. Also, an

TABLE 7 | Summary of recent advancements in 2D material-based triboelectric energy harvesters.

2D materials	Substrate	V_{oc} (V)	Load ($M\Omega$)	Power density	I_{sc} ($\mu A\ cm^{-2}$)	References
GO	PET	322.44	50	192 mW	78.22	[341]
GQDs	PDMS	128	5	$1.6\ W\ m^{-2}$	23	[342]
MXene	Silk	118.4	5	$1.09\ W\ m^{-2}$	—	[343]
MoS ₂ /PVA	Fabric	500	—	—	60	[339]
WS ₂	PET	12.2	10–25	$138\ mW\ m^{-2}$	—	[213]
MXene/Si	Al	600	5	$14.9\ W\ m^{-2}$	167	[344]
BP	PET	1860	100	$0.52\ mW\ cm^{-2}$	1.1	[345]
MoS ₂ /PVDF	PET	36	10	$445\ \mu W\ cm^{-2}$	2.8	[346]
ZnS/ZnO	PDMS	156	40	$2.98\ mW\ m^{-2}$	18.9	[347]
Graphene/NR	PET	13.5	2	$0.42\ W\ m^{-2}$	32	[348]
Graphene/perovskite	ITO	245	100	$11.23\ W\ m^{-2}$	2	[349]
Graphene/PVDF	—	110	7	$53.57\ mW\ m^{-2}$	100	[350]
S/MXene	Polymeric film	1717	8	$3.80\ W\ m^{-2}$	129	[351]
MXene/TUP	Cu foil	83	200	$231\ mW\ m^{-2}$	10	[352]
WS ₂ /p-Si	Silicon Wafer	—	40	$2.46\ \mu W\ m^{-2}$	11 nA	[353]
Plastic/h-BN	—	68	6.6	$460\ mW\ m^{-2}$	6.1	[354]
ChNCs/MXene	PDMS	56	20	$99.5\ mW\ m^{-2}$	3.2	[355]
MoS ₂ /PVDF	PDMS	189	—	$104.5\ \mu W\ cm^{-2}$	1.61	[356]
MXene/PVA/NaCl	PET	390	10	$6.66\ W\ m^{-2}$	96	[357]
RTV@MoS ₂ /PVC	Ni/Cu	1055	—	—	112	[358]
MoS ₂ @ PAN	Al	245.3	6 N	$1.75\ W\ m^{-2}$	5.12	[359]
MXene@ SnSe ₂	Polyimide	202.31	50	$19.77\ W\ m^{-2}$	108.55	[360]
V ₂ CT _x @ PVDF	Acrylic	982	50 N	$18.2\ W\ m^{-2}$	—	[361]

Abbreviations: ChNCs, chitin nanocrystals; NR, natural rubber; PAN, polyacrylonitrile. PVC, polyvinyl chloride; RTV, silicon rubber; TUP, polyurethane.

in-depth and comprehensive understanding of the fundamental mechanism underlying triboelectric effects is essential for optimizing device performance and enhancing energy conversion efficiency.

9.5 | RF Energy Harvesting

9.5.1 | Mechanism and Performance

The swift progress in telecommunication and the corresponding growth of photonic integrated circuits underscore RF energy harvesting as a highly noteworthy approach for powering different applications, including the IoT, smart city infrastructure, health observing, and wearable and biomedical devices, offering the potential to run these systems without reliance on traditional batteries. RF energy harvesting technology has several distinctive features, like reliability, predictability, controllability, and the capability to supply energy to multiple nodes simultaneously, making it a viable choice for many applications [362]. Furthermore, electromagnetic signals functional in the RF range, like Wi-Fi (2.4–2.8 GHz) and cellular telecom (700 MHz–2.7 GHz), are becoming more prevalent and perfect sources to harvest power for future electronics. Generally, harvesting RF energy involves capturing RF signals from the

radio environment and utilizing them to power low-voltage electronic devices. A typical RF energy harvesting circuit consists of a rectifier, a voltage multiplier, an antenna, and a device for energy storage (Figure 12A). The conversion of RF energy into direct current (DC) is commonly accomplished using rectennas, comprising an antenna design for gathering electromagnetic radiation, particularly AC waves from the surrounding environment, playing a critical role in harvesting RF energy for different applications. These electromagnetic waves are subsequently directed into an energetic diode with nonlinear current–voltage properties because nonlinearity plays a vital role in rectifying RF signals, which is crucial for generating the DC power essential to power electronic circuits. This DC power (rectified) can be either stored or directly used to supply power to a load. Therefore, the total efficiency of RF-DC power conversion is calculated by multiplying the efficiencies of each component within the system [367, 368].

Due to their mechanical and electrical properties, 2D materials are promising for incorporation into flexible electronics, and their atomic thickness, along with the potential for mass production, makes them ideal for emerging “smart skin” applications [369]. GR and MoS₂, as leading 2D materials, offer exceptional potential for developing high-performance flexible RF diodes, crucial for advancing rectenna-based energy

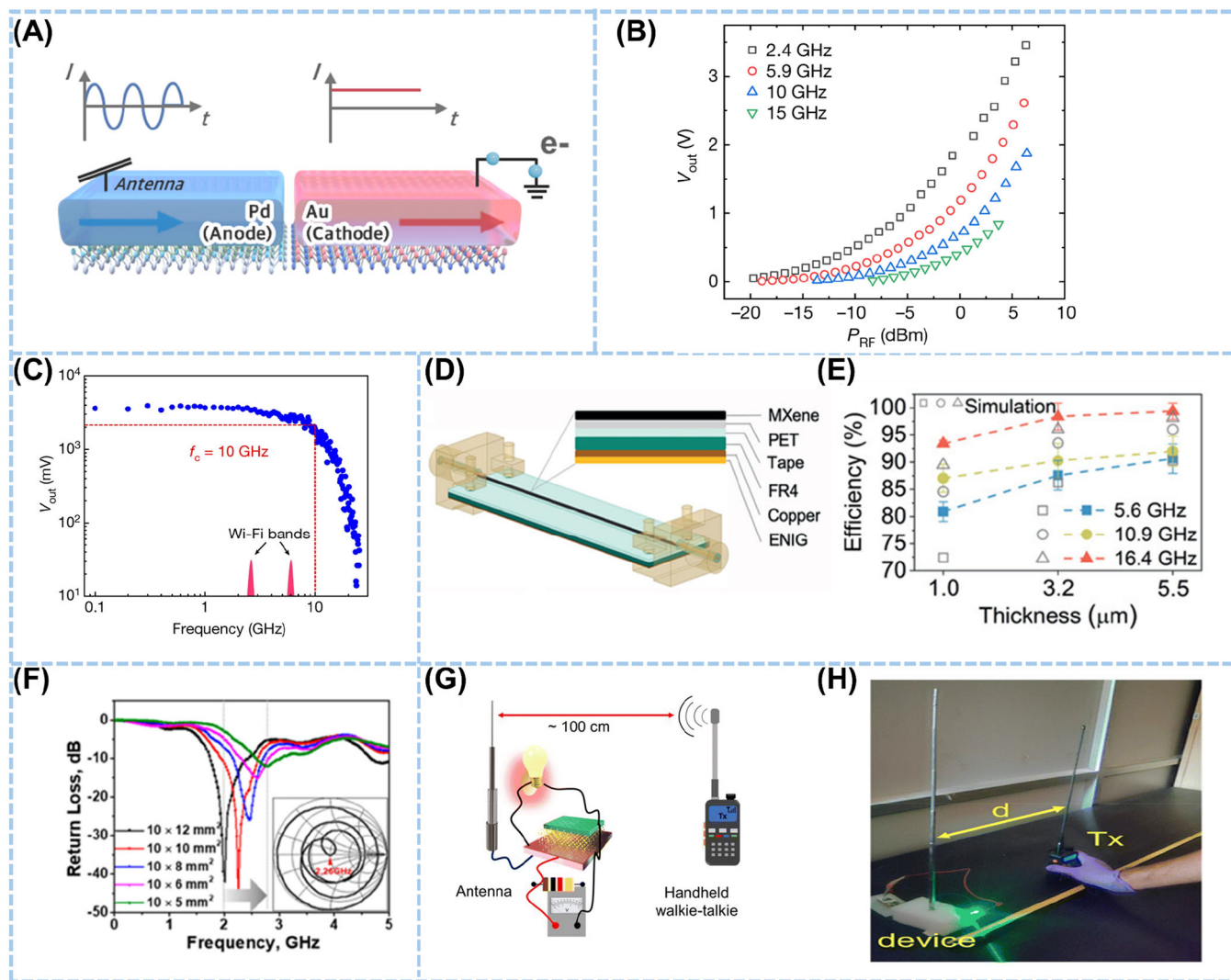


FIGURE 12 | 2D materials for radiofrequency energy harvesting. (A) A schematic representation of radiofrequency energy harvesting. (B) Variation of the output voltage to the applied RF input power measured across four distinct operating frequencies. (C) Graphical representation of achieving output voltage (3.5 V) by using an input RF power (5 mW). The graph also shows that with increasing input RF frequency, there is reduction of output voltage. Reproduced with permission: Copyright 2019, Nature Publishing Group [363]. (D) Schematic representation of the MXene microstrip transition line, which enables minimal energy loss and facilitates the operation of high-power radiation antennas operating at frequencies between 5.6 and 16.4 GHz. (E) Experimentally calculated and simulated efficiency of the MXene-based antenna as a function of different thicknesses. Reproduced with permission: Copyright 2020, John Wiley & Sons [364]. (F) NbSe₂-based antenna with variable-length return loss spectra, with the inset displaying a Smith chart impedance diagram. Reproduced with permission: Copyright 2019, American Chemical Society [365]. (G) RF energy harvesting setup based on PbS. (H) Experimental setup displaying an illumination of an LED within a radius of 100, utilizing a handheld walkie-talkie as an RF source. Reproduced with permission: Copyright 2023, The Royal Society of Chemistry [366].

harvesting systems [370]. Here are the key advantages of using 2D materials like GR or TMDCs in energy harvesting systems:

- **High Mobility:** GR's high electron mobility enables operation at THz frequencies and even optical frequencies, making it ideal for high-speed applications.
- **Smaller Antennas:** GR allows the creation of antennas with smaller dimensions than traditional metals, supporting high-density integration in compact systems.
- **Tunable Frequency:** The Fermi energy of GR can be controlled through an electric field, enabling the tunability of the antenna's oscillation frequency.

- **Mechanical Strength and Flexibility:** The combination of GR's mechanical strength, flexibility, and thin-film technology enables its use on various substrates, expanding its applications in wearable electronics beyond traditional semiconductor devices.

Recently, a significant advancement was achieved in the development of a thin and flexible rectenna utilizing a MoS₂ semiconductor heterojunction, which showcased a remarkable cutoff frequency of 10 GHz, marking a tenfold improvement in speed over existing cutting-edge flexible rectifiers [363, 371]. A comprehensive investigation was conducted by this researcher, where the output voltage was

plotted against the input RF power, across four distinct frequencies. In this way, they engineered a completely flexible and integrated rectenna capable of wireless energy harvesting from electromagnetic radiation within the Wi-Fi band, realizing without the external bias (battery-free) by combining an energetic MoS₂ rectifier with a Wi-Fi antenna. By using this innovative rectenna configuration, they achieved an output voltage (3.5 V) with an input (5 mW) RF power, demonstrating a maximum output efficiency of 40%. Additionally, this fabricated rectifier can function within the X-band (8–12 GHz), thanks to the shown cut-off frequency of 10 GHz for MoS₂, surpassing the limitations of firm silicon-based Schottky diodes (Figure 12B,C). MXene also shows promise as a 2D material for integrated RF components in diverse electronic devices because of its versatility across a wide frequency range, flexible nature, scalability property, and solution processing ability. Thanks to these characteristics, very recently, a Ti₃C₂T_x MXene microstrip (solution-processed) has been documented, enabling minimal energy loss and facilitating the operation of high-power radiation antennas operating at frequencies between 5.6 and 16.4 GHz (Figure 12D) [364], resulting in a radiation efficiency of 99% at 16.4 GHz with a 5.5 μ m thick MXene antenna, which is comparable to that of a conventional 35 μ m thick copper-based antenna (Figure 12E). A 2D NbSe₂-based monopole RF antenna also displays exceptional performance, which is marked by a minor return loss (−46.5 dB), excellent radiation proficiency (70.6%), along a tunable frequency (2.01–2.80 GHz), even though its thickness is below that of metal antennas' skin depth (Figure 12F). This method offers a facile approach to fabricating 2D materials for wearable communication systems [365]. Furthermore, a 2D galena (PbS) Schottky diode was recently used to harvest energy from RF signals, which is showcased through the utilization of a handheld walkie-talkie operating at a carrier frequency ranging from 140 to 170 MHz (Figure 12G). This device effectively captures RF energy, resulting in a maximum output DC voltage (1.8 V) and a subsequent output power (38 mW) at 150 MHz. Additionally, it illuminates an LED inside a radius of 100 cm and displays 19% PCE at 150 MHz (Figure 12H) [366].

9.5.2 | Strengths and Challenges

The operational frequency of a rectenna system is constrained by the RC (R = series resistance, C = parasitic capacitance) time constant. Due to their high sheet resistance, 2D materials face greater resistance than bulk materials, with additional resistance contributed by the weak vdW bonds at the metal contacts. Although this limits system frequency, 2D materials offer an advantage with their lower parasitic capacitance, which can enhance device performance in certain applications. The production of 2D materials must be scaled up to achieve manufacturability, meaning that the process technology needs to meet semiconductor industry standards in terms of quality, consistency, and reliability. Moreover, to achieve the lowest possible sheet resistance, enhancing the quality of 2D materials produced through scalable techniques like CVD or epitaxy is essential.

9.6 | 2D Material-Based Hybrid Composites in Energy Harvesting

The rapid advancement in smart materials and hybrid materials design is largely driven by the urgent need to address energy shortages and environmental challenges. Although the development of diverse nanomaterial classes has been remarkable, the true potential lies in engineering hybrid materials with tunable properties. This integration aims to synergistically combine the unique properties of distinct 2D materials, leading to the development of an advanced hybrid composite with superior functional attributes. Generally, different methodologies are used for 2D hybrid composites synthesis, such as top-down and bottom-up approaches, self-assembly, chemical bonding, in situ growth, interface interactions, and surface functionalization, depending on the end goals. One such approach involves incorporating MOF with MXene to enhance stability and performance. This approach takes advantage of MOF's intrinsic hydrophobic microporous structure, which serves as a protective barrier, preventing water molecules from penetrating and interacting with MXene, thereby mitigating oxidation. Additionally, metal ions within the MOF framework can bind to the dangling bonds on the MXene surface, further inhibiting oxidative degradation. Selecting MOFs with inherently hydrophobic properties can further enhance surface hydrophobicity, ensuring stable performance in humid environments. By using this concept, the triboelectric properties of PDMS were recently significantly enhanced through the synergistic integration of MOF and MXene [372]. The MOF/MXene hybrid was initially synthesized via vacuum filtration (Figure 13A), followed by the fabrication of a MOF/MXene@PDMS composite film using a blade-coating method. The final hybrid was applied to a PET substrate through scraper-coating. This hybrid material demonstrated notable triboelectric performance, effectively capturing energy from water droplets and directly powering LED panels and calculators. As shown in Figure 13B, the electron transfer mechanism between materials reveals that before contact with water, electrons remained confined to their respective potential wells, preventing their escape. Upon contact, the overlapping electron clouds between the two materials reduced the potential barrier, allowing some electrons to migrate from water molecules to the composite. After separation, the transferred charges remained as surface static charges, facilitating effective charge retention within the materials. In optoelectronic devices, optimizing interface energy alignment and defect passivation, particularly in perovskite films, is critical for enhancing the efficiency and stability of perovskite solar cells. Interface engineering of Ti₃C₂T_x MXene in perovskite solar cells has garnered significant attention due to its exceptional photoelectric properties and tunable surface chemistry. Their electrical conductivity and charge mobility facilitate efficient carrier transport along perovskite grain boundaries, whereas diverse terminal groups enable the modulation of interfacial characteristics. Although oxygen plasma treatment adjusts the WF and reduces the defect density, its effectiveness is constrained as it only modifies the existing surface groups. Therefore, a more promising approach is the functionalization of MXene surfaces with tailored molecular or atomic species to further enhance the

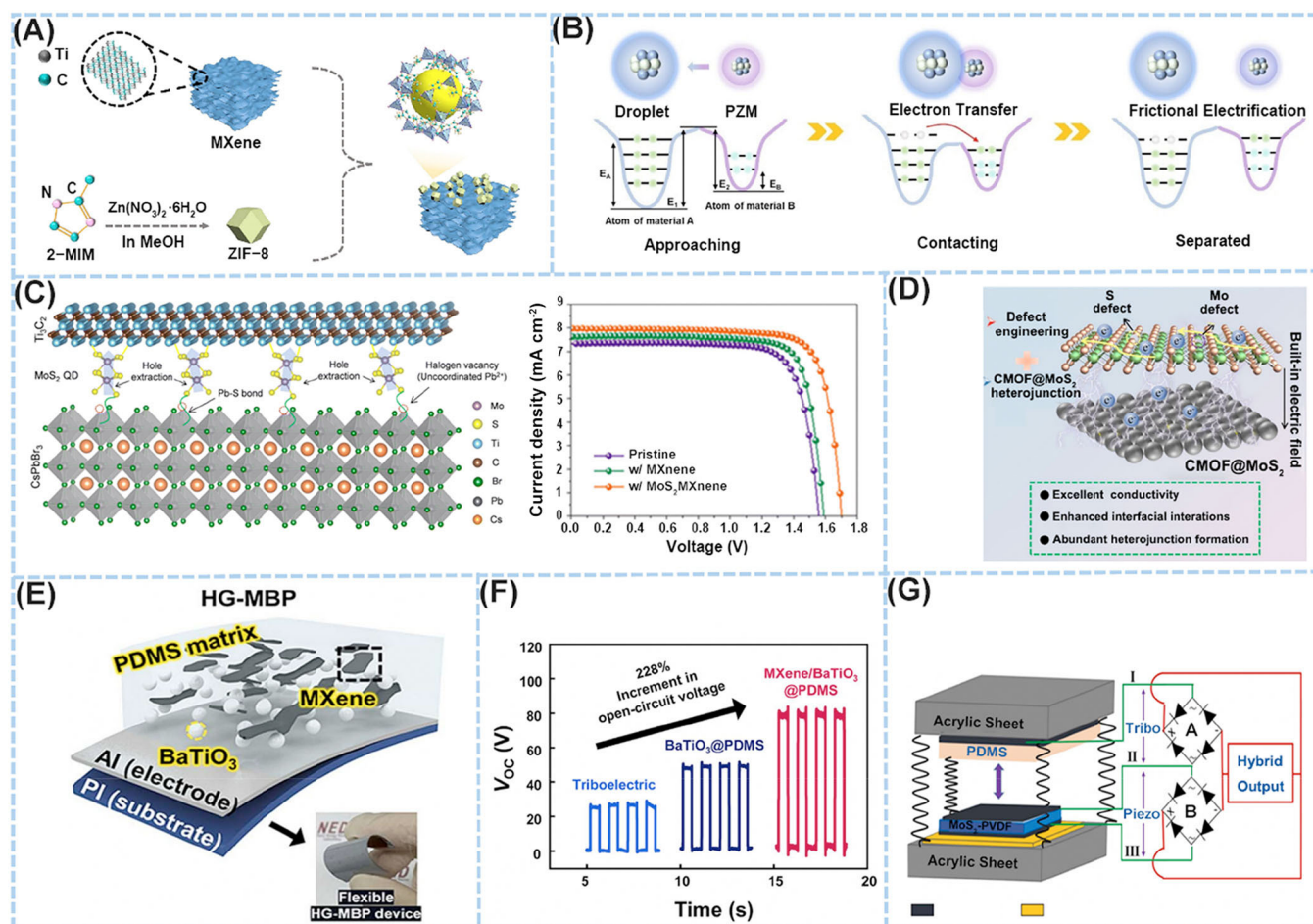


FIGURE 13 | 2D hybrid composite and hybrid nanogenerators for energy harvesting. (A) Schematic representation of MOF and MXene composite. (B) The electron cloud potential well model in an optimized TENG, which provides a theoretical framework for understanding the mechanism of electron transfer at the material interface. Reproduce with permission: Copyright 2025, Elsevier [372]. (C) An illustration of defect passivation and charge transport at the interface between the CsPbBr₃ film and MoS₂@MMXene. The device highlights the role of unsaturated sulfur atoms in MoS₂ quantum dots (QDs), which form Ti-S bonds with the MXene surface, facilitating efficient charge transfer and the current density-voltage (*J-V*) characteristics showcasing the maximum open-circuit voltage achieved in single-junction solar cell devices. Reproduced with permission: Copyright 2025, The Royal Society of Chemistry [373]. (D) Schematic representation of the CMOF@MoS₂ hybrid composite, highlighting the combined effects of defect modulation and heterostructure engineering. Reproduced with permission: Copyright 2024, Elsevier [374]. (E) Schematic representation of the fabrication process of the MXene/BaTiO₃@PDMS device, where the composite layer is spin-coated onto an aluminum electrode supported by a PI substrate. (F) Comparative voltage output (*V_{OC}*) waveform of the PDMS-based TENG (light blue), hybrid composite (deep blue), and modified hybrid generator (pink), respectively. Reproduce with permission: Copyright 2023, Elsevier [329]. (G) Illustration of the electrical circuit configuration and schematic representation of the piezo-triboelectric hybrid nanogenerator incorporating MoS₂-PVDF and PDMS layers. Reproduce with permission: Copyright 2023, Elsevier.

charge transport and overall device performance. To achieve this, MoS₂ quantum dot-grafted MXene nanoflakes were introduced to modify CsPbBr₃ perovskite films with hole-transport-layer-free carbon-based perovskite solar cells, leveraging MoS₂'s efficient hole extraction and energy-level compatibility [373]. This hybrid structure facilitates charge transport, suppresses interfacial nonradiative recombination, and enhances moisture resistance. In this configuration, the unsaturated sulfur atoms in MoS₂ establish Ti-S and Pb-S bonds with MXene and perovskite and effectively stabilize both for improved performance (Figure 13C). Recently, the integration of TENGs, SCs, and sensors into self-powered microsystems has garnered significant research interest due to their potential for wearable energy harvesting. In such systems, biomechanical energy is efficiently captured by

TENGs and stored in SCs, which subsequently act as a green energy source for sensors. Theoretically, the formation of a strong built-in electric field at the interface of carbonized MOF (CMOF) and MoS₂ facilitates rapid charge transfer and enhances the separation of photo-generated carriers, leading to improved sensing performance and PCE. In this scenario, to leverage the synergistic properties of MOF and MoS₂, a CMOF@MoS₂ hybrid composite with abundant defect sites was strategically engineered via in situ growth for hydrogel-based applications [374]. The presence of defect-rich MoS₂ layers accelerates charge transport while enhancing the electrical conductivity of the hybrid hydrogel (Figure 13D). Furthermore, the spontaneous electron transfer from MoS₂ to MOF, driven by the difference in their WFs, establishes an interfacial electric field that promotes efficient electron-hole

separation in photoexcited MoS₂, significantly improving PCE and sensing performance.

9.7 | 2D Material-Based Hybrid Nanogenerators in Energy Harvesting

Hybrid energy harvesters incorporating 2D materials offer an effective strategy for harnessing multiple environmental energy sources by integrating diverse working mechanisms. These hybrid systems optimize energy conversion by simultaneously capturing mechanical, thermal, and solar energy, ensuring a higher power output suitable for distributed energy applications. Various hybrid configurations have been explored, including mechanical-solar, mechanical-thermal, thermal-solar, and fully integrated mechanical-thermal-solar energy harvesters. However, designing efficient circuitry is crucial, as thermoelectric and PV generators produce DC, whereas piezoelectric, triboelectric, and pyroelectric devices generate AC, necessitating rectification for seamless integration. Additionally, impedance matching tailored to material properties and operating frequencies is essential for maximizing power output. A notable example of this approach is a hybrid nanogenerator fabricated by embedding MXene and a barium titanate (BaTiO₃) ceramic filler within a PDMS matrix [329]. This system integrates piezoelectric and triboelectric effects by incorporating Al as an electrode, polyimide (PI) as a substrate, and a composite MXene/BaTiO₃@PDMS film serving as both the piezoelectric and triboelectric layers (Figure 13E). Operating in a vertical contact-separation mode, this hybrid nanogenerator effectively harvests mechanical energy through contact electrification and deformation-induced polarization. The device uses two bridge rectifiers to combine the outputs of the piezoelectric and triboelectric components, achieving an open-circuit voltage of 80 V (Figure 13F), a short-circuit current of 14 μ A, and an average power density of 13.5 W m⁻². Its high sensitivity and rapid response highlight its potential for human motion monitoring and material sensing systems. The PVDF is a highly flexible piezoelectric polymer known for its excellent piezoelectric properties, ease of processing, low cost, and non-toxicity, making it a promising candidate for hybrid energy harvesting systems. When semiconducting nanofillers are incorporated, their piezoelectric output further improves, along with an enhancement in the dielectric properties and surface roughness of the PVDF matrix, leading to a significant increase in triboelectric performance. This synergistic enhancement also enables the development of hybrid nanogenerators that effectively combine piezoelectric and triboelectric mechanisms to achieve superior energy conversion for powering various electronic devices. Given the above advantages, a hybrid nanogenerator was fabricated using MoS₂-doped PVDF as one layer and a PDMS film as the second layer, operating in a vertical contact-separation mode. The composite PVDF-MoS₂ film was coated on both sides, whereas the PDMS layer was deposited on the top surface with an Al coating via thermal evaporation (Figure 13G). The combined output of the piezoelectric and TENGs was rectified using two bridge rectifiers, yielding an open-circuit voltage of 35.3 V. The device successfully demonstrated its capability by lighting up 21 LED bulbs through finger tapping and charging a 10 μ F capacitor to approximately 9 V within 400 s, showcasing its potential for self-powered electronic applications.

10 | Summary and Prospects

In summary, 2D materials offer unprecedented prospects for energy harvesting around mechanical, optical, electrical, and thermal domains. Their distinctive combination of properties marks them as an attractive candidate for addressing the increasing demand for sustainable energy solutions. Through the resolution of significant issues and leveraging interdisciplinary collaborations, 2D materials are poised to revolutionize energy harvesting technologies, paving the way to an energy-efficient and more sustainable future. Moreover, these materials have great flexibility and mechanical resilience, along with the ability to effectively tune electrical and optical properties, which in turn make them a viable candidate for energy harvesting devices. These materials then offer innovative solutions for powering wearable electronics, IoT devices, smart sensors, and other emerging technologies. Yet, challenges remain, but opportunities abound; therefore, to further enhance the 2D material-based energy harvesting module, more work needs to be done despite significant efforts and achievements in related domains. For those researchers who are interested in developing and altering material characteristics through process control, there is still a challenge in certain cases regarding the process-structure-properties-performance relationship of new 2D materials beyond GR and MoS₂. Here, we look ahead to the research challenges and an appealing direction, accompanied by abundant research prospects (Figure 14).

10.1 | Scalability and Manufacturing Techniques

The advancement of sustainable energy harvesting modules relies greatly on the scalable production of large-area, good-quality 2D materials, achieved at a low cost while maintaining precise control of their shape and dimensions. However, current manufacturing techniques face challenges like varying product quality, low process yield, and the need for precise control of growth conditions. On the one hand, exfoliation and hydro/solvothermal synthesis approaches result in inconsistent 2D material size and thicknesses, whereas on the other hand, growth approaches such as CVD and PVD result in low yields and require a substrate and specific growth environment. To overcome these challenges, a holistic approach is required, combining fundamental knowledge with technological developments, that is, addressing manufacturing obstacles like defect control, scalability, and reducing substrate constraints. The synthesis of high-quality, large-area, atomically thin 2D flakes is critical for unlocking the full potential of 2D materials in practical applications. Despite notable successes with well-studied materials such as GR, h-BN, and TMDCs, current fabrication methods remain limited by their complexity, high costs, and resource-intensive processes. To enable broader adoption and scalability, it is essential to develop more efficient and economically viable synthesis techniques while maintaining the structural integrity and quality of the resulting 2D materials. Furthermore, phase engineering of 2D materials holds promise for investigating new material features and expanding their applications in sustainable and wearable energy harvesting. Encapsulation and structural optimization are also pivotal to address the instability of some 2D materials (e.g., MXenes) in an

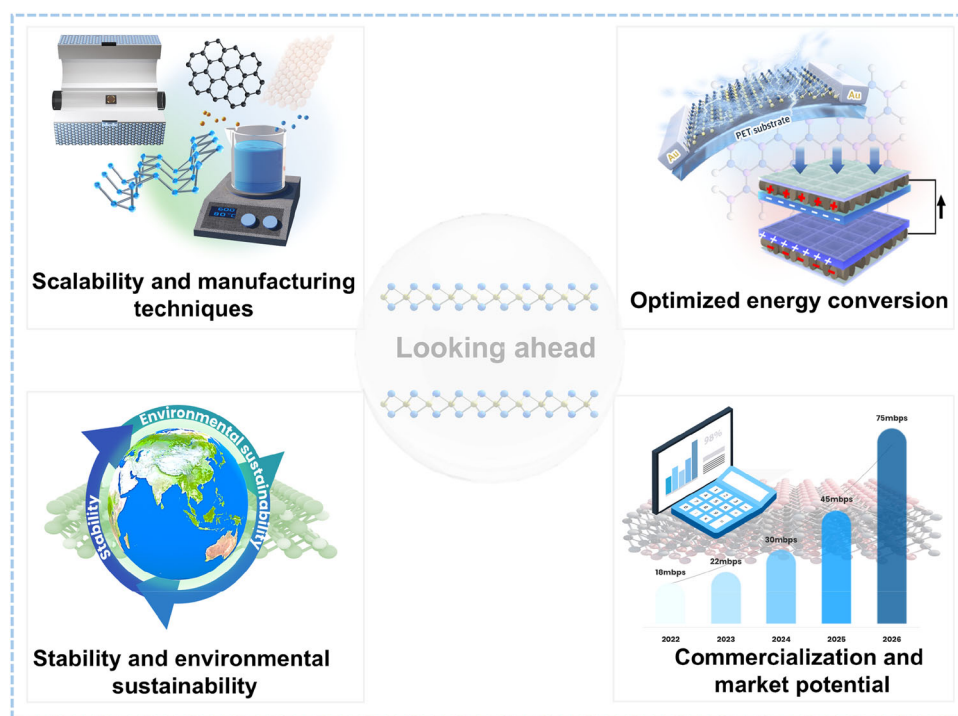


FIGURE 14 | 2D materials for flexible energy harvesting: The panel on “Scalability and manufacturing techniques” demonstrates that the sustainability of energy harvesting modules relies greatly on scalable production at low cost; hence, a holistic approach is required to combine fundamental knowledge with technological developments. A panel on “Stability and environmental sustainability” demonstrates what will happen to these materials and gadgets when their useful lives are over. Thus, searching for energy harvesting tools and materials that are easily recyclable from cradle to grave should be a top goal in the future. A panel on “Optimized energy conversion” demonstrates that exploring novel 2D materials and optimizing device structures by combining various types of energy harvesters to elaborate hybrid systems are the key focus areas in the future to obtain high-performance energy harvesters. Panel “Commercialization and market potential” demonstrates that 2D materials have prospects for effective energy harvesting, considering the growing demand for self-powered portable and wearable devices in a variety of industries.

environment. Inclusively, a comprehensive understanding of the relationship between manufacturing processes and material properties is crucial for producing 2D materials with precise dimensions, compositions, phases, and properties, which in turn are critical for sustainable and wearable energy harvesting.

10.2 | Stability and Environmental Sustainability

The durability and stability of 2D material-based energy harvesters must be prioritized, given their prolonged contact with human skin. As the device performance is impacted by daily movements exerting different stresses, alongside various factors like light exposure, moisture, and temperature variations, these also affect its long-term stability. Due to their remarkable mechanical strength, 2D materials are becoming increasingly more popular for integration into the production of wearable energy harvesters. However, their ultra-thin nature also renders them highly vulnerable to environmental degradation, particularly oxidative damage from exposure to oxygen and moisture. The primary cause of this deterioration is reactive oxygen species, which are highly oxidizing and contribute to the gradual breakdown of 2D materials' properties when stored in ambient conditions. In such circumstances, materials such as MoS_2 , MXene, and BP, for example, show instability, which raises questions regarding their durability and reliability in wearable devices. Therefore, to solve these problems and improve the stability and durability of 2D materials for sustainable energy

harvesting applications, specific techniques like the encapsulation approach, surface functionalization, incorporation of stabilizing agents, and a doping strategy to mitigate their degradation issues must be used. It is also critical to think about what will happen to these materials and gadgets when their useful lives are over; thus, the search for energy harvesting tools and materials that are easily recyclable from cradle to grave should be a top goal. Additionally, biodegradable energy harvesters would be ideal for applications with a short lifespan. Furthermore, construction of a complete and environmentally friendly system for the fabrication, storing, and retrieval of energy harvesters based upon 2D materials should also be prioritized, as the majority of 2D materials, whether they are disposed of or prepared, contaminate the environment. To eliminate this, the researchers should focus on developing green synthetic strategies to produce 2D materials. This will involve investigating methods like water-based synthesis, microwave-assisted synthesis, supercritical fluid region, and bio-inspired approaches to reduce the environmental footprint of 2D material production, as well as using sustainable and environmentally friendly precursors and solvents.

10.3 | Optimized Energy Conversion

Despite significant attention and advancements, energy harvesters still show relatively low power generation capabilities (ranging from μW to mW), which is insufficient for powering

wearable devices. Consequently, there is still substantial research scope in enhancing these devices to optimize power densities and power conversion efficiencies. In this regard, exploring novel 2D materials and optimizing device structures are the key focus areas to obtain high-performance energy harvesters. Though considerable studies have investigated the physicochemical properties of 2D materials, a comprehensive understanding of the bottleneck science is still lacking, suggesting opportunities for further research. Additionally, maximization of output power can be accomplished by combining various types of energy harvesters to elaborate hybrid systems. For example, merging PENGs and TENGs could effectively enhance power density, and similarly, merging PENGs/TENGs with thermoelectric nanogenerators or PV could further enhance power generation abilities. Hence, despite challenges, optimizing energy conversion efficiency along with power densities remains a promising avenue for further research and development in this field.

10.4 | Commercialization and Market Potential

For practical deployment, it is essential to develop scalable energy harvesting technologies that can be efficiently produced. The focus should be on creating materials that can be synthesized using a straightforward, cost-effective process with minimal resources and energy requirements, ensuring economic feasibility and sustainability at industrial scales. Therefore, the primary goal is to create ambient energy harvesters that are cost-effective and capable of generating sufficient power for their intended applications. To ensure commercialization and market potential, the focus should be on using abundant materials and energy-efficient manufacturing techniques, even as cost estimations evolve. The distinct qualities of 2D materials and the increasing need for sustainable energy solutions make their commercialization and market potential for energy harvesting promising. 2D materials have prospects for effective energy harvesting, considering the growing demand for self-powered devices in a variety of industries, including wearable electronics, the IoT, and health care. For industrialization, nevertheless, several issues, including scalability techniques (e.g., roll-to-roll, batch-to-batch, etc.), large-area uniformity, manufacturing costs, integration with current technology, and reproducibility, must be resolved. Despite these obstacles, the market potential is still substantial due to ongoing improvements in the synthesis of 2D materials, manufacture of devices, and increased public awareness of sustainable energy sources. As research and development efforts continue, 2D materials are expected to play a crucial role in the future of energy harvesting technologies. In a nutshell, to facilitate the commercial sustainability of technologies based on 2D materials, it is imperative to tackle the obstacles encountered with the use of these materials during laboratory investigations. This entails the establishment and stabilization of a specialized production line designed specifically for manufacturing devices utilizing 2D materials.

10.5 | Future Vision of Energy Harvesting

Throughout history, early civilizations primarily harnessed natural energy sources such as wind, water currents, firewood,

and animal power to meet their energy needs. However, the advent of the Industrial Revolution marked a paradigm shift, leading to the widespread adoption of fossil fuels as the dominant energy source. Although this transition fueled unprecedented economic and technological advancements, it also raised concerns regarding resource depletion and severe environmental repercussions. The overreliance on fossil fuels has significantly contributed to the ongoing energy crisis, necessitating an urgent transition toward sustainable alternatives. Therefore, compact energy harvesters play a pivotal role in advancing the IoT by enabling a seamlessly interconnected ecosystem of smart sensors for applications such as smart homes, cities, health care, and industrial automation. With the projected deployment of trillions of IoT sensors, dependence on conventional batteries presents significant environmental and logistical challenges. In contrast, energy harvesters offer a sustainable solution by continuously powering these devices using ambient energy sources, thereby enhancing their operational efficiency and durability. Additionally, the rise of wearable electronics, particularly for health monitoring, further underscores the need for compact energy harvesters, as they can harness body-generated energy, such as heat and motion, to ensure uninterrupted functionality. Although the realization of “green energy anytime, anywhere” remains a long-term goal, energy harvesting technologies are already presenting significant opportunities. PV harvesters have been successfully commercialized for powering smart sensors, whereas advancements in triboelectric, thermoelectric, piezoelectric, and RF energy harvesting have demonstrated their potential to power wearable electronics. These emerging technologies pave the way for self-sustaining energy solutions, reducing the dependence on conventional power sources and promoting the development of autonomous, low-maintenance electronic systems. Ocean wave energy harvesting has recently emerged as a promising approach for multimodal energy generation from marine-based renewable sources. Ocean waves represent an abundant and consistent energy resource, largely unaffected by environmental fluctuations, making them ideal for large-scale applications. However, their potential remains underutilized due to the absence of efficient energy conversion technologies. TENGs, with their inherent adaptability and capacity for seamless integration, provide a viable strategy for harvesting both wind and wave energy, positioning them as key players in sustainable energy advancements. The integration of 2D materials, including GR and its derivatives, transition-metal chalcogenides, layered oxides, and MXenes, into TENG systems presents significant opportunities for enhancing ocean wave energy harvesting. Their layered architecture facilitates improved charge transfer dynamics and enhances dielectric properties between the tribo-positive and tribo-negative layers, and their extensive surface area contributes to greater charge accumulation. Specifically, MXenes, with their exceptional electrical conductivity and surface termination-induced electronegativity, serve as effective electronegative fillers, boosting charge density and optimizing the performance of TENGs for ocean energy conversion applications. Moreover, integrating human motion-based energy harvesting systems presents a viable and sustainable alternative to conventional renewable energy sources, offering significant environmental, social, and economic benefits within the framework of smart cities. Given the high frequency of mechanical interactions in urban environments

(airports, shopping malls, railway stations, amusement parks, and pedestrian areas), these settings provide an ideal platform for energy harvesting. By incorporating 2D material-based TENGs into flooring, handrails, and other public contact places, the wasted biochemical energy from human activities can be effectively captured to convert it into electricity, paving the way for the development of self-powered urban environments.

Overall, potential and obstacles coexist in this intriguing field, but we believe that 2D materials will emerge as a novel material central for energy harvesting in the next decade, and energy harvesting devices will be one of the key technologies for resolving the environmental and energy crises in the future era.

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Conflicts of Interest

The authors declare no conflicts of interest.

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